# ISOENERGETIC KAM, CONFINEMENT AND THE RESTRICTED THREE-BODY PROBLEM: BLUEPRINT

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Abstract. A proof plan for the four frozen statements of track I: the isoenergetic a-posteriori KAM theorem (I-G1), all-time action confinement for two degrees of freedom (I-HEAD), the Kepler bordered determinant (I-KEPLER), and the near-circular region of the restricted circular planar three-body problem (I-R3BP). Results of the blueprint of track K are cited by their labels [K:`bp:p1:...`] and [K:`bp:p2:...`]. Every result is labelled and cites its pinned source.

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## 1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the four frozen theorems of track I and the frozen definitions they use: the fifteen definitions of `KAM/Frozen/Iso/Defs/` (owner approval D20, 27 October 2026) and the definitions of `KAM/Frozen/Defs/` of track K. The Lean declarations are authoritative; the statements below are faithful transcriptions, with the quantifier order and the order in which constants are chosen exactly as in Lean. Results of track K's blueprint are cited as [K:`bp:p1:...`] and [K:`bp:p2:...`]; they are not re-proved here.

### 1.1. **Conventions.**

- Ambient spaces are $\mathbb{R}^m = (\texttt{Fin}\ m \to \mathbb{R})$ and $\mathbb{C}^m = (\texttt{Fin}\ m \to \mathbb{C})$. Angles are $\theta \in \mathbb{C}^n$, actions $I \in \mathbb{C}^n$, phase points $z = (\theta, I) \in \mathbb{C}^n \times \mathbb{C}^n$. Functions of $\theta$ are 1-periodic, as in Part 0 of track K ([K:`bp:p0:conventions`]).
- *Norms.* $|v| = \|v\| = \max_i |v_i|$ on $\mathbb{R}^m$ and $\mathbb{C}^m$; on a product the maximum of the two component norms. This is the norm of `Metric.ball`, of `dist` in `actionNbhd`, of $\|DH\|$ (operator norm of a $\mathbb{C}$-linear map, `fderiv`), and of $\|D^2H\|$ (norm of the bilinear map `iteratedFDeriv`). A row-sum bound $\sum_j |M_{ij}| \le c$ for every $i$ is written $|M| \le c$. The norm of a complex number is its modulus.
- The strip $T_\rho = \{\theta \in \mathbb{C}^n : |\operatorname{Im} \theta_i| < \rho\ \forall i\}$ is open; for $\rho > 0$ it contains $\mathbb{R}^n$, and it is invariant under $\mathbb{Z}^n$ and $\mathbb{R}^n$ translations. The set `Set.pi univ (Icc 0 1)` $= [0,1]^n$ carries Lebesgue measure, of total mass one; $\langle u \rangle = \int_{[0,1]^n} u(\theta)\, d\theta$ for a function evaluated at real angles. Averages of vectors or matrices are entrywise.
- A map is *holomorphic on an open set* $U$ if it is `DifferentiableOn` $\mathbb{C}$ there. On an open set all derivatives `fderiv`, `iteratedFDeriv` used below are genuine (Lean totality, Remark 1.13).
- Real and complex points. A real vector $x \in \mathbb{R}^m$ is identified with $(x_i) \in \mathbb{C}^m$; "$(I : \mathbb{C}) \in S$" for $S \subset \mathbb{C}^n$ means this identification lies in $S$. The real phase points of $D \subset \mathbb{C}^n \times \mathbb{C}^n$ are the $(\theta, I) \in \mathbb{R}^n \times \mathbb{R}^n$ whose image lies in $D$.
- *Two phase-space conventions.* In I-G1 and I-HEAD, a phase point is $(\theta, I)$ (angles first) and $X_H = (\partial_I H, -\partial_\theta H)$. In the restricted three-body problem a phase point is $z = (q, p)$ with position $q \in \mathbb{R}^2$ and canonical momentum $p \in \mathbb{R}^2$ (position first, as in the Lean carrier $(\texttt{Fin}\ 2 \to \mathbb{R}) \times (\texttt{Fin}\ 2 \to \mathbb{R})$); there `radius` is the Euclidean length, not the max norm. The norm $\|\cdot\|$ of I-HEAD is the max norm; the bounds in the conclusion of I-R3BP use rad.

### 1.2. **Frozen definitions of track K used by track I.**

**Definition 1.1** (Strip, action neighbourhood)**.** $T_\rho = \{\theta \in \mathbb{C}^n : \forall i,\ |\mathrm{Im}\,\theta_i| < \rho\}$ (`complexStrip` $n$ $\rho$). For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ \|I - J\| < r\}$ (`actionNbhd`); it is open. Thus for real $I$: $I \in B_r$ iff $\exists J \in B$ with $\|I - J\| < r$.
Frozen Lean declaration: `KAM.complexStrip, KAM.actionNbhd`
Source: HL19 §2.1; see [K:`bp:p0:domains`].

**Definition 1.2** (Diophantine vectors)**.** $\omega \in \mathbb{R}^n$ is $(\gamma, \tau)$-Diophantine if for every $k \in \mathbb{Z}^n \setminus \{0\}$ $\gamma/\big(\sum_i |k_i|\big)^\tau \le |k \cdot \omega|$ (real power).
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; see [K:`bp:p0:diophantine`].

**Definition 1.3** (Hamiltonian vector fields)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and a real phase point $z$: $X_H(z) = \big(i \mapsto DH(z)(0, e_i),\ i \mapsto -DH(z)(e_i, 0)\big)$, with $D$ the real Fréchet derivative. For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ the same formula with the complex derivative gives $X_H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$. Thus $X_H = (\partial_I H, -\partial_\theta H)$.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.hamiltonianVectorFieldℂ`
Source: HL19 Rem. 2.1, p. 6.

**Definition 1.4** (Real Hamiltonian, frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and a real phase point $(\theta, I)$, $\mathcal{H}_{h,f}(\theta, I) = \mathrm{Re}\big(h(I) + f(\theta, I)\big)$ (`realHamiltonian`). The frequency map is $\nabla h(I)_i = D\big(J \mapsto \mathrm{Re}\, h(J)\big)(I)\, e_i$ for real $I$ (`frequencyMap`), a real function $\mathbb{R}^n \to \mathbb{R}^n$.
Frozen Lean declaration: `KAM.realHamiltonian, KAM.frequencyMap`
Source: see [K:`bp:p0:nearlyintegrable`].

**Definition 1.5** (Torus data on strips)**.** Let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$.

- $K$ is *primary on* $T_\rho$ if $K(\theta + k) = K(\theta) + (k, 0)$ for $\theta \in T_\rho$, $k \in \mathbb{Z}^n$ (`IsPrimaryOnStrip`); *real on reals* if every $K(\theta)$, $\theta \in \mathbb{R}^n$, is a real phase point (`IsRealOnReals`).
- Bilinear dot product $u \cdot v = \sum_i (u_{1,i} v_{1,i} + u_{2,i} v_{2,i})$ on $\mathbb{C}^n \times \mathbb{C}^n$ (`dotℂ`), $\Omega_0(a, b) = (-b, a)$ (`omegaZero`).
- Invariance error $E_{H,K,\omega}(\theta) = X_H(K(\theta)) - DK(\theta)\omega$ (`invarianceError`).
- Gram matrix $G_K(\theta)_{ij} = DK(\theta)e_i \cdot DK(\theta)e_j$ (`gramMatrix`), a complex $n \times n$ matrix.
- Normal frame $N_K(\theta)_{\cdot j} = \sum_k (G_K^{-1})_{kj}\, \Omega_0\big(DK(\theta)e_k\big) \in \mathbb{C}^n \times \mathbb{C}^n$ (`normalFrame`), with the matrix inverse of Lean.
- Torsion $T_{H,K,\omega}(\theta)_{ij} = N_{\cdot i}(\theta) \cdot \Omega_0\big(DX_H(K(\theta))N_{\cdot j}(\theta) - D_\theta(N_{\cdot j})(\theta)\,\omega\big)$ (`torsion`); $N_{\cdot j}$ is regarded as a function of $\theta$.

Frozen Lean declaration: `KAM.IsPrimaryOnStrip, KAM.IsRealOnReals, KAM.dotℂ, KAM.omegaZero, KAM.invarianceError, KAM.gramMatrix, KAM.normalFrame, KAM.torsion`
Source: HL19 (2.2), (3.2)–(3.9), pp. 7, 16–17; see [K:`bp:p0:stripdata`].

### 1.3. **Frozen definitions of track I.**

**Definition 1.6** (Energy average and extended torsion)**.** For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $K$ as above, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$, with index set $\{1, \dots, n\} \sqcup \{*\}$:

$$\bar{E}_H(K) = \int_{[0,1]^n} H\big(K(\theta)\big)\, d\theta \in \mathbb{C}, \qquad \mathsf{T}_H(\theta) = \begin{pmatrix} T_{H,K,\omega}(\theta) & \omega \\ \big(DH(K(\theta))N_{\cdot b}(\theta)\big)_b & 0 \end{pmatrix},$$

the $(n+1) \times (n+1)$ complex matrix whose upper-left block is the torsion, last column $(\omega, 0)$ (real entries, viewed in $\mathbb{C}$), and last row $b \mapsto DH(K(\theta))\big(N_{\cdot b}(\theta)\big)$ (complex derivative of $H$ applied to the normal frame), last entry 0. The averaged extended torsion is $\langle \mathsf{T}_H \rangle = \big(\int_{[0,1]^n} \mathsf{T}_H(\theta)_{ij}\, d\theta\big)_{ij}$, the integral taken over real angles.
Frozen Lean declaration: `KAM.Iso.energyAverage, KAM.Iso.extendedTorsion, KAM.Iso.extendedTorsionAverage`
Source: HL19 (3.13)–(3.14), p. 19, with $c = H$, $d = n$ (Case III), $\hat{\omega} = \omega$; the last row is $DH(K)N$ even off invariance, not $\omega^\top$ (HL19 (3.17) is the invariant case, Rem. 3.7, p. 19).

**Definition 1.7** (Bordered Hessian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and real $I$, the real $(n+1)\times(n+1)$ matrix

$$\mathsf{B}_h(I) = \begin{pmatrix} \big(\partial_b(\nabla h)_a(I)\big)_{a,b} & \nabla h(I) \\ \nabla h(I)^\top & 0 \end{pmatrix},$$

where $\partial_b(\nabla h)_a(I) = D(\nabla h)(I)e_b$ read at $a$, the real derivative of the frequency map (`borderedHessian`). No symmetry is assumed in the definition.

Frozen Lean declaration: `KAM.Iso.borderedHessian`

Source: Chi05 p. 17 (display after (3.43)); HL19 Rem. 3.7, p. 19.

**Definition 1.8** (Hamiltonian solutions with action in $B$)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $B \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, write $\mathrm{Sol}(H, B, J, z)$ if for all $t \in J$: $z(t)_2 \in B$, $H$ is real differentiable at $z(t)$, and $z$ is differentiable at $t$ with $z'(t) = X_H(z(t))$ (`HasDerivAt`).

Frozen Lean declaration: `KAM.Iso.IsHamiltonianSolutionOn`

Source: ours (D20); the domain $B$ and the differentiability of $H$ are part of the solution concept.

**Definition 1.9** (Planar Euclidean length, the primaries, collisions)**.** For $q \in \mathbb{R}^2$: $\rho_2(q)^2 = q_0^2 + q_1^2$ (`radiusSq`) and $\mathrm{rad}(q) = \sqrt{q_0^2+q_1^2}$ (`radius`; the radicand is nonnegative). For $\mu \in \mathbb{R}$: $a_1(\mu) = (-\mu, 0)$, $a_2(\mu) = (1-\mu, 0)$ (`primary`$_1$, `primary`$_2$). A phase point $z = (q,p) \in \mathbb{R}^2 \times \mathbb{R}^2$ is *collision free* (`CollisionFree` $\mu$) if $\mathrm{rad}(q - a_1(\mu)) > 0$ and $\mathrm{rad}(q - a_2(\mu)) > 0$.

Frozen Lean declaration: `KAM.Iso.radiusSq`, `KAM.Iso.radius`, `KAM.Iso.primary`$_1$, `KAM.Iso.primary`$_2$, `KAM.Iso.CollisionFree`

Source: Chi05 (3.34), (3.37), p. 12–13 (barycentric rotating frame, unit separation, total mass one); ours.

**Definition 1.10** (Restricted Hamiltonian and field)**.** On collision-free points $z = (q,p)$, with $r_j = \mathrm{rad}(q - a_j(\mu))$:

$$\mathcal{H}^{\mathrm{R3}}_\mu(q,p) = \tfrac{1}{2}|p|_2^2 + q_1p_0 - q_0p_1 - \frac{1-\mu}{r_1} - \frac{\mu}{r_2}, \qquad |p|_2^2 := p_0^2 + p_1^2,$$

$$V_\mu(q,p) = \Big((p_0+q_1,\ p_1-q_0),\ (p_1,-p_0) - \frac{1-\mu}{r_1^3}(q-a_1) - \frac{\mu}{r_2^3}(q-a_2)\Big).$$

Thus $V_\mu = (\partial_p \mathcal{H}^{\mathrm{R3}}_\mu, -\partial_q \mathcal{H}^{\mathrm{R3}}_\mu)$ on the open collision-free set. A function $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ is a *restricted solution on* $J$, $\mathrm{RSol}_\mu(J, z)$, if for every $t \in J$ the point $z(t)$ is collision free and $z$ is differentiable at $t$ with $z'(t) = V_\mu(z(t))$.

Frozen Lean declaration: `KAM.Iso.restrictedHamiltonian`, `KAM.Iso.restrictedVectorField`, `KAM.Iso.IsRestrictedSolutionOn`

Source: Chi05 (1.2), p. 2 and (3.34)–(3.39), pp. 12–14, in barycentric rotating coordinates with canonical momentum (not rotating velocity); the equations $V_\mu$ are the Hamilton equations of $\mathcal{H}^{\mathrm{R3}}_\mu$ (the check is part of the proofs, see PLAN.md).

**Definition 1.11** (Near-circular region)**.** For $a_-, a_+, e_-, e_+ \in \mathbb{R}$, $\mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is the set of $(q,p)$ for which there exist $r, a, e \in \mathbb{R}$ with

$$r > 0,\ r^2 = q_0^2 + q_1^2, \quad a > 0, \quad 0 < e < 1, \quad a_- < a < a_+, \quad e_- < e < e_+,$$

$$\tfrac{1}{2}|p|_2^2 - \frac{1}{r} = -\frac{1}{2a}, \qquad q_0p_1 - q_1p_0 > 0, \qquad (q_0p_1 - q_1p_0)^2 = a(1-e^2).$$

(`NearCircular`.) The three numbers $r, a, e$ are determined by $(q,p)$, so no choice is involved.

Frozen Lean declaration: `KAM.Iso.NearCircular`

Source: Chi05 (2.24)–(2.26), p. 9: $E = -1/(2a)$, $\Gamma^2 = a(1-e^2)$ in the units $\mu = M = 1$; referred to $\mu = 0$ (primary at the origin).

**Definition 1.12** (The Kepler bordered matrix)**.** For $\kappa, \Omega \in \mathbb{R}$ and $L > 0$,

$$\mathsf{K}_{\kappa,\Omega}(L) = \begin{pmatrix} -3\kappa/L^4 & 0 & \kappa/L^3 \\ 0 & 0 & -\Omega \\ \kappa/L^3 & -\Omega & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3},$$

the bordered Hessian $\left(\begin{smallmatrix} D^2h & \nabla h \\ \nabla h^\top & 0 \end{smallmatrix}\right)$ of $h(L,G) = -\kappa/(2L^2) - \Omega G$ (`keplerBorder`, $L$ in the subtype $\{L : \mathbb{R} \mid L > 0\}$).
Frozen Lean declaration: `KAM.Iso.keplerBorder`
Source: Chi05 p. 17 ($\kappa = \Omega = 1$).

### 1.4. **Lean totality.**

*Remark* 1.13 (Lean totality). The frozen definitions use `fderiv` (zero where a function is not differentiable), the matrix inverse (zero for a singular matrix), the Bochner integral (zero for a non-integrable function), division, `Real.sqrt` (zero for a negative argument) and real powers. The statements guard them as follows; each proof must establish the guarded facts before using the defining formulas.

(T1) *Derivatives of $H$ in I-G1.* $D$ is open, $H$ is holomorphic on $D$, so $DH$, $X_H$, $DX_H$ and $D^2H$ are genuine derivatives at every point of $D$; $K$ is holomorphic on the open strip $T_\rho$ and $B_\mu(K(\theta)) \subset D$, so $H \circ K$, $X_H \circ K$ and $DK$ are genuine on $T_\rho$.

(T2) *Inverses in I-G1.* The Gram matrix inverse is used only under the hypothesis $\det G_K(\theta) \in \mathbb{C}^\times$ on $T_\rho$, and the inverse of the averaged extended torsion only under $\det\langle \mathsf{T}_H \rangle \in \mathbb{C}^\times$. Then $N_K$ is genuine and holomorphic on $T_\rho$ (by the holomorphic inverse, [K:`bp:p1:holomorphic-inverse`]), so $DN_{\cdot j}$ and $T$ are genuine. Gram invertibility must be proved before using a normal frame or torsion for $K_\infty$; no such fact is needed to interpret the frozen final invariance error or energy equality, which use only $DK_\infty$ and $X_H$.

(T3) *Integrals in I-G1.* The cube $[0,1]^n \subset \mathbb{R}^n \subset T_\rho$ is compact, $\rho > 0$, and the integrands $H \circ K$ and the entries of $\mathsf{T}_H$ are continuous on it by (T1)–(T2), hence integrable ([K:`bp:p1:ext-integral`]). Integrability is not a hypothesis; it must be proved.

(T4) *Powers and division in I-G1.* $\gamma > 0$ and $\rho > 0$, so $\gamma^4$, $\gamma^2$, $\rho^{4\tau}$, $\rho^{2\tau}$ are positive and the quotients are genuine; $\gamma/(\sum|k_i|)^\tau$ has a positive base since $k \neq 0$.

(T5) *I-HEAD.* $h$ is holomorphic on the open set $B_{r_0}$, so $\nabla h$ and $D(\nabla h)$ are genuine at real points of $B_{r_0}$ (an open subset of $\mathbb{R}^2$) and $\mathsf{B}_h$ is genuine. The solution predicate Sol includes differentiability of $\mathcal{H}_{h,f}$ at each visited point, which also guards $X_{\mathcal{H}}$; it holds for $z(t)_2 \in B \subset B_{r_0}$ and real $\theta$ because $f$ is holomorphic on $T_{s_0} \times B_{r_0}$.

(T6) *I-KEPLER, I-R3BP.* $L > 0$ guards every denominator of $\mathsf{K}_{\kappa,\Omega}(L)$. Every inverse distance in $\mathcal{H}^{\mathrm{R3}}_\mu$, $V_\mu$ is guarded by collision freedom, and the radicand of `radius` is a sum of squares. In $\mathcal{N}$, $r > 0$ and $a > 0$ precede the energy equation.

### 1.5. **The frozen theorems.**

**Theorem 1.14** (I-G1: isoenergetic a-posteriori KAM)**.** *Let $n \geq 2$ and $\tau \in \mathbb{R}$ with $\tau \geq n-1$; let $\rho, \rho_\infty \in \mathbb{R}$ with $0 < \rho_\infty < \rho$; let $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \sigma_\omega, \nu \in \mathbb{R}$ with $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma > 0$, $\sigma_\omega > 1$, $\nu > 0$; let $\omega^* \in \mathbb{R}^n$, $\omega^* \neq 0$. Then there exist $C_1, C_2, C_3 > 0$ such that for all $D \subset \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, and $s, \gamma, e, E_0 \in \mathbb{R}$ satisfying*

*(I1) $D$ is open and $(\theta + k, I) \in D$ whenever $(\theta, I) \in D$, $k \in \mathbb{Z}^n$;*
*(I2) $H$ is holomorphic on $D$, and $H(\theta + k, I) = H(\theta, I)$ on $D$ for $k \in \mathbb{Z}^n$;*
*(I3) for every real phase point $(\theta, I)$ lying in $D$, $\operatorname{Im} H(\theta, I) = 0$;*
*(I4) for every $z \in D$: $\|DH(z)\| \leq M$, $\|X_H(z)\| \leq M$, $\|D(X_H)(z)\| \leq M$, $\|D^2(X_H)(z)\| \leq M$, $\|D^2H(z)\| \leq M$;*
*(I5) $K$ is holomorphic on $T_\rho$, primary on $T_\rho$, and real on reals;*
*(I6) for every $\theta \in T_\rho$: $B_\mu(K(\theta)) \subset D$ (open max-norm ball) and $\|DK(\theta)\| \leq \sigma_K - \mu$;*
*(I7) for every $\theta \in T_\rho$: $\det G_K(\theta) \neq 0$ and $|G_K(\theta)^{-1}| \leq \sigma_B - \mu$ (maximal row sum);*
*(I8) $\det\langle \mathsf{T}_H \rangle \neq 0$ with $\mathsf{T}_H$ built from $(H, K, s\omega^*)$, and $|\langle \mathsf{T}_H \rangle^{-1}| \leq \sigma_T - \mu$ (maximal row sum);*
*(I9) $1 < s < \sigma_\omega$, $\nu \leq (s-1)\|\omega^*\|$, $\nu \leq (\sigma_\omega - s)\|\omega^*\|$;*
*(I10) $0 < \gamma \leq \bar\gamma$, $\omega^* \in \mathcal{D}_{\gamma,\tau}$, $0 \leq e$;*
*(I11) $\|E_{H,K,s\omega^*}(\theta)\| \leq e$ for every $\theta \in T_\rho$;*

*(I12)* $|\bar{E}_H(K) - E_0| \le e$;
*(I13)* $C_1 e/(\gamma^4 \rho^{4\tau}) < 1$,

*there are* $K_\infty : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ *and* $s_\infty \in \mathbb{R}$ *with* $1 < s_\infty < \sigma_\omega$ *such that*

*(O1)* $K_\infty$ *is holomorphic on* $T_{\rho_\infty}$*, primary on* $T_{\rho_\infty}$ *and real on reals;*
*(O2)* $E_{H,K_\infty,s_\infty\omega^*}(\theta) = 0$ *for all* $\theta \in T_{\rho_\infty}$*;*
*(O3)* $H(K_\infty(\theta)) = E_0$ *for all* $\theta \in T_{\rho_\infty}$ *(pointwise);*
*(O4)* $B_{\mu/2}(K_\infty(\theta)) \subset D$ *and* $\|DK_\infty(\theta)\| \le \sigma_K - \mu/2$ *for* $\theta \in T_{\rho_\infty}$*;*
*(O5)* $\|K_\infty(\theta) - K(\theta)\| \le C_2 e/(\gamma^2 \rho^{2\tau})$ *for* $\theta \in T_{\rho_\infty}$*;*
*(O6)* $\|s_\infty\omega^* - s\omega^*\| \le C_3 e/(\gamma^2 \rho^{2\tau})$.

*Here* $C_1, C_2, C_3$ *are chosen after all of* $n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma}, \sigma_\omega, \nu, \omega^*$ *and before* $D, H, K, s, \gamma, e, E_0$*; hence they may depend on the former, and on none of the latter.*

Frozen Lean declaration: `KAM.Iso.isoenergetic_kam`

Source: *HL19 Theorem 3.6 and Rem. 3.7, pp. 18–19, in the canonical case* $d = n$, $c = h$*; proof pp. 45–53. Deviations (all conservative, D-I-G1-UNIFORM, D-I-ENERGY, notes.md): ray data,* $\bar{\gamma}$ *and the margins are fixed first; the strict inequalities of (3.16) are replaced by* $\le$ *(candidate erratum: with* $E_c = 0$ *the strict bounds read* $0 < 0$*); the energy conclusion is pointwise.*

*Remark* 1.15 (Reading of I-G1). (a) The frequency of the approximate torus is $\omega = s\omega^*$ on the open ray $\Theta = \{s'\omega^* : 1 < s' < \sigma_\omega\}$. The Diophantine condition is imposed on $\omega^*$, hence (by homogeneity of $k \mapsto k \cdot \omega$) $s'\omega^*$ is nonresonant for every $s' > 0$, with constant $\gamma s'$. (b) $\nu$ enters only through (I9) and through the choice of $C_1, C_2, C_3$; it measures the distance of $s\omega^*$ to the two *endpoints* $\omega^*$ and $\sigma_\omega\omega^*$ of the segment $\Theta$, i.e. $\min(s - 1, \sigma_\omega - s)|\omega^*| \ge \nu$ (HL19 $\mathrm{H}'_5$, $\operatorname{dist}(\omega, \partial\Theta) > 0$, made quantitative, with $\partial\Theta$ the boundary *relative to the line* $\mathbb{R}\omega^*$; for $n \ge 2$ the boundary of $\Theta$ in $\mathbb{R}^n$ is all of $\bar{\Theta}$, so the ambient reading would be vacuous). The frozen Lean states the two endpoint inequalities explicitly. (c) Taking $e = 0$ gives the exact-torus case; then (I13) holds for every $C_1$, and (O5), (O6) force $K_\infty = K$ on $T_{\rho_\infty}$ and $s_\infty = s$. (d) The row-sum bound in (I7)–(I8) is the max-norm operator norm on $\mathbb{C}^n$, $\mathbb{C}^{n+1}$ respectively.

**Theorem 1.16** (I-HEAD: all-time action confinement for two degrees of freedom)**.** *Let* $B \subset \mathbb{R}^2$ *be open, bounded and nonempty;* $s_0, r_0 > 0$*;* $h$ *holomorphic on* $B_{r_0}$ *with* $\operatorname{Im} h(I) = 0$ *for every real* $I$ *with* $I \in B_{r_0}$ *and*

$$\det \mathsf{B}_h(I) \neq 0 \qquad \text{for every real } I \in B_{r_0}.$$

*Let* $B' \subset \mathbb{R}^2$ *be open with* $\overline{B'}$ *compact and* $\overline{B'} \subset B$*, and let* $\eta > 0$*. Then there is* $\varepsilon_0 > 0$ *such that for every* $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ *and* $\varepsilon \in \mathbb{R}$ *with*

- $f$ *holomorphic on* $T_{s_0} \times B_{r_0}$*,* $f(\theta + k, I) = f(\theta, I)$ *there for* $k \in \mathbb{Z}^2$*, and* $\operatorname{Im} f(\theta, I) = 0$ *for real* $\theta$ *and real* $I \in B_{r_0}$*;*
- $0 \le \varepsilon < \varepsilon_0$ *and* $|f(z)| \le \varepsilon$ *on* $T_{s_0} \times B_{r_0}$*,*

*and every* $z_0 = (\theta_0, I_0) \in \mathbb{R}^2 \times \mathbb{R}^2$ *with* $I_0 \in B'$*, there is* $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ *with* $z(0) = z_0$*,* $\operatorname{Sol}(\mathcal{H}_{h,f}, B, \mathbb{R}, z)$*,* $\|z(t)_2 - I_0\| < \eta$ *for all* $t \in \mathbb{R}$*, and: for all* $a < 0 < b$ *and all* $y : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ *with* $y(0) = z_0$ *and* $\operatorname{Sol}(\mathcal{H}_{h,f}, B, (a, b), y)$ *one has* $y(t) = z(t)$ *for all* $t \in (a, b)$.

Frozen Lean declaration: `KAM.Iso.all_time_confinement`

Source: *ours (D-I-CONFINEMENT): assembled from I-G1, separation in an energy cylinder (Arnold-type barriers; cf. CC03 Theorem 1.1, p. 7, a numerical one-energy precedent with a truncated potential, not used) and ODE continuation.*

*Remark* 1.17 (Reading of I-HEAD). (a) The constant $\varepsilon_0$ depends on $B, s_0, r_0, h, B', \eta$ only (it may not depend on $f$, $\varepsilon$, $z_0$). (b) $\mathsf{B}_h(I)$ is the matrix of Definition 1.7 with $n = 2$; its determinant is the isoenergetic nondegeneracy of HL19 $\mathrm{H}'_4$ at the torus $\theta \mapsto (\theta, I)$: for $H = h$, $K(\theta) = (\theta, I)$, $\omega = \nabla h(I)$ one computes $N_{\cdot b} = (0, e_b)$, $T = D(\nabla h)(I)$, last row $\nabla h(I)^\top$, so $\mathsf{T}_h = \mathsf{B}_h(I)$ exactly (to be proved in part p1, see PLAN.md). (c) The conclusion contains global existence for all real $t$ and

uniqueness among every solution on any interval $(a,b) \ni 0$ that remains in $\{I \in B\}$; in particular it contains uniqueness of $z$ among solutions confined to $B$. (d) The norm $\|\cdot\|$ is the max norm on $\mathbb{R}^2$.

**Theorem 1.18** (I-KEPLER: the classical isoenergetic determinant)**.** *For all $\kappa, \Omega > 0$ and $L > 0$:* $\det \mathsf{K}_{\kappa,\Omega}(L) = 3\kappa\Omega^2/L^4$ *and* $\det \mathsf{K}_{\kappa,\Omega}(L) > 0$.
*Frozen Lean declaration:* `KAM.Iso.kepler_bordered_det`
*Source: Chi05 p. 17 ($\kappa = \Omega = 1$: $3/L^4$); general $\kappa, \Omega$ is the same cofactor computation (ours).*

**Theorem 1.19** (I-R3BP: near-circular region of the restricted three-body problem)**.** *Let $a_-, a_+, e_-, e_+ \in \mathbb{R}$ with $0 < a_- < a_+$, $0 < e_- < e_+ < 1/4$ and $a_+(1+e_+) < 1$. Then the set $\mathcal{N} = \mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is open and nonempty, and there are $\mu_0, d, R \in \mathbb{R}$ with $0 < \mu_0 \le 1/2$, $d > 0$, $R > 0$ such that for every $\mu \in [0, \mu_0)$ and every $z_0 \in \mathcal{N}$ there is $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $z(0) = z_0$,* $\mathrm{RSol}_\mu(\mathbb{R}, z)$ *and for every $t \in \mathbb{R}$*

$$\operatorname{rad}\big(q(t) - a_1(\mu)\big) \ge d, \quad \operatorname{rad}\big(q(t) - a_2(\mu)\big) \ge d,$$
$$\operatorname{rad}(q(t)) \le R, \quad \operatorname{rad}(p(t)) \le R, \qquad z(t) = (q(t), p(t)),$$

*and: for all $a < 0 < b$ and $y$ with $y(0) = z_0$ and* $\mathrm{RSol}_\mu((a,b), y)$, *$y(t) = z(t)$ on $(a,b)$. Here $\mu_0, d, R$ depend on $a_\pm, e_\pm$ only, not on $\mu$ or $z_0$.*
*Frozen Lean declaration:* `KAM.Iso.restricted_near_circular`
*Source: ours (D-I-PHYSICAL), derived from I-HEAD, I-KEPLER and the Delaunay coordinates of Chi05 §2–3, pp. 8–9, 12–17; not the numerical Victoria theorem of CC03.*

*Remark* 1.20 (Reading of I-R3BP). (a) $\mu = 0$ is included (a massless secondary at $(1,0)$). (b) The hypothesis $a_+(1+e_+) < 1$ says that the apocentre radius $a(1+e)$ of every orbit of the region is below $1-$margin, so the orbits stay inside the orbit of the secondary; $e_+ < 1/4$ and $e_- > 0$ keep the orbits away from the singularities $e = 0$, $e = 1$ of the Delaunay chart. (c) The region is $\mu$-independent and referenced to $\mu = 0$ (the primary at the origin); the primaries are at $a_1(\mu)$, $a_2(\mu)$ in the dynamics. (d) Collision exclusion for all times is part of the conclusion (distances $\ge d$).

## 2. Extended torsion and the bordered triangular system

This part collects the linear-algebra and first-order geometry of the isoenergetic scheme of HL19 §6 (pp. 45–51), for the frozen objects $\bar{E}_H$, $\mathsf{T}_H$, $\langle \mathsf{T}_H \rangle$ and $\mathsf{B}_h$ of Definitions 1.6 and 1.7. It proves five things.

(1) $\mathsf{T}_H = \mathsf{B}_h(I)$ exactly at the trivial torus $\theta \mapsto (\theta, I)$ (Lemma 2.6); this is why the nondegeneracy hypothesis of I-HEAD is the isoenergetic condition of I-G1.
(2) The last row of $\mathsf{T}_H$ is $\omega^\top - (\eta^L)^\top$ exactly, hence $\omega^\top$ on an invariant torus (Lemma 2.8); HL19 Rem. 3.7 only states the invariant case, with a sign slip.
(3) The bordered triangular system of HL19 Lemma 6.1 with its estimates (Lemma 2.12), and its exact origin as the reduction of the linearized isoenergetic equation in the frame (Lemma 2.14). *This is the only genuinely new analytic lemma*; HL19 print only "analogous to Lemma 5.1" for its proof, so the full proof is supplied.
(4) The approximate energy identities (Lemma 2.9): $D(H \circ K) = DH(K)DK$, $\partial_\omega(H \circ K) = -DH(K)E$, the compatibility $\langle \eta^N \rangle = 0$, and the size of the energy defect.
(5) Energy conservation of an invariant torus (Lemma 2.10): $E = 0$ and $\langle H \circ K \rangle = E_0$ give $H \circ K = E_0$ pointwise, the pointwise energy conclusion (O3) of I-G1.

Everything about the frame $L = DK$, $N$, $P$, $Q$, the symplectic defects and the cohomological equation is *cited* from track K's Part 2 ([K:`bp:p2:frame-identities`], [K:`bp:p2:exactness`], [K:`bp:p2:geometry`], [K:`bp:p2:cohomological`], [K:`bp:p2:triangular`], ...) and Part 1, not reproved: the canonical Case III objects of the frozen definitions are those of [K:`bp:p2:objects`] with $h$ replaced by $H$.

### 2.1. **Setting, sign dictionary, external facts.**

**Standing assumption 2.1** (Standing setting of Part 2)**.** Let $n \geq 2$ and $\tau \geq 0$ (I-G1 has $\tau \geq n-1$; only $\tau \geq 0$ is used here), $M > 0$. Let $D \subset \mathbb{C}^n \times \mathbb{C}^n$ be open with $(\theta + k, I) \in D$ for $(\theta, I) \in D$, $k \in \mathbb{Z}^n$ (I1); let $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ be holomorphic on $D$, 1-periodic in $\theta$, real at real phase points of $D$ (I2, I3), with $\|DH\|, \|X_H\|, \|DX_H\|, \|D^2 X_H\| \leq M$ on $D$ (I4). Let $\gamma > 0$ and $\omega \in \mathbb{R}^n \cap \mathcal{D}_{\gamma,\tau}$. Let $r > 0$ and let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ be holomorphic, primary and real on reals on $T_r$, with $K(T_r) \subset D$ (I5, I6). Where the frame is used, $\det G_K \neq 0$ on $T_r$.

This is exactly Assumption [K:`bp:p2:standing`] with $h := H$, $\mathfrak{B} := D$, together with the bound $\|DH\| \leq M$, which is used only in Lemma 2.9(e) and Lemma 2.5(b). All objects $L, G, B, N, P, X_N, T, E, A, \Omega_K, E_{\mathrm{sym}}, Q, \eta$ and the averages are those of [K:`bp:p2:objects`], with $h := H$; [K:`bp:p2:frozen-agreement`] identifies them with the frozen `gramMatrix`, `normalFrame`, `torsion`, `invarianceError`. The norms are those of Part 1: max norms, row-sum operator norms $|M| = \max_i \sum_j |M_{ij}|$ (so a $1 \times n$ matrix has the norm $\sum_j |M_{1j}|$, which is the norm of $DH(z)$ in (I4)). Write $\Gamma_\delta := c_R \gamma^{-1} \delta^{-\varkappa}$, $\varkappa = \tau + n$, as in [K:`bp:p2:constants`].

*Remark* 2.2 (Rays: $s\omega^*$ is Diophantine with the constant of $\omega^*$). In I-G1, $\omega = s\omega^*$ with $s > 1$ and $\omega^* \in \mathcal{D}_{\gamma,\tau}$. For $k \neq 0$, $|k \cdot s\omega^*| = s|k \cdot \omega^*| \geq \gamma / \big(\sum_i |k_i|\big)^\tau$, so $\omega \in \mathcal{D}_{\gamma,\tau}$ with the *same* $\gamma$. Hence every result of K's Part 1 and Part 2 that requires $\omega \in \mathcal{D}_{\gamma,\tau}$ applies to $\omega = s\omega^*$ with $\Gamma_\delta$ as above.

*Remark* 2.3 (Lean totality for this part). Following Remark 1.13: every statement below that uses $N$, $T$, $\mathsf{T}_H$ assumes $\det G_K \neq 0$ on $T_r$ first, so that $B = G^{-1}$ is the true inverse, holomorphic ([K:`bp:p1:holomorphic-inverse`]); every use of $\langle \mathsf{T}_H \rangle$ is preceded by continuity of $\mathsf{T}_H$ on $\mathbb{R}^n$ (Lemma 2.5(a)), hence integrability ([K:`bp:p1:ext-integral`]); every derivative of $H$, $X_H$ is taken at a point of the open set $D$ where $H$ is holomorphic, so `fderiv` is the genuine derivative; the frozen $\mathsf{T}_H$ entry `fderiv` $\mathbb{C}$ $H$ $(K\theta)$ $(N_{\cdot b})$ is then the matrix product $DH(K(\theta))N_{\cdot b}(\theta)$.

*Remark* 2.4 (Sign dictionary HL19 $\leftrightarrow$ this blueprint). HL19 use $\mathfrak{L}_\omega = -\sum_i \omega_i \partial_{\theta_i} = -\partial_\omega$ (4.1), p. 20, and the right operator $\mathfrak{R}_\omega$ of (4.2), $\widehat{(\mathfrak{R}_\omega v)}_k = -\hat{v}_k/(2\pi i k \cdot \omega)$. Track K's operator $\mathcal{R}$ ([K:`bp:p2:cohomological`]) solves $\partial_\omega u = g$, $\langle u \rangle = 0$, i.e. $\widehat{(\mathcal{R}g)}_k = \hat{g}_k/(2\pi i k \cdot \omega)$. Thus

$$\mathfrak{L}_\omega = -\partial_\omega, \qquad \mathfrak{R}_\omega = -\mathcal{R} \quad \text{(on the data of (4.1))}.$$

With the canonical $\Omega = \Omega_0$ and $\Omega X_h = (Dh)^\top$ (Rem. 2.1, p. 6; our [K:`bp:p2:hamiltonian`](d)), HL19's invariance error is $E = X_h(K) - DK\,\omega$ (2.2), p. 7, which p. 45 writes as $X_h(K) + \mathfrak{L}_\omega K$. The linearized system (6.1), p. 45, with $\bar{\omega} = \omega + \Delta\omega$, $\Delta\omega = -\omega\xi^\omega$ (6.3), is therefore

$$A\,\Delta K - \partial_\omega \Delta K - DK\,\Delta\omega = A\,\Delta K - \partial_\omega \Delta K + L\omega\,\xi^\omega = -E, \qquad \langle DH(K)\Delta K \rangle = -E^\omega,$$

$A = DX_H \circ K$, $L = DK$, $E^\omega = \langle H \circ K \rangle - E_0$, and $\bar{\omega} = (1 - \xi^\omega)\omega$ as on p. 50. The system (6.7) is

$$T\xi^N - \partial_\omega \xi^L + \omega\,\xi^\omega = \eta^L, \qquad -\partial_\omega \xi^N = \eta^N, \qquad \langle Y \xi^N \rangle = \eta^\omega,$$

and the solution (6.8)–(6.10) reads $\xi^N = \xi_0^N - \mathcal{R}\eta^N$, $\xi^L = \xi_0^L + \mathcal{R}\big(T\xi^N + \omega\xi^\omega - \eta^L\big)$, with (6.10) unchanged because $\mathfrak{R}_\omega = -\mathcal{R}$. Lemma 2.14 verifies that this is the equation obtained from (6.1) in the frame, signs included. Here and below $Y(\theta) := DH(K(\theta))N(\theta) \in \mathbb{C}^{1 \times n}$ is HL19's $H$ of (6.14), renamed to avoid a clash with the Hamiltonian.

External facts used in this part. Besides [K:`bp:p1:ext-integral`] (continuous functions on a compact box are integrable, integrals are linear and bounded by the supremum), the proofs use only:

- (X1) *Chain and product rules* for Fréchet derivatives of holomorphic maps, and linearity of the derivative (Mathlib `HasFDerivAt.comp`, `fderiv_add`, `fderiv_const`).
- (X2) *Complex derivative in a real direction.* If $\varphi$ is holomorphic on an open $U \subset \mathbb{C}^n$, $x \in U \cap \mathbb{R}^n$ and $v \in \mathbb{R}^n$, then $\varphi|_{U \cap \mathbb{R}^n}$ has the real directional derivative $D\varphi(x)v$ at $x$; if moreover $\varphi$ is real on $U \cap \mathbb{R}^n$, then $D\varphi(x)v \in \mathbb{R}$ (it is a limit of real difference quotients). See [K:`bp:p1:hamiltonian-real`], step [K:`step:p1:hr-derivative`].

## 2.2. **The extended torsion.**

**Lemma 2.5** (Regularity, periodicity, realness and size of the extended torsion)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* $T_r$*, and put* $Y := DH(K)N$ *on* $T_r$*. Then:*

*(a)* $Y$ *and the entries of* $\mathsf{T}_H$ *(built from* $(H, K, \omega)$*) are holomorphic and* 1*-periodic on* $T_r$ *and real at real* $\theta$*. In particular* $\mathsf{T}_H$ *is continuous on* $\mathbb{R}^n$ *and* $\langle \mathsf{T}_H \rangle$ *is defined, real, and equals the block matrix*

$$\langle \mathsf{T}_H \rangle = \begin{pmatrix} \langle T \rangle & \omega \\ \langle Y \rangle & 0 \end{pmatrix}$$

*with* $\langle T \rangle = \langle T_{H,K,\omega} \rangle$ *the torsion average of* [K:`bp:p2:objects`].

*(b) If* $|DK| \leq \sigma'_K$ *and* $|B| \leq \sigma'_B$ *on* $T_r$*, then* $|Y| \leq M\sigma'_K\sigma'_B$ *on* $T_r$*, and on any set where* $|T| \leq C_T$*,* $|\mathsf{T}_H| \leq \max\{C_T + |\omega|,\ M\sigma'_K\sigma'_B\}$ *in the row-sum norm of* $\mathbb{C}^{(n+1)\times(n+1)}$*.*

*Frozen Lean declaration:* `KAM.Iso.extendedTorsion, KAM.Iso.extendedTorsionAverage`
*Source: HL19 (3.13), p. 19 (*$c = H$*,* $d = n$*); HL19 (6.14), p. 47 for* $Y$*. Estimates: ours.*

*Proof.* **Step S2.1: Holomorphy.** $N$ is holomorphic on $T_r$ ([K:`step:p2:fb-holo`], which uses only $K(T_r) \subset D$, $\det G_K \neq 0$ and the holomorphic inverse [K:`bp:p1:holomorphic-inverse`]; no bound and no torsion condition enters), and $DH$ is holomorphic on $D$ ([K:`bp:p1:holomorphic-regularity`]), so $Y = DH \circ K \cdot N$ is a product of holomorphic functions ((X1)). The entries of $T$ and the constant block $\omega$ are holomorphic by the same step. Hence every entry of $\mathsf{T}_H$ is holomorphic, in particular continuous on $\mathbb{R}^n \subset T_r$.

**Step S2.2: Periodicity and realness.** Differentiating $H(\theta+k, I) = H(\theta, I)$ gives $DH(z+(k,0)) = DH(z)$ on $D$, and $K(\theta + k) = K(\theta) + (k, 0)$; so $DH(K(\theta + k)) = DH(K(\theta))$. $N$ and $T$ are 1-periodic by [K:`bp:p2:periodicity`]. At real $\theta$, $K(\theta)$ is a real phase point of $D$, where $DH$ is real (the argument of [K:`bp:p2:hamiltonian`](c), step [K:`step:p2:ham-real`], applies to $H$, real at real points), and $N$, $T$ are real ([K:`bp:p2:periodicity`]). Hence $Y$ is real at real points.

**Step S2.3: The average.** By Step S2.1, each entry is integrable on $[0, 1]^n$ ([K:`bp:p1:ext-integral`]), so the Bochner integral of the frozen $\langle \mathsf{T}_H \rangle$ is the entrywise integral, real because the integrand is real at real $\theta$; the entries of the border are $\langle \omega \rangle = \omega$ (the cube has volume one) and $\langle 0 \rangle = 0$. (Lean totality, Remark 2.3.)

**Step S2.4: Bounds.** $|Y| \leq |DH(K)||N| \leq M\,|\Omega_0||L||B| \leq M\sigma'_K\sigma'_B$, by [K:`bp:p2:matrix`](b), (d). The last column of $\mathsf{T}_H$ is the $n \times 1$ matrix $\omega$, of norm $|\omega|$; by [K:`bp:p2:matrix`](c) the $(n+1)\times(n+1)$ norm is at most the maximum of the two block-row bounds $C_T + |\omega|$ and $|Y|$. □

**Lemma 2.6** ($\mathsf{T}_H = \mathsf{B}_h$ at the trivial torus)**.** *Let* $U \subset \mathbb{C}^n$ *be open and* $h : U \to \mathbb{C}$ *holomorphic with* $h(J) \in \mathbb{R}$ *for every real* $J \in U$*. Let* $D = \mathbb{C}^n \times U$*,* $H(\theta, J) := h(J)$ *on* $D$*,* $I \in U \cap \mathbb{R}^n$ *and* $K(\theta) = (\theta, I)$ *(*$\theta \in \mathbb{C}^n$*). Put* $\omega := \nabla h(I) \in \mathbb{R}^n$ *(*`frequencyMap`*). Then* $K$ *is holomorphic, primary and real on reals,* $K(\mathbb{C}^n) \subset D$*,* $G_K = I_n$*,* $E_{H,K,\omega} \equiv 0$*, and for every* $\theta \in \mathbb{C}^n$

$$\mathsf{T}_H(\theta) = \mathsf{B}_h(I) \quad \textit{(the real matrix, viewed in } \mathbb{C}^{(n+1)\times(n+1)}\textit{)}, \qquad \langle \mathsf{T}_H \rangle = \mathsf{B}_h(I),$$

*the frozen* `borderedHessian`*. In particular* $\det\langle \mathsf{T}_H \rangle \neq 0$ *iff* $\det \mathsf{B}_h(I) \neq 0$*, and the (total, Lean) inverses of the two matrices correspond under the cast, so* $|\langle \mathsf{T}_H \rangle^{-1}| = |\mathsf{B}_h(I)^{-1}|$ *in both cases.*
*Frozen Lean declaration:* `KAM.Iso.extendedTorsion, KAM.Iso.extendedTorsionAverage, KAM.Iso.borderedHessian`
*Source: ours; HL19 Rem. 3.7, p. 19, and Chi05 p. 17 (determinant* $3/L^4$ *in the Kepler case). Reading remark: Remark 1.17(b).*

*Proof.* **Step S2.5: The torus.** $K(\theta+k) = (\theta+k, I) = K(\theta)+(k, 0)$, $K$ is affine, hence holomorphic, and real on real $\theta$ because $I$ is real. $DK(\theta)e_i = (e_i, 0)$, so $G_K = (e_i \cdot e_j) = I_n$ (using the bilinear dot of Definition 1.5), $\det G_K = 1$ and $B = I_n$.

**Step S2.6: The normal frame.** $N_{\cdot j} = \sum_k \delta_{kj}\Omega_0(e_k,0) = \Omega_0(e_j,0) = (0,e_j)$, a constant function of $\theta$, so $D_\theta N_{\cdot j} = 0$.

**Step S2.7: The vector field.** Since $H(\theta,J) = h(J)$, $X_H(\theta,J) = \big((\partial_a h(J))_a,\, 0\big)$, so $E_{H,K,\omega}(\theta) = X_H(K\theta) - DK\,\omega = (\nabla^{\mathbb{C}} h(I) - \omega, 0)$, where $\nabla^{\mathbb{C}} h(I) = (\partial_a h(I))_a$. By (X2) applied to $\varphi = h$ (real on $U \cap \mathbb{R}^n$), $\partial_a h(J) \in \mathbb{R}$ for real $J$, and the real function $J \mapsto \operatorname{Re} h(J) = h(J)$ on $U \cap \mathbb{R}^n$ has partial derivative $\partial_a h(J)$ in the direction $e_a$; by the definition of `frequencyMap`, $\omega_a = \partial_a h(I)$. Hence $E \equiv 0$, and more generally $\nabla h(J)_a = \partial_a h(J)$ for real $J \in U$.

**Step S2.8: The torsion.** $DX_H(\theta,I)(0,e_j) = \big((\partial_j\partial_a h(I))_a, 0\big)$, and $\Omega_0(w,0) = (0,w)$. Hence $T(\theta)_{ij} = N_{\cdot i} \cdot \Omega_0\big(DX_H N_{\cdot j} - 0\big) = (0,e_i)\cdot(0,(\partial_j\partial_a h(I))_a) = \partial_j\partial_i h(I)$.

**Step S2.9: The border.** The last column is $\omega = \nabla h(I)$, entries $\omega_a = \partial_a h(I)$. The last row is $DH(K\theta)N_{\cdot b} = DH(\theta,I)(0,e_b) = \partial_b h(I) = \omega_b$. The corner is 0.

**Step S2.10: Comparison with the bordered Hessian.** The $(a,b)$ entry of $\mathsf{B}_h(I)$ is $D(\nabla h)(I)e_b$ read at $a$, the real derivative in direction $e_b$ of $J \mapsto \nabla h(J)_a = \partial_a h(J)$ on $U \cap \mathbb{R}^n$ (Step S2.7); since $\partial_a h$ is holomorphic, this is $\partial_b\partial_a h(I)$ by (X2), a real number. So the upper left blocks agree (and are symmetric by [K:`bp:p1:holomorphic-regularity`]); the border of $\mathsf{B}_h(I)$ is $\nabla h(I)$ in the last column and row, as found above. Thus $\mathsf{T}_H(\theta) = \mathsf{B}_h(I)$ for all $\theta$; the constant has average equal to itself since $[0,1]^n$ has volume one. The statements on the determinant and the inverse follow by cases. If $\det \mathsf{B}_h(I) \neq 0$, a real matrix and its image in $\mathbb{C}^{(n+1)\times(n+1)}$ have the same determinant, and inverses correspond ([K:`bp:p1:real-inverse`]); the row-sum norm is unchanged. If $\det \mathsf{B}_h(I) = 0$, the cast has determinant 0 and both total inverses are the zero matrix, so the inverse norms are both 0 (the statement is then vacuous for (I8), which requires $\det \neq 0$). □

*Remark* 2.7 (Use). For $H = h + f$ with $f$ small, the torus $K(\theta) = (\theta, I)$ with $\omega = \nabla h(I)$ (or $s\omega^*$ with a nearby $I$, see Part 3) has $\langle \mathsf{T}_H \rangle = \mathsf{B}_h(I) + O(\|f\|)$ by Lemma 2.8 below and the Cauchy estimates; this is how (I8) of I-G1 is verified from $\det \mathsf{B}_h \neq 0$, with $\sigma_T$ from [K:`bp:p1:neumann`].

**Lemma 2.8** (The energy row of the extended torsion)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* $T_r$ *and let* $E = E_{H,K,\omega}$, $\eta^L = -N^\top \Omega_0 E$ *as in* [K:`bp:p2:objects`]. *Then on* $T_r$*:*

*(a)* $Y = DH(K)N = \omega^\top - (\eta^L)^\top$. *In particular* $|Y - \omega^\top| = \sum_b |\eta^L_b| \le n|\eta^L|$ *(row-sum norm of a* $1 \times n$ *matrix) and, entrywise,* $Y_b = \omega_b + N_{\cdot b}^\top \Omega_0 E$.

*(b) If* $E = 0$ *on* $T_r$ *(invariant torus), then* $Y = \omega^\top$ *on* $T_r$*, so*

$$\mathsf{T}_H(\theta) = \begin{pmatrix} T(\theta) & \omega \\ \omega^\top & 0 \end{pmatrix} \quad (\theta \in T_r), \qquad \langle \mathsf{T}_H \rangle = \begin{pmatrix} \langle T \rangle & \omega \\ \omega^\top & 0 \end{pmatrix}.$$

*(c) In general* $\langle \mathsf{T}_H \rangle = \mathsf{T}^{\mathrm{inv}} - \begin{pmatrix} 0 & 0 \\ \langle \eta^L \rangle^\top & 0 \end{pmatrix}$, *where* $\mathsf{T}^{\mathrm{inv}} := \left(\begin{smallmatrix} \langle T \rangle & \omega \\ \omega^\top & 0 \end{smallmatrix}\right)$ *is built from the same* $T$*; and if* $|E| \le e$ *on* $T_r$, $|L| \le \sigma'_K$, $|B| \le \sigma'_B$, *then* $|\langle \mathsf{T}_H \rangle - \mathsf{T}^{\mathrm{inv}}| \le 2n^2 \sigma'_K \sigma'_B\, e$.

*Frozen Lean declaration:* `KAM.Iso.extendedTorsion`

*Source: HL19 Rem. 3.7, p. 19 (part (b), (3.17)), proved with the sign corrected (Remark 2.17, E3); (a), (c) are ours: HL19 (3.13) defines the last row as* $Dc(K)N$ *without simplifying it off invariance.*

*Proof.* **Step S2.11: The gradient identity.** By [K:`bp:p2:hamiltonian`](d) (equivalently [K:`bp:p1:hamiltonian-real`]), $\Omega_0 X_H = (DH)^\top$ on $D$, so $DH(z) = (\Omega_0 X_H(z))^\top = X_H(z)^\top \Omega_0^\top = -X_H(z)^\top \Omega_0$.

**Step S2.12: Insert the error.** At $z = K(\theta)$, $X_H(K) = L\omega + E$ by the definition of $E$. Hence

$$Y = DH(K)N = -(L\omega + E)^\top \Omega_0 N = -\omega^\top L^\top \Omega_0 N - E^\top \Omega_0 N = \omega^\top - E^\top \Omega_0 N,$$

using $L^\top \Omega_0 N = -I_n$ ([K:`bp:p2:frame-identities`](a)). The $b$-th entry of the row $E^\top \Omega_0 N$ is the scalar $E^\top \Omega_0 N_{\cdot b} = N_{\cdot b}^\top \Omega_0^\top E = -N_{\cdot b}^\top \Omega_0 E = (\eta^L)_b$. This is (a) and the entrywise form. The

row-sum norm of the $1 \times n$ matrix $(\eta^L)^\top$ is $\sum_b |\eta^L_b| \le n|\eta^L|$, so $|Y - \omega^\top| = \sum_b |\eta^L_b| \le n|\eta^L|$ (the factor $n$ cannot be dropped: for $H = (\omega - (1,\dots,1)) \cdot I$, $K = (\theta, 0)$ one has $\eta^L = (1,\dots,1)$ and $|Y - \omega^\top| = n$).

**Step S2.13: Invariant torus and average.** (b) is (a) with $\eta^L = 0$ (since $E = 0$), together with the definition of $\mathsf{T}_H$ (Definition 1.6) and Lemma 2.5(a). For (c): averaging (a) and using linearity, $\langle Y \rangle = \omega^\top - \langle \eta^L \rangle^\top$, so the two matrices differ only in the last row, which has norm $\le n|\langle \eta^L \rangle| \le n \sup_{\mathbb{R}^n} |\eta^L|$ ([K:`bp:p2:averages`](a)), and $|\eta^L| \le |N^\top||\Omega_0||E| \le 2n\,|N||E| \le 2n\,\sigma'_K \sigma'_B\, e$ ($|N^\top| \le 2n|N|$) ([K:`bp:p2:matrix`](b), (d), with $|N| \le |L||B|$). Hence the bound $2n^2 \sigma'_K \sigma'_B e$. □

## 2.3. **Energy identities.**

**Lemma 2.9** (Approximate energy identities)**.** *Assume 2.1 and let $E = E_{H,K,\omega}$, $E^{\mathrm{en}} := H \circ K - \langle H \circ K \rangle$ on $T_r$. Then:*

- *(a) $H \circ K$ is holomorphic, 1-periodic on $T_r$, real on reals, and $D(H \circ K) = DH(K)\,DK$; the average $\langle H \circ K \rangle = \bar{E}_H(K)$ is the frozen `energyAverage`, real.*
- *(b) $\partial_\omega (H \circ K) = D(H \circ K)\omega = -DH(K)\,E$ on $T_r$.*
- *(c) $L^\top \Omega_0 E = (D(H \circ K))^\top - \Omega_K \omega$ on $T_r$. If $\det G_K \neq 0$, then $\eta^N = L^\top \Omega_0 E$.*
- *(d) $\langle D(H \circ K) \rangle = 0$, $\langle \Omega_K \rangle = 0$ and $\langle \eta^N \rangle = 0$ (the compatibility condition of Lemma 2.12).*
- *(e) If $|E| \le \varepsilon$ on $T_r$ and $0 < \delta < r$, then $|E^{\mathrm{en}}| \le M \Gamma_\delta \varepsilon$ on $T_{r-\delta}$, and for $0 < \delta' < r - \delta$, $|D(H \circ K)| \le nM\Gamma_\delta \varepsilon / \delta'$ on $T_{r-\delta-\delta'}$.*
- *(f) (Energy defect.) Let $\det G_K \neq 0$ on $T_r$, let $\xi = (\xi^L, \xi^N)$ be holomorphic on a strip $T_{r'}$ with $0 < r' \le r$ (so $\xi$ is continuous on the real cube), and $\Delta K = P\xi$. Then on $T_{r'}$*

$$DH(K)\,\Delta K = Y\xi^N + D(H \circ K)\,\xi^L = Y\xi^N + (\eta^N)^\top \xi^L - \omega^\top \Omega_K\, \xi^L.$$

*Consequently, if $\langle Y \xi^N \rangle = \eta^\omega$, then $\langle DH(K) \Delta K \rangle = \eta^\omega + E^\omega_{\mathrm{lin}}$ with $E^\omega_{\mathrm{lin}} := \langle D(H \circ K)\,\xi^L \rangle$, and $|E^\omega_{\mathrm{lin}}| \le \sup_{[0,1]^n} |D(H \circ K)| \sup_{[0,1]^n} |\xi^L| \le n\big(\sup_{[0,1]^n} |\eta^N| + |\omega| \sup_{[0,1]^n} |\Omega_K|\big) \sup_{[0,1]^n} |\xi^L|$, all suprema finite by continuity on the compact cube.*

*Frozen Lean declaration: `KAM.Iso.energyAverage`*

*Source: HL19 Lemma 4.2, (4.3)–(4.5), p. 21, and (4.9) ($\mathfrak{L}_\omega(c \circ K) = Dc(K)E$, equivalent to (b) with $\mathfrak{L}_\omega = -\partial_\omega$); p. 46 ($Dc(K)L = D(c \circ K)$, $E^\omega_{\mathrm{lin}}$ of p. 49); (6.11)–(6.14), p. 47; compatibility $\langle \eta^N \rangle = 0$ from Lemma 4.6 as recalled on p. 47.*

*Proof.* **Step S2.14: Regularity and chain rule.** $K$ is holomorphic on $T_r$ with values in $D$, and $H$ is holomorphic on $D$, so $H \circ K$ is holomorphic with $D(H \circ K) = DH(K)DK$ ((X1)). It is 1-periodic since $H(K(\theta + k)) = H(K(\theta) + (k, 0)) = H(K(\theta))$, and real at real $\theta$ because $K(\theta)$ is a real phase point of $D$ (I3). The average is $\int_{[0,1]^n} H(K(\theta))d\theta = \bar{E}_H(K)$ by definition, integrable by [K:`bp:p2:averages`](a). This is (a).

**Step S2.15: The derivative along $\omega$.** $D(H \circ K)\omega = DH(K)L\omega = DH(K)\big(X_H(K) - E\big)$. By [K:`bp:p2:hamiltonian`](d), $DH = (\Omega_0 X_H)^\top$, so $DH(z)X_H(z) = X_H^\top \Omega_0^\top X_H = -X_H^\top \Omega_0 X_H = 0$ ($\Omega_0$ is antisymmetric). Hence $D(H \circ K)\omega = -DH(K)E$, which is (b).

**Step S2.16: The gradient identity.** This repeats the gradient step of [K:`bp:p2:exactness`]: $L^\top \Omega_0 X_H(K) = (DH(K)L)^\top = (D(H \circ K))^\top$, hence $L^\top \Omega_0 E = (D(H \circ K))^\top - L^\top \Omega_0 L\omega = (D(H \circ K))^\top - \Omega_K \omega$. The identification $\eta^N = L^\top \Omega_0 E$ is the definition of $\eta$ in [K:`bp:p2:objects`]. This is (c).

**Step S2.17: Averages.** $D(H \circ K)$ has the entries $\partial_{\theta_\ell}(H \circ K)$ of a holomorphic periodic function, so the average vanishes ([K:`bp:p2:averages`](b)); $\langle \Omega_K \rangle = 0$ and $\langle \eta^N \rangle = 0$ are [K:`bp:p2:exactness`]. (Also directly: $\langle \eta^N \rangle = \langle D(H \circ K) \rangle^\top - \langle \Omega_K \rangle \omega = 0$ by (c).)

**Step S2.18: The size of the energy fluctuation.** $E^{\mathrm{en}}$ is holomorphic and 1-periodic on $T_r$, has zero average, and by (b) $|\partial_\omega E^{\mathrm{en}}| = |DH(K)E| \le |DH(K)||E| \le M\varepsilon$ on $T_r$ (here (I4), $K(T_r) \subset D$,

is used). [K:bp:p2:cohomological](b), on $T_r$ with loss $\delta$, gives $|E^{\mathrm{en}}| \le \Gamma_\delta M\varepsilon$ on $T_{r-\delta}$. The Cauchy estimate [K:bp:p1:cauchy-strip] for the bounded holomorphic function $E^{\mathrm{en}}$ on $T_{r-\delta}$ with loss $\delta'$ gives $|DE^{\mathrm{en}}| = |D(H\circ K)| \le n\Gamma_\delta M\varepsilon/\delta'$ on $T_{r-\delta-\delta'}$. This is (e).

**Step S2.19: The energy defect.** $P = (L\ N)$, so $\Delta K = L\xi^L + N\xi^N$ and $DH(K)\Delta K = DH(K)L\xi^L + DH(K)N\xi^N = D(H\circ K)\xi^L + Y\xi^N$ by (a). By (c), $D(H\circ K) = (\eta^N + \Omega_K\omega)^\top = (\eta^N)^\top + \omega^\top\Omega_K^\top = (\eta^N)^\top - \omega^\top\Omega_K$ since $\Omega_K^\top = -\Omega_K$ ($\Omega_K = L^\top\Omega_0 L$, $\Omega_0^\top = -\Omega_0$). Since $r' > 0$, every term is continuous on $[0,1]^n \subset T_{r'}$, hence integrable ([K:bp:p1:ext-integral]), and the average of the sum splits linearly. For the bound: $|\langle D(H\circ K)\xi^L\rangle| \le \sup_{[0,1]^n}|D(H\circ K)\xi^L|$ ([K:bp:p1:ext-integral], volume one), and $|D(H\circ K)| \le |(\eta^N)^\top| + |\omega^\top\Omega_K| \le n|\eta^N| + n|\omega||\Omega_K|$, with [K:bp:p2:matrix](a) for transposes (a $1\times n$ row $v^\top$ has $|v^\top| = \sum|v_i| \le n|v|$). □

**Lemma 2.10** (Energy conservation of an invariant torus)**.** *Assume 2.1 on a strip $T_r$ (with $\omega \in \mathcal{D}_{\gamma,\tau}$) and $E_{H,K,\omega}(\theta) = 0$ for all $\theta \in T_r$. Then:*

*(a) $H\circ K \equiv \bar E_H(K)$ is constant on $T_r$.*
*(b) If $E_0 \in \mathbb{R}$ (or $\mathbb{C}$) and $\bar E_H(K) = E_0$, then $H(K(\theta)) = E_0$ for every $\theta \in T_r$.*
*(c) Quantitatively, if $E_{H,K,\omega}(\theta) = 0$ on $T_r$ then $\sup_{T_r}|H\circ K - E_0| = |\bar E_H(K) - E_0|$.*

*Frozen Lean declaration: `KAM.Iso.energyAverage`*
*Source: HL19 Rem. 2.2, p. 7 ("quasi-periodicity implies $h(K(\theta)) = \langle h\circ K\rangle$"); here proved. This is the pointwise energy conclusion (O3) of I-G1 (decision D-I-ENERGY).*

*Proof.* **Step S2.20: The derivative vanishes along $\omega$.** Let $u := H\circ K - \langle H\circ K\rangle$, holomorphic and 1-periodic on $T_r$ with $\langle u\rangle = 0$ (Lemma 2.9(a)). By Lemma 2.9(b), $Du\,\omega = -DH(K)E = 0$ on $T_r$.

**Step S2.21: Uniqueness for the cohomological equation.** For every $0 < \delta < r$, $u|_{T_{r-\delta}}$ is holomorphic, 1-periodic, of zero average, and solves $Du\,\omega = g$ with $g = 0$ (holomorphic, bounded, periodic, $\langle g\rangle = 0$ on $T_r$). By the uniqueness clause of [K:bp:p1:cohomological] (no boundedness of the solution assumed), $u = 0$ on $T_{r-\delta}$. As $\delta \in (0,r)$ is arbitrary, $u = 0$ on $T_r = \bigcup_\delta T_{r-\delta}$, which is (a).

**Step S2.22: Pointwise energy.** (b) and (c) are (a): $H\circ K \equiv \bar E_H(K)$, so $|H(K(\theta)) - E_0| = |\bar E_H(K) - E_0|$ for every $\theta$. □

*Remark* 2.11 (Where (O3) of I-G1 comes from). In the proof of I-G1 (Part 2), $E_{H,K_\infty,s_\infty\omega^*} = 0$ on $T_{\rho_\infty}$ and $\bar E_H(K_\infty) = E_0$: the second follows from $|\bar E_H(K_j) - E_0| \to 0$ along the iteration and the uniform convergence $K_j \to K_\infty$ (continuity of $H$ and of the integral on the compact cube), which is part of Part 2; then Lemma 2.10(b), with $\omega = s_\infty\omega^* \in \mathcal{D}_{\gamma,\tau}$ (Remark 2.2), gives $H(K_\infty) = E_0$ on $T_{\rho_\infty}$. Neither the Gram determinant of $K_\infty$ nor the frame is used (Remark 1.13(T2)).

2.4. **The bordered triangular system.** Notation: for $T : T_r \to \mathbb{C}^{n\times n}$ and $Y : T_r \to \mathbb{C}^{1\times n}$ holomorphic and 1-periodic put $\mathcal{T} := \begin{pmatrix}\langle T\rangle & \omega\\ \langle Y\rangle & 0\end{pmatrix} \in \mathbb{C}^{(n+1)\times(n+1)}$; with $T = T_{H,K,\omega}$, $Y = DH(K)N$, this is $\langle \mathsf{T}_H\rangle$ (Lemma 2.5(a)). Let $\mathcal{R}$ be the solution operator of [K:bp:p2:cohomological](a): for $g$ holomorphic, periodic with $\langle g\rangle = 0$, $\mathcal{R}g$ is the unique zero-average periodic solution of $\partial_\omega u = g$.

**Lemma 2.12** (Bordered triangular equations (HL19 Lemma 6.1))**.** *Let $\gamma > 0$, $\tau \ge 0$, $\omega \in \mathbb{R}^n \cap \mathcal{D}_{\gamma,\tau}$, $r > 0$, $\delta > 0$ with $2\delta < r$, and $\Gamma_\delta = c_R\gamma^{-1}\delta^{-\varkappa}$. Let $\eta^L, \eta^N : T_r \to \mathbb{C}^n$ be holomorphic, 1-periodic, $|\eta^L| \le H_L$ and $|\eta^N| \le H_N$ on $T_r$, with $\langle\eta^N\rangle = 0$, and let $\eta^\omega \in \mathbb{C}$. Let $T : T_r \to \mathbb{C}^{n\times n}$ and $Y : T_r \to \mathbb{C}^{1\times n}$ be holomorphic and 1-periodic with $|T| \le H_T$, $|Y| \le H_Y$ on $T_{r-\delta}$. Assume*

$\det \mathcal{T} \neq 0$ *and* $|\mathcal{T}^{-1}| \le \sigma$ *(row-sum norm of* $\mathbb{C}^{(n+1)\times(n+1)}$*). Fix* $\xi_0^L \in \mathbb{C}^n$ *and define, in this order,*

$$\begin{aligned} u^N &:= \mathcal{R}\eta^N \quad \text{on } T_{r-\delta},\\ \begin{pmatrix}\xi_0\\ \xi^\omega\end{pmatrix} &:= \mathcal{T}^{-1}\begin{pmatrix}\langle \eta^L\rangle + \langle Tu^N\rangle\\ \eta^\omega + \langle Yu^N\rangle\end{pmatrix} \in \mathbb{C}^n\times\mathbb{C},\\ \xi^N &:= \xi_0 - u^N, \qquad g^L := T\xi^N + \omega\,\xi^\omega - \eta^L \quad \text{on } T_{r-\delta},\\ \xi^L &:= \xi_0^L + \mathcal{R}g^L \quad \text{on } T_{r-2\delta}. \end{aligned}$$

*Then:*

(a) *(Well defined.)* $u^N$, $\xi^N$, $g^L$ *are holomorphic and* 1*-periodic on* $T_{r-\delta}$, $\langle g^L\rangle = 0$, *so* $\xi^L$ *is defined and holomorphic,* 1*-periodic on* $T_{r-2\delta}$*; the averages* $\langle Tu^N\rangle$, $\langle Yu^N\rangle$ *exist.*

(b) *(The system.) On* $T_{r-2\delta}$,

$$T\xi^N - \partial_\omega \xi^L + \omega\,\xi^\omega = \eta^L, \qquad -\partial_\omega \xi^N = \eta^N, \qquad \langle Y\xi^N\rangle = \eta^\omega,$$

*i.e.* $\Lambda\xi - \partial_\omega \xi + \binom{\omega}{0}\xi^\omega = \eta$ *with* $\Lambda = \left(\begin{smallmatrix}0 & T\\ 0 & 0\end{smallmatrix}\right)$, $\xi = (\xi^L, \xi^N)$, $\eta = (\eta^L, \eta^N)$.

(c) *(Estimates.) With* $S_* := \sigma\max\{H_L + H_T\Gamma_\delta H_N,\ |\eta^\omega| + H_Y\Gamma_\delta H_N\}$*:*

$$|\xi_0|,\ |\xi^\omega| \le |(\xi_0,\xi^\omega)| \le S_*, \qquad \sup_{T_{r-\delta}}|\xi^N| \le S_* + \Gamma_\delta H_N,$$

$$\sup_{T_{r-\delta}}|g^L| \le H_T(S_* + \Gamma_\delta H_N) + |\omega|S_* + H_L, \qquad \sup_{T_{r-2\delta}}|\xi^L - \xi_0^L| \le \Gamma_\delta \sup_{T_{r-\delta}}|g^L|.$$

(d) *(Uniqueness.) If* $(\tilde\xi^L, \tilde\xi^N, \tilde\xi^\omega)$ *are holomorphic, periodic on* $T_{r-2\delta}$ ($\tilde\xi^\omega \in \mathbb{C}$) *and satisfy the three equations of (b) on* $T_{r-2\delta}$*, then* $\tilde\xi^\omega = \xi^\omega$, $\tilde\xi^N = \xi^N$ *and* $\tilde\xi^L - \xi^L$ *is constant. Thus the solution is unique up to* $\langle \xi^L\rangle$ *(the free phase direction, HL19 Rem. 3.5).*

(e) *(Realness.) If* $\omega$ *is real and* $\eta^L, \eta^N, T, Y$ *are real on reals,* $\eta^\omega \in \mathbb{R}$, $\mathcal{T}$ *real (automatic), and* $\xi_0^L \in \mathbb{R}^n$*, then* $\xi_0$, $\xi^\omega$, $\xi^N$, $\xi^L$, $g^L$ *are real on reals.*

*Source: HL19 Lemma 6.1, pp. 46–47, (6.7)–(6.10), with signs converted by Remark 2.4; the estimates are those on p. 46, with the additional term* $|\omega|S_*$ *in* $g^L$ *(see Remark 2.13). Structure as in* [K:bp:p2:triangular] *(HL19 Lemma 5.1, pp. 31–32).*

*Source gap: HL19 prove Lemma 6.1 in four lines ("It is analogous to Lemma 5.1…"), and Lemma 5.1 is itself only referred to ("directly adapted from Lemma 4.14 in [33]"). The proof below is supplied here in full; the new points relative to* [K:bp:p2:triangular] *are Steps S2.25, S2.26 and S2.27, and uniqueness (d).*

*Proof.* **Step S2.23: First cohomological equation.** $\eta^N$ is holomorphic, periodic on $T_r$, bounded by $H_N$, with $\langle \eta^N\rangle = 0$. [K:bp:p2:cohomological](a) gives $u^N = \mathcal{R}\eta^N$ holomorphic and 1-periodic on $T_{r-\delta}$, $\langle u^N\rangle = 0$, $\partial_\omega u^N = \eta^N$, $|u^N| \le \Gamma_\delta H_N$ on $T_{r-\delta}$, real on reals if $\eta^N$ is.

**Step S2.24: Averages of the products.** $Tu^N$ and $Yu^N$ are holomorphic and periodic on $T_{r-\delta}$, hence continuous on $\mathbb{R}^n$, so their averages exist ([K:bp:p2:averages](a)) and $|\langle Tu^N\rangle| \le H_T\Gamma_\delta H_N$, $|\langle Yu^N\rangle| \le H_Y\Gamma_\delta H_N$, $|\langle \eta^L\rangle| \le H_L$. Let $w = (\langle \eta^L\rangle + \langle Tu^N\rangle,\ \eta^\omega + \langle Yu^N\rangle) \in \mathbb{C}^{n+1}$; in the max norm of $\mathbb{C}^{n+1}$, $|w| \le \max\{H_L + H_T\Gamma_\delta H_N, |\eta^\omega| + H_Y\Gamma_\delta H_N\}$.

**Step S2.25: The bordered average system.** *(new relative to* [K:bp:p2:triangular]*.)* Since $\det\mathcal{T} \neq 0$, $(\xi_0, \xi^\omega) := \mathcal{T}^{-1}w$ is the unique solution of

$$\langle T\rangle\xi_0 + \omega\,\xi^\omega = \langle \eta^L\rangle + \langle Tu^N\rangle, \qquad \langle Y\rangle\xi_0 = \eta^\omega + \langle Yu^N\rangle, \tag{1}$$

and $|(\xi_0,\xi^\omega)| \le |\mathcal{T}^{-1}||w| \le S_*$ ([K:bp:p2:matrix](b)). This is (6.10) of HL19 in our sign convention (Remark 2.4). Note that $\langle T\rangle$ alone need not be invertible: the matrix $\mathcal{T}$ determines the constant $\xi_0$ and the frequency correction $\xi^\omega$ together.

**Step S2.26: The energy equation and $\xi^N$.** By Step S2.23, $\xi^N = \xi_0 - u^N$ is holomorphic, periodic on $T_{r-\delta}$ with $\partial_\omega\xi^N = -\eta^N$ and $|\xi^N| \le |\xi_0| + \Gamma_\delta H_N \le S_* + \Gamma_\delta H_N$. By linearity of the

average and the second equation of (1), $\langle Y\xi^N\rangle = \langle Y\rangle\xi_0 - \langle Yu^N\rangle = \eta^\omega$. This is the third equation of (b) and the second one.

**Step S2.27: The second cohomological equation.** *(new relative to* [K:`bp:p2:triangular`]*.)* $g^L = T\xi^N + \omega\xi^\omega - \eta^L$ is holomorphic and periodic on $T_{r-\delta}$ with $|g^L| \le H_T(S_* + \Gamma_\delta H_N) + |\omega|S_* + H_L$. Its average is, by linearity, $\langle T\rangle\xi_0 - \langle Tu^N\rangle + \omega\xi^\omega - \langle\eta^L\rangle = 0$ by the first equation of (1). This is the point of the border: the frequency correction $\xi^\omega$ absorbs the part of the mean of $T\xi^N - \eta^L$ that the constant $\xi_0$ cannot adjust once it is spent on the energy equation. [K:`bp:p2:cohomological`](a) on $T_{r-\delta}$ with loss $\delta$ (valid since $\delta < r-\delta$) gives $\mathcal{R}g^L$ holomorphic, periodic on $T_{r-2\delta}$, with $\partial_\omega\mathcal{R}g^L = g^L$ and $|\mathcal{R}g^L| \le \Gamma_\delta \sup|g^L|$. Then $T\xi^N - \partial_\omega\xi^L + \omega\xi^\omega = T\xi^N - g^L + \omega\xi^\omega = \eta^L$. With Steps S2.23 and S2.26 this is (a), (b), (c).

**Step S2.28: Uniqueness.** Let $(\tilde\xi^L, \tilde\xi^N, \tilde\xi^\omega)$ be as in (d). Then $\partial_\omega(\tilde\xi^N + u^N) = -\eta^N + \eta^N = 0$ on $T_{r-2\delta}$, and $\tilde\xi^N + u^N - \tilde c$, $\tilde c := \langle\tilde\xi^N + u^N\rangle$, is a zero-average holomorphic periodic solution of $Du\,\omega = 0$; by the uniqueness in [K:`bp:p1:cohomological`] (as in Step S2.21) it is 0. So $\tilde\xi^N = \tilde c - u^N$. Averaging the first equation of (b) for the tilde-solution ($\langle\partial_\omega\tilde\xi^L\rangle = 0$, [K:`bp:p2:averages`](b)) and using the third gives exactly (1) for $(\tilde c, \tilde\xi^\omega)$, whence $(\tilde c, \tilde\xi^\omega) = (\xi_0, \xi^\omega)$ by invertibility of $\mathcal{T}$, and $\tilde\xi^N = \xi^N$. Then $\partial_\omega(\tilde\xi^L - \xi^L) = 0$, so $\tilde\xi^L - \xi^L$ is constant, by the same uniqueness lemma applied to $\tilde\xi^L - \xi^L - \langle\tilde\xi^L - \xi^L\rangle$.

**Step S2.29: Realness.** If the data are real on reals, $\langle T\rangle, \langle Y\rangle, \langle\eta^L\rangle, \langle Tu^N\rangle, \langle Yu^N\rangle$ are real ([K:`bp:p2:averages`](a)), $\mathcal{T}$ is a real matrix with real inverse ([K:`bp:p1:real-inverse`]), so $\xi_0, \xi^\omega$ are real; $\mathcal{R}$ preserves realness ([K:`bp:p2:cohomological`](a)), so $u^N, \xi^N, g^L, \xi^L$ are real on reals. □

*Remark* 2.13 (Differences from HL19's printed statement). (i) HL19's $\mathfrak{R}_\omega$ discards the average of its argument, so their (6.9) $\xi^L = \xi_0^L + \mathfrak{R}_\omega(\eta^L - T\xi^N)$ omits the constant vector $\omega\xi^\omega$, which has no effect on non-zero Fourier modes; in the operator $\mathcal{R}$ of [K:`bp:p2:cohomological`] (defined on zero-average data) the term is carried explicitly in $g^L$, which costs the extra summand $|\omega|S_*$ in the bound of $\sup|g^L|$. HL19's bound for $\xi^L$ (p. 46, last line) is true but is not obtained from [K:`bp:p2:cohomological`] as stated; we do not need it, because $|\omega| \le \sigma_\omega|\omega^*|$ is available in the application. (ii) HL19 have real data and $\xi_0^L \in \mathbb{R}^n$; our lemma holds for complex data and records realness separately. (iii) The estimates of HL19 are in max of the two $\mathbb{C}^{n+1}$ block components, as here. (iv) Uniqueness (d) is not stated in HL19.

**Lemma 2.14** (The linearized isoenergetic equation in the frame)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* $T_r$ *and* $\omega \in \mathbb{R}^n$ *(any). Let* $\xi = (\xi^L, \xi^N)$ *be holomorphic on* $T_{r'}$, $0 < r' \le r$, *and* $\xi^\omega \in \mathbb{C}$*; put* $\Delta K = P\xi$, $\Delta\omega = -\omega\xi^\omega$. *Then on* $T_{r'}$*:*

*(a)* $QL = \binom{I_n}{-\Omega_K}$, *hence* $QL\omega\,\xi^\omega = \binom{\omega}{-\Omega_K\omega}\xi^\omega$.

*(b) (Projected identity, unconditional.)* $Q\big(A\,\Delta K - \partial_\omega\Delta K - DK\,\Delta\omega\big) + QE = \Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega - \eta + E_{\rm lin}$, *where*

$$E_{\rm lin} := E_{\rm red}\xi + \Omega_0 E_{\rm sym}\,\partial_\omega\xi - \begin{pmatrix}0\\ \Omega_K\omega\end{pmatrix}\xi^\omega.$$

*(c) (Residual.) Put* $\mathrm{Res} := A\Delta K - \partial_\omega\Delta K + L\omega\xi^\omega + E$ *(the residual of the linearized equation). If the bordered system* $\Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega = \eta$ *of Lemma 2.12(b) holds on* $T_{r'}$*, then* $Q\,\mathrm{Res} = E_{\rm lin}$ *there, and* $|E_{\rm lin}| \le |E_{\rm red}||\xi| + |E_{\rm sym}||\partial_\omega\xi| + |\Omega_K||\omega||\xi^\omega|$.

*(d) (Conditional equivalence.) If moreover* $\det P \neq 0$ *on* $T_{r'}$ *(equivalently* $\det Q \neq 0$*; this holds when* $|E_{\rm sym}| < 1$*, see the proof, and must be proved in the application from its smallness conditions: it does not follow from* $\det G_K \neq 0$*), then* $\mathrm{Res} = Q^{-1}E_{\rm lin}$*; and for any* $\xi, \xi^\omega$*,* $\mathrm{Res} = 0 \iff \Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega = \eta - E_{\rm lin}$.

*The data are $\eta^L, \eta^N$ of* [K:bp:p2:objects]*, $T$ the torsion and $Y = DH(K)N$. The defects are quadratic in the total error only on shrunken strips where the bounds of Lemma 2.15 and of Lemma 2.12(c) hold.*
*Source: HL19 (6.4)–(6.5), p. 45, with $\Omega = \Omega_0$, $A = 0$ (canonical Case III); (6.34), p. 49 for $E_{\mathrm{lin}}$. Frame identities:* [K:bp:p2:frame-identities].

*Proof.* **Step S2.30: The product $QL$.** $Q = -\Omega_0 P^\top \Omega_0$ and $P^\top \Omega_0 L = \binom{L^\top \Omega_0 L}{N^\top \Omega_0 L} = \binom{\Omega_K}{I_n}$ by [K:bp:p2:frame-identities](a). By [K:bp:p2:matrix](d), $\Omega_0 \binom{U_1}{U_2} = \binom{-U_2}{U_1}$, so $QL = -\binom{-I_n}{\Omega_K} = \binom{I_n}{-\Omega_K}$. This is (a).

**Step S2.31: Expanding the linearized operator.** $\partial_\omega(P\xi) = (\partial_\omega P)\xi + P\partial_\omega \xi$, so $A\Delta K - \partial_\omega \Delta K = (AP - \partial_\omega P)\xi - P\partial_\omega \xi$, and $-DK\,\Delta\omega = L\omega\,\xi^\omega$. By [K:bp:p2:frame-identities](e), $Q(AP - \partial_\omega P) = \Lambda + E_{\mathrm{red}}$; by (f), $QP = I_{2n} - \Omega_0 E_{\mathrm{sym}}$; by (a), $QL\omega\xi^\omega = \binom{\omega}{0}\xi^\omega - \binom{0}{\Omega_K \omega}\xi^\omega$. Adding $QE = -\eta$: $Q(\dots) + QE = (\Lambda + E_{\mathrm{red}})\xi - \partial_\omega \xi + \Omega_0 E_{\mathrm{sym}} \partial_\omega \xi + \binom{\omega}{0}\xi^\omega - \binom{0}{\Omega_K \omega}\xi^\omega - \eta$, which is (b).

**Step S2.32: Consequence.** If the border system holds, (b) gives $Q\,\mathrm{Res} = E_{\mathrm{lin}}$ (c). For (d): $Q = -\Omega_0 P^\top \Omega_0$ is invertible iff $P$ is, and $\det(P)^2 \det \Omega_0 = \det(P^\top \Omega_0 P) = \det(\Omega_0 + E_{\mathrm{sym}}) \neq 0$ when $|E_{\mathrm{sym}}| < 1$ (Neumann series [K:bp:p1:neumann] for $I - \Omega_0 E_{\mathrm{sym}}$, $|\Omega_0 E_{\mathrm{sym}}| = |E_{\mathrm{sym}}|$, [K:bp:p1:matrix-norms]). With $Q$ invertible, (b) reads $Q\,\mathrm{Res} = [\Lambda\xi - \partial_\omega \xi + \binom{\omega}{0}\xi^\omega - \eta] + E_{\mathrm{lin}}$ for every $\xi, \xi^\omega$, which gives both claims of (d). Without invertibility of $Q$ multiplying by $Q$ loses information (a primary real torus with $\det G_K \neq 0$ and singular $P$, $Q$ at a real angle exists). The bound on $E_{\mathrm{lin}}$ is [K:bp:p2:matrix](b), (c). □

2.5. **Geometry and frame perturbation without the torsion hypothesis.** Track K's Lemmas [K:bp:p2:geometry] and [K:bp:p2:perturbation] are stated for $(r, \beta, \varepsilon)$-admissible tori, which include [K:bp:p2:A5]: invertibility of the *unbordered* averaged torsion $\langle T \rangle$. In the isoenergetic setting only the *bordered* matrix $\langle \mathsf{T}_H \rangle$ is assumed invertible; $\langle T \rangle$ may be singular. The two lemmas below restate the conclusions of K's lemmas that do not involve $\langle T \rangle$, under explicit hypotheses without any torsion condition, and prove them from the cited K tools (the torsion hypothesis of K is used there only for its item (f)). They are cited by Part 3 and *replace* the K labels everywhere in the isoenergetic scheme. The regularity facts of [K:bp:p2:frame-basic](a)–(d) used in the proofs need only [K:bp:p2:A2]–[K:bp:p2:A4] (and [K:bp:p2:A6] for (c)); no condition on any torsion average occurs there.

**Lemma 2.15** (Geometry of an approximately invariant torus, without [K:bp:p2:A5])**.** *Let $\omega \in \mathcal{D}_{\gamma,\tau}$, $0 < \delta \leq 1$, $2\delta < r$, $0 \leq \varepsilon \leq 1$, and let $K$ be holomorphic, primary and real on reals on $T_r$ with $B_\beta(K(\theta)) \subset D$, $|DK| \leq \sigma_K$, $\det G_K \neq 0$, $|B| \leq \sigma_B$ and $|E_{H,K,\omega}| \leq \varepsilon$ on $T_r$ (for some $\beta > 0$; no condition on any torsion average, bordered or not; $\mathfrak{m}, \Theta, C_*$ are the constants of* [K:bp:p2:constants]*, depending on $(n, \tau, M, \sigma_K, \sigma_B, \sigma_T)$, with $\sigma_T$ entering only through $\mathfrak{m} = \max\{1, M, \sigma_K, \sigma_B, \sigma_T\}$). Then the conclusions of* [K:bp:p2:geometry] *hold: $|DE| \leq n\varepsilon/\delta$, $|\partial_\omega \Omega_K| \leq 2n(n+1)\mathfrak{m}\varepsilon/\delta$, $|E_{\mathrm{red}}| \leq C_{\mathrm{red}}\varepsilon/\delta$ on $T_{r-\delta}$, and $|\Omega_K| \leq C_\Omega \varepsilon/(\gamma\delta^{\varkappa+1})$, $|E_{\mathrm{sym}}| \leq C_{\mathrm{sym}}\varepsilon/(\gamma\delta^{\varkappa+1})$ on $T_{r-2\delta}$. Also, on $T_r$: $|\eta^L| \leq 2n\mathfrak{m}^2\varepsilon$, $|\eta^N| \leq 2n\mathfrak{m}\varepsilon$ and $|L| \leq \mathfrak{m}$, $|B| \leq \mathfrak{m}$, $|N| \leq \mathfrak{m}^2$, $|P| \leq 2\mathfrak{m}^2$; and if $|\omega| \leq \Theta$, then $|X_N| \leq C_X/\delta$, $|T| \leq C_T/\delta$ on $T_{r-\delta}$.*
*Source: HL19 Lemmas 4.3, 4.6, 4.10, pp. 22–30, as in* [K:bp:p2:geometry]*; ours (restated without the torsion hypothesis).*

*Proof.* The last sentence is [K:bp:p2:frame-basic](b)–(d), whose hypotheses are [K:bp:p2:A2]–[K:bp:p2:A4] (and [K:bp:p2:A6] for (c)); they contain no torsion condition.

**Step S2.33: Derivatives of the error.** $E$ is holomorphic with $|E| \leq \varepsilon$ on $T_r$ ([K:bp:p2:frame-basic](a)); [K:bp:p1:cauchy-strip] gives $|DE| \leq n\varepsilon/\delta$ on $T_{r-\delta}$.

**Step S2.34: $\partial_\omega\Omega_K$ and $E_{\rm red}$.** By [K:bp:p2:frame-identities](d), (e), [K:bp:p2:matrix] and [K:bp:p2:frame-basic](b), as in [K:step:p2:geo-domegak], [K:step:p2:geo-ered]: $|\partial_\omega\Omega_K| \le (2n\mathfrak{m}+2n^2\mathfrak{m})\varepsilon/\delta$ and $|E_{\rm red}| \le C_{\rm red}\varepsilon/\delta$ on $T_{r-\delta}$.

**Step S2.35: Lagrangian defect and symplecticity defect.** $\Omega_K$ is holomorphic, 1-periodic on $T_{r-\delta}$ with $\langle\Omega_K\rangle = 0$ ([K:bp:p2:exactness]; $K$ primary). [K:bp:p2:cohomological](b) on $T_{r-\delta}$ with loss $\delta < r-\delta$ gives the bound of $\Omega_K$ on $T_{r-2\delta}$, and $|E_{\rm sym}| \le \mathfrak{m}^2|\Omega_K|$ by [K:bp:p2:frame-identities](b). □

**Lemma 2.16** (Perturbation of the frame, without [K:bp:p2:A5])**.** *Let $r>0$, $0<\delta\le 1$, $2\delta<r$, $|\omega|\le\Theta$, $\omega\in\mathbb{R}^n$. Let $K$ be holomorphic on $T_r$ with $B_\beta(K(\theta))\subset D$, $|DK|\le\sigma_K$, $\det G\neq 0$, $|B|\le\sigma_B$ on $T_r$. Let $\tilde K$ be holomorphic on $T_r$ with $|\tilde K-K|\le d$ on $T_r$, where $d<\beta$ and $2\mathfrak{m}C_{\Delta T}\,d/\delta^2\le 1$. Then (a)–(e) of* [K:bp:p2:perturbation] *hold:*

*(a) $B_{\beta-d}(\tilde K(\theta))\subset D$ on $T_r$ and $|\tilde A-A|\le\mathfrak{m}d$;*
*(b) $|D\tilde K-DK|\le nd/\delta$ on $T_{r-\delta}$;*
*(c) $\det\tilde G\neq 0$, $|\tilde B|\le 2\mathfrak{m}$, $|\tilde B-B|\le C_B d/\delta$ on $T_{r-\delta}$;*
*(d) $|\tilde N-N|\le C_{\Delta N}d/\delta$, $|\tilde N|\le 2\mathfrak{m}^2$, $|\tilde N^\top|\le 4n\mathfrak{m}^2$ on $T_{r-\delta}$;*
*(e) $|\tilde T-T|\le C_{\Delta T}d/\delta^2$ on $T_{r-2\delta}$ (here $\tilde T$ is the torsion of $\tilde K$ for the same $\omega$).*

*Source: HL19 Lemma 5.2, Steps 3–7, pp. 36–42, as in* [K:bp:p2:perturbation]*; the torsion hypothesis there is used only for its item (f).*

*Proof.* **Step S2.36: Smallness.** From the definitions of [K:bp:p2:constants], $C_{\Delta T}\ge\max\{n,C_G,C_{\Delta N}\}$ ([K:step:p2:pt-small], pure arithmetic on constants). With $d/\delta\le d/\delta^2\le 1/(2\mathfrak{m}C_{\Delta T})$: $nd/\delta\le\mathfrak{m}$, $\mathfrak{m}C_G d/\delta\le\frac12$, $C_{\Delta N}d/\delta\le\mathfrak{m}^2$, $\mathfrak{m}C_{\Delta T}d/\delta^2\le\frac12$.

**Step S2.37: Domain.** $|z-\tilde K(\theta)|<\beta-d$ implies $|z-K(\theta)|<\beta$, so $z\in D$; the segment $[K,\tilde K]$ lies in $B_\beta(K(\theta))\subset D$, and [K:bp:p2:hamiltonian](e) gives $|\tilde A-A|\le Md\le\mathfrak{m}d$.

**Step S2.38: Frame.** [K:bp:p1:cauchy-strip], [K:bp:p1:cauchy-corollaries] ($p=2n$) on $\tilde K-K$: $|\tilde L-L|\le nd/\delta$, $|(\tilde L-L)^\top|\le 2nd/\delta$ on $T_{r-\delta}$; so $|\tilde L|\le 2\mathfrak{m}$, $|\tilde L^\top|\le 4n\mathfrak{m}$. Then $\tilde G-G=\tilde L^\top(\tilde L-L)+(\tilde L-L)^\top L$ has norm $\le C_G d/\delta$, $|B||\tilde G-G|\le\frac12$, and [K:bp:p1:neumann] gives $\tilde G$ invertible, $|\tilde B|\le 2\mathfrak{m}$, $|\tilde B-B|\le C_B d/\delta$ (Gram invertibility before any use of $\tilde N$, $\tilde T$: Remark 1.13(T2)).

**Step S2.39: Normal frame and torsion.** $\tilde N-N=\Omega_0\big((\tilde L-L)\tilde B+L(\tilde B-B)\big)$ gives (d). On $T_{r-2\delta}$, $\tilde X_{\tilde N}-X_N=(\tilde A-A)\tilde N+A(\tilde N-N)-\partial_\omega(\tilde N-N)$, with the first two terms $\le 2\mathfrak{m}^3 d+\mathfrak{m}C_{\Delta N}d/\delta$ and the third $\le n\Theta C_{\Delta N}d/\delta^2$ ([K:bp:p1:cauchy-corollaries] with $|\omega|_1\le n\Theta$); so $|\tilde X_{\tilde N}-X_N|\le C_{\Delta X}d/\delta^2$. Finally $\tilde T-T=\tilde N^\top\Omega_0(\tilde X_{\tilde N}-X_N)+(\tilde N-N)^\top\Omega_0 X_N$ and $|X_N|\le C_X/\delta$ on $T_{r-\delta}$ ([K:bp:p2:frame-basic](d), whose hypotheses [K:bp:p2:A2]–[K:bp:p2:A4] and $|\omega|\le\Theta$ are exactly those of this lemma; no error or torsion condition) give $C_{\Delta T}d/\delta^2$. □

## 2.6. Suspected errata and sign checks.

*Remark* 2.17 (Candidate errata of HL19, checked against the pinned PDF (`arxiv_1711.05515.pdf`, v1)). All three were checked on the rendered pages 18, 19, 46 (not only on the text layer).

**E1 (strict inequalities).:** Thm 3.6, (3.16), p. 19: "$\|K_\infty-K\|_{\rho_\infty} < \dfrac{\mathfrak{C}_2\|E_c\|_\rho}{\gamma^2\rho^{2\tau}}$, $|\omega_\infty-\omega| < \dfrac{\mathfrak{C}_3\|E_c\|_\rho}{\gamma^2\rho^{2\tau}}$"; likewise (3.11), p. 18 for Thm 3.1. With $E_c\equiv 0$ the right side is 0, and (3.15) holds, but no $K_\infty$ satisfies $\|K_\infty-K\|<0$. The statements must read "$\le$" (the frozen I-G1 does).

**E2 (sign in (3.12)).:** p. 18: "(3.12) $X_h(K(\theta)) = \mathfrak{L}_\omega K(\theta)$, $\langle c \circ K\rangle = c_0$". With $\mathfrak{L}_\omega = -\partial_\omega$ (4.1), p. 20, this reads $X_h(K) = -DK\omega$, contradicting the invariance equation (2.1), p. 6, $X_h(K) = DK\,\omega$, and the error (2.2), p. 7, $E = X_h(K) - DK\omega$. The correct form is $X_h(K) + \mathfrak{L}_\omega K = 0$, which is how p. 45 (the total error before (6.1)) and p. 48 (Lemma 6.2, $\bar{E}_c$) write it. Our Remark 2.4 uses the consistent convention.

**E3 (Rem. 3.7, first equality).:** p. 19: "$Dh(K(\theta))N(\theta) = X_h(K(\theta))^\top \Omega(K(\theta))N(\theta) = -\omega^\top DK(\theta)^\top \Omega(K(\theta))N(\theta) = (\omega^\top\ 0_{n-d}^\top) = \hat{\omega}^\top$". By Rem. 2.1, p. 6, $\Omega X_h = (Dh)^\top$, so $Dh = (\Omega X_h)^\top = -X_h^\top \Omega$: the first equality misses a minus sign. The second equality is likewise wrong for $X_h = +DK\omega$ (it would need $X_h = -DK\omega$, the E2 convention). The two slips cancel and the final value $Dh\,N = \hat{\omega}^\top$ is correct, as Lemma 2.8(a) shows (using $L^\top \Omega_0 N = -I_n$, [K:`bp:p2:frame-identities`](a)); an independent check at the trivial torus is Lemma 2.6, Step S2.9. The correct chain is $Dh\,N = -X_h^\top \Omega N = -\omega^\top DK^\top \Omega N = \omega^\top$.

Other points checked and found consistent (no erratum): (6.7)–(6.10), p. 46, with $\mathfrak{L}_\omega = -\partial_\omega$, $\mathfrak{R}_\omega = -\mathcal{R}$ (Remark 2.4); (6.3) and $\bar{\omega} = (1 - \xi^\omega)\omega$ (p. 50, (6.32)); the equation $QL\omega = \binom{\omega}{0} - \binom{0}{\Omega_K \omega}$ against the display on p. 46 (Lemma 2.14(a)); the sign $\Delta\omega = -\omega\xi^\omega$. HL19's (6.9) omits the constant $\omega\xi^\omega$ (Remark 2.13(i)), which is harmless.

*Remark* 2.18 (Doubts about the frozen statements). None of the frozen statements used in this part looks false. Two reading points. (a) The frozen $\mathsf{B}_h$ uses the real derivative of $\mathrm{Re}\, h$ via `frequencyMap`; Lemma 2.6 identifies it with the complex Hessian of $h$ only under "$h$ real on the real points of a neighbourhood of $I$", which I-HEAD assumes on all of $B_{r_0}$; without that hypothesis $\mathsf{T}_h$ and $\mathsf{B}_h$ can differ by the imaginary part. (b) The frozen last row of $\mathsf{T}_H$ is $DH(K)N$, not $\omega^\top$ (Definition 1.6); Lemma 2.8 shows the difference is $(\eta^L)^\top = O(e)$, so the bordered matrices with the frozen row and with the invariant row are perturbatively comparable once a Neumann bound ([K:`bp:p1:neumann`]) and the associated smallness and margin conditions are proved; this is not automatic, because $\sigma_T$ is fixed before the error in I-G1. The frozen row $DH(K)N$ is retained throughout.

## 3. One isoenergetic step, the iteration, and the proof of I-G1

This part proves Theorem 1.14 (I-G1). It adapts the quasi-Newton scheme of track K's Part 2 ([K:`bp:p2:standing`]–[K:`bp:p2:g1`]) to HL19's isoenergetic scheme (§6.2–6.3, pp. 47–53). The unknown of one step is now the pair $(\Delta K, \Delta\omega)$ with $\Delta\omega$ *parallel to the ray* $\mathbb{R}\omega^*$, determined together with $\Delta K$ by the bordered system of Part 2 (Lemma 2.12): the frequency correction $\xi^\omega$ absorbs the mean of the torsion equation that the constant $\xi_0$ cannot, once $\xi_0$ is spent on the *normal* energy equation $\langle Y\xi^N\rangle = -E^\omega$ ($Y = DH(K)N$). The remaining tangential part $\langle D(H\circ K)\xi^L\rangle$ of $\langle DH(K)\Delta K\rangle$ is not imposed; it is a quadratic defect, estimated in Lemma 3.8.

What changes with respect to track K.

(1) The frequency of the $s$-th iterate is $\omega_s = s_s\omega^*$ with $s_s$ real and moving; the Diophantine condition is inherited from $\omega^*$ (Remark 2.2). The distance to the two endpoints of the segment (hypothesis (I9), reading remark 1.15(b)) is carried as a margin that decreases from $\nu$ to at least $\nu/2$.

(2) The invariance error and the energy error $E^\omega = \langle H\circ K\rangle - E_0$ are controlled together, and the averaged $(n+1)\times(n+1)$ extended torsion replaces $\langle T\rangle$. Its inverse is perturbed by the Neumann series ([K:`bp:p1:neumann`]), which is HL19 Lemma A.1, p. 55, in the max-row-sum norm.

(3) *Open point of the plan: does* [K:`bp:p2:newton-error`] *apply?* No, and not merely for lack of hypotheses (see Remark 3.9): for the frozen frequency $\omega$ the first bracket $X_H(K+\Delta K) - D(K+\Delta K)\omega$ has a *first-order* defect $L\Delta\omega$, removed only by the change of frequency. The new error is therefore computed at $\bar{\omega} = \omega + \Delta\omega$ in Lemma 3.8, with the extra term $-D\Delta K\,\Delta\omega$ (HL19 p. 50, $L_{\Delta\omega}\Delta K$).

(4) Several lemmas of track K are stated for tori that are $(r,\beta,\varepsilon)$-admissible, a notion that includes invertibility of the *torsion* average $\langle T\rangle$ (condition [K:bp:p2:A5]). In the isoenergetic setting $\langle T\rangle$ itself need not be invertible, so those lemmas are not cited; their conclusions that do not involve $\langle T\rangle$ are restated (Lemmas 2.15, 2.16) and proved without that hypothesis.

No other result of track K is changed. Constants depend on $(n,\tau,M,\sigma_K,\sigma_B,\sigma_T,\mu,\nu)$ only, *not* on $\omega^*$, $\sigma_\omega$, $\bar\gamma$ (except through $C_1$, see Theorem 3.17); the hypotheses (I9), (I10) are used qualitatively.

### 3.1. **Standing data, the ray, constants.**

**Standing assumption 3.1** (Standing data of Part 3)**.** Fix $n\ge 2$, $\tau\ge n-1$, $M,\sigma_K,\sigma_B,\sigma_T>0$, $\sigma_\omega>1$, $\nu>0$, $\omega^*\in\mathbb{R}^n\setminus\{0\}$. Let $D\subset\mathbb{C}^n\times\mathbb{C}^n$ and $H:\mathbb{C}^n\times\mathbb{C}^n\to\mathbb{C}$ satisfy (I1)–(I4) of Theorem 1.14 (open translation-invariant $D$; $H$ holomorphic, 1-periodic in $\theta$, real at real points of $D$; $\|DH\|,\|X_H\|,\|DX_H\|,\|D^2X_H\|,\|D^2H\|\le M$ on $D$). Let $E_0\in\mathbb{R}$, $\gamma>0$ and $\omega^*\in\mathcal{D}_{\gamma,\tau}$.

Taking $\mathfrak{B}:=D$, $h:=H$ this contains Assumption [K:bp:p2:standing] (which does not use $\|DH\|\le M$ or $\|D^2H\|\le M$); hence [K:bp:p2:hamiltonian] applies to $H$ on $D$, and Assumption 2.1 holds for every $\omega=s\omega^*$ with $s>1$. All objects $L,G,B,N,P,A,T,E,\Omega_K,E_{\mathrm{sym}},E_{\mathrm{red}},Q,\eta$ are those of [K:bp:p2:objects] with $h:=H$; $Y:=DH(K)N$; $\mathcal{T}:=\langle\mathsf{T}_H\rangle$. The energy error of a torus $K$ is $E^\omega(K):=\bar E_H(K)-E_0=\langle H\circ K\rangle-E_0$.

**Lemma 3.2** (Frequencies on the ray)**.** *Let $s,\bar s\in\mathbb{R}$ and $\omega=s\omega^*$, $\bar\omega=\bar s\omega^*$.*

*(a) If $s>0$, then $\omega\in\mathcal{D}_{\gamma s,\tau}$; if $s\ge 1$, then $\omega\in\mathcal{D}_{\gamma,\tau}$. Moreover $\gamma\le|\omega_1^*|\le|\omega^*|$, and $|\bar\omega-\omega|=|\bar s-s||\omega^*|$.*

*(b) Put $\nu(s):=\min\{s-1,\sigma_\omega-s\}|\omega^*|$. Then $\nu(\bar s)\ge\nu(s)-|\bar\omega-\omega|$. In particular, if $\nu(s)\ge\nu'$ and $|\bar\omega-\omega|\le d<\nu'$, then $1<\bar s<\sigma_\omega$ and $\nu(\bar s)\ge\nu'-d>0$.*

*Source: HL19 ($H_5'$), p. 19 and (6.22), p. 48; ours.*

*Proof.* **Step S3.1: Diophantine.** For $k\ne 0$, $|k\cdot s\omega^*|=|s||k\cdot\omega^*|\ge|s|\gamma(\sum|k_i|)^{-\tau}$ (Remark 2.2 for $s\ge1$). Taking $k=e_1$ gives $\gamma\le|\omega_1^*|$.

**Step S3.2: Margins.** $\bar s-1\ge s-1-|\bar s-s|$ and $\sigma_\omega-\bar s\ge\sigma_\omega-s-|\bar s-s|$; multiply by $|\omega^*|$. □

**Definition 3.3** (Constants)**.** Let $\varkappa,a,c_R,\mathfrak{m},\Theta,C_X,C_T,C_\Omega,C_{\mathrm{sym}},C_{\mathrm{red}},C_{\xi_0},C_G,C_B,C_{\Delta N},C_{\Delta X},C_{\Delta T}$ be the constants of [K:bp:p2:constants]; they depend only on $(n,\tau,M,\sigma_K,\sigma_B,\sigma_T)$, and $\varkappa=\tau+n\ge 2n-1\ge3$, $a=2\varkappa+2\ge8$. Define, in this order,

$$\hat C_S:=\max\{C_{\xi_0},\ \mathfrak{m}\Theta+2nc_R\mathfrak{m}^5\},\qquad \hat C_{\xi N}:=\hat C_S+2nc_R\mathfrak{m},$$
$$\hat C_g:=C_T\hat C_{\xi N}+\Theta\hat C_S+2n\mathfrak{m}^2\Theta,\qquad \hat C_\xi:=\max\{c_R\hat C_g,\ \Theta\hat C_{\xi N}\},$$
$$\hat C_{\Delta K}:=2\mathfrak{m}^2\hat C_\xi,\qquad \hat C_\Delta:=\max\{\hat C_{\Delta K},\ \Theta^2\hat C_S\},$$
$$C_P:=\mathfrak{m}^2+n+\mathfrak{m}^3+C_X,\qquad \hat C_\partial:=C_P\hat C_\xi+2\mathfrak{m}^2\Theta\hat C_g,$$
$$\hat C_{\mathrm{lin}}:=C_{\mathrm{red}}\hat C_\xi+C_{\mathrm{sym}}\hat C_g+\Theta C_\Omega\hat C_S,\qquad \hat C_{E1}:=4\mathfrak{m}^2\Theta^2\hat C_{\mathrm{lin}}+\mathfrak{m}\hat C_{\Delta K}^2+\Theta\hat C_S\hat C_\partial,$$
$$\hat C_{E2}:=n\Theta^2(2n\mathfrak{m}+C_\Omega)c_R\hat C_g+\mathfrak{m}\hat C_{\Delta K}^2,\qquad \hat C_E:=\max\{\hat C_{E1},\hat C_{E2}\},$$
$$C_Y:=n\mathfrak{m}(2\mathfrak{m}^2+C_{\Delta N}),\qquad C_{\Delta\mathsf{T}}:=C_{\Delta T}+8n^2\mathfrak{m}^4+1+C_Y,$$
$$\hat{\mathfrak{c}}_{\mathrm{step}}:=2\Theta^3\max\{C_{\mathrm{sym}},\ \mathfrak{m}C_{\Delta\mathsf{T}}\hat C_\Delta\},\qquad \hat{\mathfrak{c}}_{\mathrm{sum}}:=4\Theta^2\hat C_\Delta\max\{n,\ C_B,\ \mathfrak{m}^2C_{\Delta\mathsf{T}}\}.$$

For $\mu,\nu>0$ and $0<\rho_\infty<\rho$ put

$$\hat{\mathfrak{c}}_*:=\max\Big\{2^{2a}\hat C_E,\ \hat{\mathfrak{c}}_{\mathrm{step}}\max\{1,2/\mu\},\ \frac{2\hat{\mathfrak{c}}_{\mathrm{sum}}}{\mu},\ \frac{4\Theta^2\hat C_\Delta}{\nu}\Big\},\qquad \delta_0:=\min\Big\{1,\frac{\rho-\rho_\infty}{8}\Big\}.$$

All are finite, $\ge1$ unless stated otherwise, depend only on $(n,\tau,M,\sigma_K,\sigma_B,\sigma_T)$, and $\hat{\mathfrak{c}}_*$ in addition on $(\mu,\nu)$, $\delta_0$ on $(\rho,\rho_\infty)$. Each is introduced in one result (register, Remark 3.18).

Source: HL19 Table 4, Table 5, p. 59, (6.45)–(6.51), pp. 52–53 (their constants depend on the current torus; here, as in [K:bp:p2:constants-register], through $\mathfrak{m}$ and the margin $\mu$).

**Definition 3.4** (Isoenergetic admissible parameterization)**.** Let $r, \beta > 0$, $0 \le \varepsilon \le 1$, $s > 1$, $\omega := s\omega^*$. A map $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ is $(r, \beta, \varepsilon, s)$-*admissible* if

(B1) $K$ is holomorphic on $T_r$, primary on $T_r$, and real on reals;
(B2) $B_\beta(K(\theta)) \subset D$ for $\theta \in T_r$;
(B3) $|DK| \le \sigma_K$ on $T_r$;
(B4) $\det G_K \neq 0$ and $|B| \le \sigma_B$ on $T_r$;
(B5) $\det\langle \mathsf{T}_H \rangle \neq 0$ ($\mathsf{T}_H$ built from $(H, K, \omega)$; defined by Lemma 2.5) and $|\langle \mathsf{T}_H \rangle^{-1}| \le \sigma_T$;
(B6) $|E_{H,K,\omega}| \le \varepsilon$ on $T_r$;
(B7) $|\bar{E}_H(K) - E_0| \le \varepsilon$.

For such $K$ the frequency bound of [K:bp:p2:frequency] (hypotheses [K:bp:p2:A3], [K:bp:p2:A4], [K:bp:p2:A6] are (B3), (B4), (B6); $\omega \in \mathcal{D}_{\gamma,\tau}$ by Lemma 3.2) gives

$$\gamma \le |\omega| \le \Theta, \tag{2}$$

and $\mathfrak{m}, \Theta, \gamma$ satisfy $\gamma\delta^j \le \Theta$ for $\delta \le 1$, $j \ge 0$: absorption steps below use this exactly as [K:step:p2:cor-absorb].

The hypotheses (I5)–(I12) of I-G1 say that $K$ is $(\rho, \mu, e, s)$-admissible with the sharper constants $\sigma_K - \mu, \sigma_B - \mu, \sigma_T - \mu$ in (B3)–(B5), once $e \le 1$ is known (Step S3.33).

## 3.2. Geometry and frame perturbation without the torsion hypothesis (moved).

*Remark* 3.5 (Geometry and frame perturbation: moved to Part 2). The statement and proof of the geometry lemma without [K:bp:p2:A5], formerly here, are Lemma 2.15 (Part 2); its hypotheses are (B1)–(B4), (B6). All uses of "Lemma 2.15" below mean that lemma.

*Remark* 3.6 (Frame perturbation: moved to Part 2). The perturbation lemma without [K:bp:p2:A5] (items (a)–(e)) is Lemma 2.16; all uses of "Lemma 2.16" below mean that lemma.

## 3.3. The isoenergetic correction.

**Lemma 3.7** (Size of the isoenergetic correction)**.** *Let $s > 1$, $\omega = s\omega^*$, $0 < \delta \le 1$, $2\delta < r$, $0 \le \varepsilon \le 1$, and let $K$ be $(r, \beta, \varepsilon, s)$-admissible. Apply Lemma 2.12 with*

$$\eta^L = -N^\top \Omega_0 E, \quad \eta^N = L^\top \Omega_0 E, \quad \eta^\omega := -E^\omega(K),$$
$$T = T_{H,K,\omega}, \quad Y = DH(K)N, \quad \sigma = \sigma_T, \quad \xi_0^L := 0,$$

*and put $\xi = (\xi^L, \xi^N)$, $\xi^\omega \in \mathbb{R}$, $\Delta K := P\xi$ on $T_{r-2\delta}$, $\Delta\omega := -\omega\xi^\omega$, $\bar{\omega} := \omega + \Delta\omega = \bar{s}\omega^*$ with $\bar{s} = s(1 - \xi^\omega)$. Then the lemma applies, $\xi, \Delta K$ are holomorphic, 1-periodic and real on reals, $\xi^\omega \in \mathbb{R}$, and with $\Gamma_\delta = c_R\gamma^{-1}\delta^{-\varkappa}$:*

$$|\xi_0|, |\xi^\omega| \le \frac{\hat{C}_S\varepsilon}{\gamma\delta^{\varkappa+1}}, \qquad |\xi^N| \le \frac{\hat{C}_{\xi N}\varepsilon}{\gamma\delta^{\varkappa+1}}, \qquad |g^L| \le \frac{\hat{C}_g\varepsilon}{\gamma\delta^{\varkappa+2}} \qquad \text{on } T_{r-\delta},$$
$$|\xi| \le \frac{\hat{C}_\xi\varepsilon}{\gamma^2\delta^a}, \quad |\Delta K| \le \frac{\hat{C}_{\Delta K}\varepsilon}{\gamma^2\delta^a}, \quad |\partial_\omega\xi| \le \frac{\hat{C}_g\varepsilon}{\gamma\delta^{\varkappa+2}}, \quad |\partial_\omega\Delta K| \le \frac{\hat{C}_\partial\varepsilon}{\gamma^2\delta^{a+1}} \qquad \text{on } T_{r-2\delta},$$
$$|\Delta\omega| = |\bar{s} - s||\omega^*| \le \frac{\Theta^2\hat{C}_S\varepsilon}{\gamma^2\delta^a}.$$

*Consequently, with $d := \hat{C}_\Delta\varepsilon/(\gamma^2\delta^a)$: $|\Delta K| \le d$ on $T_{r-2\delta}$ and $|\Delta\omega| \le d$.*
*Source: HL19 Lemma 6.1, pp. 46–47, Lemma 6.2 Step 1, pp. 48–49, (6.29)–(6.31), Table 4, p. 59.*

*Proof.* **Step S3.3: Hypotheses of the border solve.** $\omega \in \mathcal{D}_{\gamma,\tau}$, $\gamma \le |\omega| \le \Theta$ by (2). By Lemma 2.15: $H_L = 2n\mathfrak{m}^2\varepsilon$, $H_N = 2n\mathfrak{m}\varepsilon$, $H_T = C_T/\delta$ on $T_{r-\delta}$; $\eta^L, \eta^N, T$ are holomorphic, 1-periodic, real on reals ([K:bp:p2:periodicity], [K:bp:p2:frame-basic]); $\langle \eta^N \rangle = 0$ (Lemma 2.9(d)). $Y$ is holomorphic, periodic, real on reals with $|Y| \le M\sigma_K\sigma_B \le \mathfrak{m}^3 =: H_Y$ (Lemma 2.5); $\mathcal{T} = \langle \mathsf{T}_H \rangle$ (same lemma(a)) is invertible with $|\mathcal{T}^{-1}| \le \sigma_T \le \mathfrak{m}$ by (B5); $\eta^\omega \in \mathbb{R}$ since $\langle H \circ K \rangle$ is real (Lemma 2.9(a)) and $E_0 \in \mathbb{R}$; $|\eta^\omega| \le \varepsilon$ by (B7).

**Step S3.4: The constants $\xi_0, \xi^\omega$.** In $S_* = \sigma \max\{H_L + H_T\Gamma_\delta H_N,\ |\eta^\omega| + H_Y\Gamma_\delta H_N\}$, using $\gamma\delta^j \le \Theta$: first entry $\le 2n\mathfrak{m}(\mathfrak{m}\Theta + c_R C_T)\varepsilon/(\gamma\delta^{\varkappa+1})$, so $\sigma\cdot$ it $\le C_{\xi_0}\varepsilon/(\gamma\delta^{\varkappa+1})$; second entry $\le (\Theta + 2nc_R\mathfrak{m}^4)\varepsilon/(\gamma\delta^{\varkappa})$, so $\sigma\cdot$ it $\le (\mathfrak{m}\Theta + 2nc_R\mathfrak{m}^5)\varepsilon/(\gamma\delta^{\varkappa+1})$. Hence $S_* \le \hat{C}_S\varepsilon/(\gamma\delta^{\varkappa+1})$.

**Step S3.5: $\xi^N, g^L, \xi^L$.** Lemma 2.12(c): $|\xi^N| \le S_* + \Gamma_\delta H_N \le \hat{C}_{\xi N}\varepsilon/(\gamma\delta^{\varkappa+1})$; $|g^L| \le H_T(S_* + \Gamma_\delta H_N) + |\omega|S_* + H_L \le \hat{C}_g\varepsilon/(\gamma\delta^{\varkappa+2})$ (absorbing $\varepsilon \le \Theta\varepsilon/(\gamma\delta^{\varkappa+2})$); $|\xi^L| = |\xi^L - \xi_0^L| \le \Gamma_\delta \sup|g^L| = c_R\hat{C}_g\varepsilon/(\gamma^2\delta^{2\varkappa+2})$, and $|\xi^N| \le \Theta\hat{C}_{\xi N}\varepsilon/(\gamma^2\delta^a)$. So $|\xi| \le \hat{C}_\xi\varepsilon/(\gamma^2\delta^a)$ and $|\Delta K| \le |P||\xi| \le \hat{C}_{\Delta K}\varepsilon/(\gamma^2\delta^a)$. By Lemma 2.12(b), $\partial_\omega\xi = (g^L, -\eta^N)$ and $|\eta^N| \le 2n\mathfrak{m}\varepsilon \le \hat{C}_g\varepsilon/(\gamma\delta^{\varkappa+2})$.

**Step S3.6: The frequency correction.** $|\Delta\omega| = |\omega||\xi^\omega| \le \Theta\hat{C}_S\varepsilon/(\gamma\delta^{\varkappa+1}) \le \Theta^2\hat{C}_S\varepsilon/(\gamma^2\delta^a)$, and $\Delta\omega = -\xi^\omega s\,\omega^*$, so $|\Delta\omega| = |\bar{s} - s||\omega^*|$.

**Step S3.7: Derivative of the correction along $\omega$.** $\partial_\omega L = AL - DE$ ([K:bp:p2:frame-identities](c)), so $|\partial_\omega L| \le \mathfrak{m}^2 + n\varepsilon/\delta$ on $T_{r-\delta}$; $\partial_\omega N = AN - X_N$, so $|\partial_\omega N| \le \mathfrak{m}^3 + C_X/\delta$. With $\delta, \varepsilon \le 1$: $|\partial_\omega P| \le C_P/\delta$ on $T_{r-\delta}$. Then $\partial_\omega \Delta K = (\partial_\omega P)\xi + P\partial_\omega\xi$ has norm $\le C_P\hat{C}_\xi\varepsilon/(\gamma^2\delta^{a+1}) + 2\mathfrak{m}^2\hat{C}_g\varepsilon/(\gamma\delta^{\varkappa+2}) \le \hat{C}_\partial\varepsilon/(\gamma^2\delta^{a+1})$ (as $\varkappa + 2 \le a + 1$).

**Step S3.8: Regularity.** By Lemma 2.12(a), (e) $\xi$ is holomorphic, periodic, real on reals ($\xi_0^L = 0 \in \mathbb{R}^n$); $P$ is too ([K:bp:p2:periodicity]); hence $\Delta K = P\xi$. The final sentence: $\hat{C}_{\Delta K}, \Theta^2\hat{C}_S \le \hat{C}_\Delta$. □

**Lemma 3.8** (The new errors)**.** *In the setting of Lemma 3.7 assume moreover $d < \beta$ and $C_{\mathrm{sym}}\varepsilon/(\gamma\delta^{\varkappa+1}) \le \frac{1}{2}$. Put $\bar{K} := K + \Delta K$ on $T_{r-2\delta}$, $\bar{K} := K$ elsewhere. Then $\bar{K}(T_{r-2\delta}) \subset D$, $S := I_{2n} - \Omega_0 E_{\mathrm{sym}}$ is invertible with $|S^{-1}| \le 2$ on $T_{r-2\delta}$, and on $T_{r-2\delta}$*

$$\bar{E} := E_{H,\bar{K},\bar{\omega}} = PS^{-1}E_{\mathrm{lin}} + \Delta^2 X + \xi^\omega\,\partial_\omega\Delta K, \tag{3}$$

$$E_{\mathrm{lin}} := E_{\mathrm{red}}\xi + \Omega_0 E_{\mathrm{sym}}\,\partial_\omega\xi - \begin{pmatrix} 0 \\ \Omega_K\omega \end{pmatrix}\xi^\omega, \qquad \Delta^2 X := X_H(\bar{K}) - X_H(K) - A\Delta K.$$

*Moreover $\bar{E}^\omega := \bar{E}_H(\bar{K}) - E_0 = E^\omega_{\mathrm{lin}} + \langle \Delta^2 H \rangle$ with $E^\omega_{\mathrm{lin}} = \langle D(H \circ K)\xi^L \rangle$, $\Delta^2 H := H(\bar{K}) - H(K) - DH(K)\Delta K$, and*

$$|\bar{E}| \le \frac{\hat{C}_E\,\varepsilon^2}{\gamma^4\delta^{2a}} \quad \text{on } T_{r-2\delta}, \qquad |\bar{E}^\omega| \le \frac{\hat{C}_E\,\varepsilon^2}{\gamma^4\delta^{2a}}.$$

*Source: HL19 Lemma 6.2 Step 2, p. 49–50, (6.34)–(6.39); $L_{\Delta\omega}\Delta K = -\xi^\omega L_\omega \Delta K$ is the last term of (3).*

*Proof.* **Step S3.9: Invertibility and domain.** As in [K:step:p2:ne-invert], [K:step:p2:ne-domain]: $|\Omega_0 E_{\mathrm{sym}}| \le \frac{1}{2}$ (Lemma 2.15), [K:bp:p1:neumann] gives $|S^{-1}| \le 2$; $QP = S$ ([K:bp:p2:frame-identities](f)) so $Q^{-1} = PS^{-1}$. $|\Delta K| \le d < \beta$ puts $\bar{K}(\theta)$ in $B_\beta(K(\theta)) \subset D$, and the segment $[K, \bar{K}]$ too.

**Step S3.10: Expansion.** On $T_{r-2\delta}$: $\bar{E} = X_H(K + \Delta K) - (L + D\Delta K)(\omega + \Delta\omega) = E + A\Delta K - \partial_\omega\Delta K - L\Delta\omega + \Delta^2 X - D\Delta K\,\Delta\omega$, and $D\Delta K\,\Delta\omega = -\xi^\omega\partial_\omega\Delta K$.

**Step S3.11: Reduction by $Q$.** By Lemma 2.14(b), $Q(E + A\Delta K - \partial_\omega\Delta K - L\Delta\omega) = \Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega - \eta + E_{\mathrm{lin}}$, which equals $E_{\mathrm{lin}}$ by the first two equations of Lemma 2.12(b) (the data are those of the correction). Apply $Q^{-1} = PS^{-1}$; this is (3).

**Step S3.12: Estimates.** By Lemmas 2.15, 3.7 on $T_{r-2\delta}$: $|E_{\rm red}\xi| \le C_{\rm red}\hat{C}_\xi\varepsilon^2/(\gamma^2\delta^{a+1})$; $|\Omega_0 E_{\rm sym}\partial_\omega\xi| \le C_{\rm sym}\hat{C}_g\varepsilon^2/(\gamma^2\delta^{2\varkappa+3})$ with $2\varkappa+3=a+1$; $|\Omega_K\omega\xi^\omega| \le \Theta C_\Omega\hat{C}_S\varepsilon^2/(\gamma^2\delta^{2\varkappa+2}) \le$ the same with $\delta^{a+1}$. So $|E_{\rm lin}| \le \hat{C}_{\rm lin}\varepsilon^2/(\gamma^2\delta^{a+1})$. By [K:bp:p2:hamiltonian](e), $|\Delta^2 X| \le M|\Delta K|^2 \le \mathfrak{m}\hat{C}^2_{\Delta K}\varepsilon^2/(\gamma^4\delta^{2a})$. Also $|\xi^\omega\partial_\omega\Delta K| \le \hat{C}_S\hat{C}_\partial\varepsilon^2/(\gamma^3\delta^{\varkappa+a+2})$. Using $\gamma^2\delta^{a-1} \le \Theta^2$ and $\gamma\delta^{a-\varkappa-2} \le \Theta$ ($a-\varkappa-2=\varkappa\ge 0$) to put everything over $\gamma^4\delta^{2a}$, and $|P||S^{-1}| \le 4\mathfrak{m}^2$: $|\bar{E}| \le \hat{C}_{E1}\varepsilon^2/(\gamma^4\delta^{2a})$.

**Step S3.13: The energy error.** $\bar{K}$ is real on reals, continuous on $\mathbb{R}^n \subset T_{r-2\delta}$, so $H\circ\bar{K}$ is integrable. Pointwise on $\mathbb{R}^n$: $H(\bar{K}) = H(K) + DH(K)\Delta K + \Delta^2 H$, so $\bar{E}^\omega = E^\omega + \langle DH(K)\Delta K\rangle + \langle\Delta^2 H\rangle$. By Lemma 2.9(f) and the third equation of the border system, $\langle DH(K)\Delta K\rangle = \eta^\omega + E^\omega_{\rm lin} = -E^\omega + E^\omega_{\rm lin}$. By [K:bp:p1:taylor] ($F = H$ on $U = D$, $|D^2H| \le M$, segment in $D$): $|\Delta^2 H| \le \frac{M}{2}|\Delta K|^2 \le \mathfrak{m}\hat{C}^2_{\Delta K}\varepsilon^2/(\gamma^4\delta^{2a})$. By Lemma 2.9(f), on $\mathbb{R}^n$: $|D(H\circ K)| \le n|\eta^N| + n|\omega||\Omega_K| \le n\Theta(2n\mathfrak{m}+C_\Omega)\varepsilon/(\gamma\delta^{\varkappa+1})$, and $|\xi^L| \le c_R\hat{C}_g\varepsilon/(\gamma^2\delta^a)$; the product is $\le \hat{C}_{E2}\varepsilon^2/(\gamma^4\delta^{2a})$ (using $\gamma\delta^{\varkappa+1} \le \Theta$). □

*Remark* 3.9 (Why [K:bp:p2:newton-error] does not apply). Its hypothesis is that $\Delta K = P\xi$ with $\xi$ from [K:bp:p2:triangular], i.e. $\Lambda\xi - \partial_\omega\xi = \eta$. Here $\Lambda\xi - \partial_\omega\xi = \eta - \binom{\omega}{0}\xi^\omega$. For the frozen frequency the first bracket is $X_H(K+\Delta K) - D(K+\Delta K)\omega = PS^{-1}E_{\rm lin} + L\Delta\omega + \Delta^2 X$ (Step S3.11), which carries the *first-order* term $L\Delta\omega = -L\omega\xi^\omega = O(\varepsilon)$, cancelled only by the frequency change. The variant therefore has the new frequency in the error, the extra defect $-D\Delta K\,\Delta\omega$, and the extra term $\Omega_K\omega\xi^\omega$ in $E_{\rm lin}$.

### 3.4. The new extended torsion and the step.

**Lemma 3.10** (The new averaged extended torsion)**.** *Let $K$ be $(r,\beta,\varepsilon,s)$-admissible, $0<\delta\le 1$, $4\delta<r$, and let $\bar{K}$, $\bar{\omega}=\bar{s}\omega^*$ be as in Lemma 3.8. Then $\bar{\omega}\in\mathbb{R}^n$ and $\mathsf{T}_H$ for $(\bar{K},\bar{\omega})$ is defined on $T_{r-3\delta}$ (no Diophantine property of $\bar{\omega}$ is used here). Assume $d<\beta$ and $2\mathfrak{m}C_{\Delta\mathsf{T}}\,d/\delta^2 \le 1$. Then $\bar{K}$ is holomorphic on $T_{r-2\delta}$, primary there and real on reals; $\det\bar{G}\neq 0$ on $T_{r-3\delta}$ with the crude bounds $|D\bar{K}|\le 2\mathfrak{m}$, $|\bar{B}|\le 2\mathfrak{m}$ (not the bounds (B3), (B4): an admissible $K$ may saturate $\sigma_K,\sigma_B$, so absolute bounds for $\bar{K}$ need an input margin, which the iteration, Proposition 3.14(iii), (iv), carries separately); and $B_{\beta-d}(\bar{K})\subset D$ on $T_{r-2\delta}$, $|D\bar{K}-DK|\le nd/\delta$ and $|\bar{B}-B|\le C_B d/\delta$ on $T_{r-3\delta}$; the matrix $\langle\bar{\mathsf{T}}\rangle := \langle\mathsf{T}_H(\bar{K},\bar{\omega})\rangle$ is defined, invertible, and*

$$|\langle\bar{\mathsf{T}}\rangle - \langle\mathsf{T}\rangle| \le C_{\Delta\mathsf{T}}\frac{d}{\delta^2}, \qquad |\langle\bar{\mathsf{T}}\rangle^{-1} - \langle\mathsf{T}\rangle^{-1}| \le 2\mathfrak{m}^2 C_{\Delta\mathsf{T}}\frac{d}{\delta^2}.$$

*Source: HL19 Lemma 6.2 Step 7, p. 51, (6.43)–(6.44); Lemma A.1, p. 55.*

*Proof.* **Step S3.14: Frame.** Apply Lemma 2.16 on $T_{r-2\delta}$ (so $2\delta<r-2\delta$) with $\tilde{K}=\bar{K}$, $|\bar{K}-K|\le d$ on $T_{r-2\delta}$ (Lemma 3.7); the smallness holds as $C_{\Delta\mathsf{T}}\ge C_{\Delta T}$. This gives the stated ball, derivative and Gram-inverse bounds, $|\bar{N}-N|\le C_{\Delta N}d/\delta$, $|\bar{N}|\le 2\mathfrak{m}^2$ on $T_{r-3\delta}$ and $|\bar{T}_\omega - T_\omega|\le C_{\Delta T}d/\delta^2$ on $T_{r-4\delta}$; $\bar{N}$, $\bar{T}_\omega$ are holomorphic on $T_{r-3\delta}\supset\mathbb{R}^n$, so $\mathsf{T}_H(\bar{K},\bar{\omega})$ is continuous on $\mathbb{R}^n$ and its average is defined (Lemma 2.5, applied to $\bar{K}$ on $T_{r-3\delta}$).

**Step S3.15: Effect of the frequency on the torsion block.** The torsion entries are $\bar{N}_i\cdot\Omega_0(\bar{A}\bar{N}_j - D\bar{N}_j\,\omega')$ with $\omega'=\omega$ or $\bar{\omega}$; so $T_{\bar{\omega}}(\bar{K}) - T_\omega(\bar{K}) = -\bar{N}^\top\Omega_0\,[D\bar{N}_j\Delta\omega]_j$. By [K:bp:p1:cauchy-corollaries] on $T_{r-3\delta}$ with loss $\delta$, $|\partial_{\Delta\omega}\bar{N}| \le n|\Delta\omega|\,2\mathfrak{m}^2/\delta$ on $T_{r-4\delta}$, and $|\bar{N}^\top|\le 4n\mathfrak{m}^2$; hence the difference has norm $\le 8n^2\mathfrak{m}^4 d/\delta$ on $T_{r-4\delta}$ (as $|\Delta\omega|\le d$).

**Step S3.16: Effect on the energy row.** $\bar{Y}-Y = (DH(\bar{K})-DH(K))\bar{N} + DH(K)(\bar{N}-N)$. By [K:bp:p1:increments-open] applied to $F=DH$ ($|DF|=|D^2H|\le M$, segment in $D$), $|DH(\bar{K})-DH(K)|\le Md$. Each entry satisfies $|\bar{Y}_b - Y_b| \le Md\,2\mathfrak{m}^2 + M\,C_{\Delta N}d/\delta$, and the row-sum norm of the $1\times n$ row is at most $n$ times this: $|\bar{Y}-Y|\le C_Y d/\delta$ on $T_{r-3\delta}$ (here $\delta\le 1$).

**Step S3.17: The average.** The difference $\langle\bar{\mathsf{T}}\rangle - \langle\mathsf{T}\rangle$ is the average ([K:bp:p2:averages](a)) of $\begin{pmatrix} \bar{T}_{\bar\omega} - T_\omega & \Delta\omega \\ \bar{Y} - Y & 0 \end{pmatrix}$ on $\mathbb{R}^n \subset T_{r-4\delta}$; by [K:bp:p2:matrix](c) its norm is $\le \max\{C_{\Delta T} d/\delta^2 + 8n^2\mathfrak{m}^4 d/\delta + d,\ C_Y d/\delta\} \le C_{\Delta\mathsf{T}} d/\delta^2$.

**Step S3.18: Inverse (Lemma A.1).** $|\langle\mathsf{T}\rangle^{-1}| \le \sigma_T \le \mathfrak{m}$, so $|\langle\mathsf{T}\rangle^{-1}||\langle\bar{\mathsf{T}}\rangle - \langle\mathsf{T}\rangle| \le \frac{1}{2}$, and [K:bp:p1:neumann] (the max-row-sum norm of $\mathbb{C}^{(n+1)\times(n+1)}$; HL19 Lemma A.1) gives invertibility and the difference bound $2\sigma_T^2 \le 2\mathfrak{m}^2$. □

**Proposition 3.11** (The isoenergetic step)**.** *Let $r > 0$, $0 < \delta \le 1$ with $4\delta < r$, $\beta > 0$, $0 \le \varepsilon \le 1$, $s > 1$ and let $K$ be $(r, \beta, \varepsilon, s)$-admissible with*

$$\hat{\mathfrak{c}}_{\mathrm{step}} \frac{\varepsilon}{\gamma^4 \delta^{2a}} \le \min\{1, \beta\}.$$

*Let $\bar{K}, \bar{\omega} = \bar{s}\omega^*$ be as in Lemmas 3.7, 3.8, and $d := \hat{C}_\Delta \varepsilon/(\gamma^2\delta^a)$. Then*

*(a) $\bar{K}$ is holomorphic, primary on $T_{r-2\delta}$, real on reals;*
*(b) $|\bar{K} - K| \le d \le \beta/2$ on $T_{r-2\delta}$, $B_{\beta-d}(\bar{K}(\theta)) \subset D$ there, and $|\bar{s} - s||\omega^*| \le d$;*
*(c) $|E_{H,\bar{K},\bar{\omega}}| \le \hat{C}_E \varepsilon^2/(\gamma^4\delta^{2a})$ on $T_{r-2\delta}$ and $|\bar{E}_H(\bar{K}) - E_0| \le \hat{C}_E\varepsilon^2/(\gamma^4\delta^{2a})$;*
*(d) $|D\bar{K} - DK| \le nd/\delta$ and $\det \bar{G} \ne 0$, $|\bar{B} - B| \le C_B d/\delta$ on $T_{r-3\delta}$;*
*(e) $\langle \mathsf{T}_H(\bar{K}, \bar{\omega})\rangle$ is invertible and $|\langle\bar{\mathsf{T}}\rangle^{-1} - \langle\mathsf{T}\rangle^{-1}| \le 2\mathfrak{m}^2 C_{\Delta\mathsf{T}} d/\delta^2$.*

*Source: HL19 Lemma 6.2, pp. 47–51.*

*Proof.* **Step S3.19: Smallness.** Put $X := \varepsilon/(\gamma^4\delta^{2a}) \le \min\{1, \beta\}/\hat{\mathfrak{c}}_{\mathrm{step}}$. With $\gamma \le \Theta$ (2) and $\delta \le 1$: $C_{\mathrm{sym}}\varepsilon/(\gamma\delta^{\varkappa+1}) \le \Theta^3 C_{\mathrm{sym}} X \le \frac{1}{2}$; $d \le \Theta^2 \hat{C}_\Delta X \le \beta/2$; $2\mathfrak{m} C_{\Delta\mathsf{T}} d/\delta^2 \le 2\mathfrak{m}\Theta^2 C_{\Delta\mathsf{T}} \hat{C}_\Delta X \le 1$ (as $\gamma^2\delta^{a-2} \le \Theta^2$).

**Step S3.20: Correction and errors.** Lemmas 3.7 (with $2\delta < r$) and 3.8 give (a) ($\bar{K}(\theta + k) = K(\theta) + (k, 0) + \Delta K(\theta)$ by periodicity of $\Delta K$), (b) except the ball, and (c) (note $\hat{C}_E$ bounds both errors).

**Step S3.21: Ball, frame, torsion.** The ball is Lemma 2.16(a); (d), (e) are Lemma 3.10. □

### 3.5. Schedule, decay, induction and limit.

**Definition 3.12** (Schedule)**.** Given $0 < \rho_\infty < \rho$, $\gamma > 0$, $e \ge 0$: $\delta_s := 2^{-s}\delta_0$, $r_s := \rho - 8\delta_0(1 - 2^{-s})$, $\varepsilon_0 := e$, $\varepsilon_{s+1} := \hat{C}_E \varepsilon_s^2/(\gamma^4\delta_s^{2a})$, $d_s := \hat{C}_\Delta \varepsilon_s/(\gamma^2\delta_s^a)$. Then $r_0 = \rho$, $r_{s+1} = r_s - 4\delta_s$, $\rho_\infty \le \rho - 8\delta_0 < r_s$, $\delta_s \le 1$, $4\delta_s < r_s$ (as [K:bp:p2:schedule]).
Source: HL19 §6.3, p. 52.

**Lemma 3.13** (Decay)**.** *If $q := 2^{2a}\hat{C}_E e/(\gamma^4\delta_0^{2a}) \le 1$, then $\varepsilon_s \le q^{2^s - 1} 2^{-2as} e \le 2^{-2as} e$; hence $\varepsilon_s/\delta_s^j \le e/\delta_0^j$ for $0 \le j \le 2a$ and $\sum_s \varepsilon_s/\delta_s^j \le 2e/\delta_0^j$ for $0 \le j \le 2a - 1$.*

*Proof.* Identical to [K:bp:p2:error-decay] (it uses only $\varepsilon_{s+1} = Q' 2^{2as}\varepsilon_s^2$ with $Q' = \hat{C}_E/(\gamma^4\delta_0^{2a})$ and [K:bp:p1:iteration] with $p = 2a$; the geometric series as there). □

**Proposition 3.14** (The induction)**.** *Let $K$, $s = s_0 > 1$ satisfy (I5)–(I12) of Theorem 1.14 on $T_\rho$ with the margin $\mu$: (B1); $B_\mu(K) \subset D$; $|DK| \le \sigma_K - \mu$; $\det G \ne 0$, $|B| \le \sigma_B - \mu$; $\det\langle\mathsf{T}_H\rangle \ne 0$, $|\langle\mathsf{T}_H\rangle^{-1}| \le \sigma_T - \mu$; $|E_{H,K,s\omega^*}| \le e$, $|\bar{E}_H(K) - E_0| \le e$; and $\nu(s_0) \ge \nu$ (I9). Assume*

$$e \le 1, \qquad \hat{\mathfrak{c}}_* \frac{e}{\gamma^4\delta_0^{2a}} \le 1.$$

*Then there are $K_j$, $s_j \in \mathbb{R}$ ($K_0 = K$) such that with $D_j := \sum_{i<j} d_i$, for every $j$:*

*(i) $K_j$ is $(r_j, \mu - D_j, \varepsilon_j, s_j)$-admissible and $\varepsilon_j \le e \le 1$;*

*(ii)* $D_j \le \mu/4$ *and* $D_j \le \nu/2$; $|K_j - K| \le D_j$ *on* $T_{r_j}$; $|s_j - s_0||\omega^*| \le D_j$ *and* $\nu(s_j) \ge \nu - D_j \ge \nu/2$, *so* $1 < s_j < \sigma_\omega$ *(Lemma 3.2);*
*(iii)* $|DK_j| \le \sigma_K - \mu + \sum_{i<j} n d_i/\delta_i \le \sigma_K - \mu/2$ *on* $T_{r_j}$;
*(iv)* $|B_j| \le \sigma_B - \mu + \sum_{i<j} C_B d_i/\delta_i \le \sigma_B - \mu/2$ *on* $T_{r_j}$;
*(v)* $|\langle \mathsf{T}_j\rangle^{-1}| \le \sigma_T - \mu + \sum_{i<j} 2\mathfrak{m}^2 C_{\Delta\mathsf{T}} d_i/\delta_i^2 \le \sigma_T - \mu/2$, $\mathsf{T}_j := \mathsf{T}_H(K_j, s_j\omega^*)$;
*(vi)* $|K_{j+1} - K_j| \le d_j$, $|DK_{j+1} - DK_j| \le n d_j/\delta_j$ *on* $T_{r_{j+1}}$, $|s_{j+1} - s_j||\omega^*| \le d_j$.

*Source: HL19 §6.3, p. 52 (the eighth term of the maximum (6.16) is checked there as here, via the margin).*

*Proof.* **Step S3.22: Summed bounds.** $K$ is $(\rho, \mu, e, s_0)$-admissible, so $\gamma \le \Theta$ by (2). Put $X := e/(\gamma^4\delta_0^{2a}) \le 1/\hat{\mathfrak{c}}_*$. Since $e/(\gamma^2\delta_0^i) \le \Theta^2 X$ for $i \le 2a$, Lemma 3.13 ($\hat{\mathfrak{c}}_* \ge 2^{2a}\hat{C}_E$; $a+2 \le 2a-1$ as $a \ge 8$) gives $\sum_i d_i/\delta_i^k \le 2\Theta^2\hat{C}_\Delta X$ for $k = 0, 1, 2$. By the choice of $\hat{\mathfrak{c}}_*$: $\sum d_i \le \min\{\mu/4, \nu/2\}$, $\sum n d_i/\delta_i \le \mu/4$, $\sum C_B d_i/\delta_i \le \mu/4$, $\sum 2\mathfrak{m}^2 C_{\Delta\mathsf{T}} d_i/\delta_i^2 \le \mu/2$ (these use $\hat{\mathfrak{c}}_* \ge 2\hat{\mathfrak{c}}_{\mathrm{sum}}/\mu$ and $\ge 4\Theta^2\hat{C}_\Delta/\nu$).

**Step S3.23: Step smallness.** $\varepsilon_j \le e \le 1$, $\varepsilon_j/\delta_j^{2a} \le e/\delta_0^{2a}$, so $\hat{\mathfrak{c}}_{\mathrm{step}}\varepsilon_j/(\gamma^4\delta_j^{2a}) \le \hat{\mathfrak{c}}_{\mathrm{step}} X \le \min\{1, \mu/2\} \le \min\{1, \mu - D_j\}$.

**Step S3.24: Base and induction.** $j = 0$: $K_0 = K$ is $(\rho, \mu, e, s_0)$-admissible; (ii)–(v) hold with empty sums. Assume (i)–(v) at $j$. Proposition 3.11 applies (with $r_j, \delta_j, \mu - D_j, \varepsilon_j, s_j$) and yields $\bar{K}_j$, $\bar{s}_j$ with $|\bar{s}_j - s_j||\omega^*| \le d_j$. Since $d_j \le D_{j+1} - D_j \le \nu/2 - D_j < \nu - D_j \le \nu(s_j)$ (Step S3.22), Lemma 3.2(b) gives $1 < \bar{s}_j < \sigma_\omega$ and $\nu(\bar{s}_j) \ge \nu(s_j) - d_j \ge \nu - D_{j+1}$. Set $K_{j+1} := \bar{K}_j$, $s_{j+1} := \bar{s}_j$. By (b)–(e) on $T_{r_{j+1}} \subset T_{r_j - 3\delta_j}$: (B1); $B_{\mu - D_{j+1}}(K_{j+1}) \subset D$; $|E_{j+1}|$, $|E^\omega_{j+1}| \le \varepsilon_{j+1}$ (B6, B7); (vi); $|K_{j+1} - K| \le D_j + d_j$; $|DK_{j+1}| \le |DK_j| + n d_j/\delta_j$; $|B_{j+1}| \le |B_j| + C_B d_j/\delta_j$; $\det\langle \mathsf{T}_{j+1}\rangle \ne 0$ with the stated inverse bound; and by Lemma 3.2(b), $\nu(s_{j+1}) \ge \nu(s_j) - d_j \ge \nu - D_{j+1}$. With the summed bounds this is (i)–(v) at $j+1$ ((B3)–(B5) follow as in [K:`step:p2:it-step`]). □

**Proposition 3.15** (The limit)**.** *In the setting of Proposition 3.14 put* $K_\infty := \lim_j K_j$ *on* $T_{\rho_\infty}$, $K_\infty := K$ *elsewhere, and* $s_\infty := \lim_j s_j$. *Then:*

*(a) the limits exist;* $K_j \to K_\infty$ *and* $DK_j \to DK_\infty$ *uniformly on* $T_{\rho_\infty}$; $K_\infty$ *is holomorphic on* $T_{\rho_\infty}$, *primary there and real on reals;*
*(b)* $|K_\infty - K| \le 2\hat{C}_\Delta e/(\gamma^2\delta_0^a) \le \mu/4$ *on* $T_{\rho_\infty}$, *and* $|s_\infty\omega^* - s_0\omega^*| \le 2\hat{C}_\Delta e/(\gamma^2\delta_0^a) \le \nu/2$;
*(c)* $1 < s_\infty < \sigma_\omega$ *(indeed* $\nu(s_\infty) \ge \nu/2$*);*
*(d)* $B_{\mu/2}(K_\infty(\theta)) \subset D$ *and* $|DK_\infty(\theta)| \le \sigma_K - \mu/2$ *on* $T_{\rho_\infty}$;
*(e)* $E_{H, K_\infty, s_\infty\omega^*} = 0$ *on* $T_{\rho_\infty}$.

*Source: HL19 p. 52–53, (6.50)–(6.51).*

*Proof.* **Step S3.25: Convergence.** $T_{\rho_\infty} \subset T_{r_j}$; by (vi) $\sum_j d_j \le 2\hat{C}_\Delta e/(\gamma^2\delta_0^a)$ and $\sum_j n d_j/\delta_j \le 2n\hat{C}_\Delta e/(\gamma^2\delta_0^{a+1})$ converge (Lemma 3.13), so $K_j$, $DK_j$ are uniformly Cauchy and $(s_j)$ is Cauchy ($\omega^* \ne 0$). [K:`bp:p1:uniform-limits`] gives holomorphy, primary, real, and $DK_\infty = \lim DK_j$ (this is (a)).

**Step S3.26: Closeness and frequency.** $|K_\infty - K| \le \sup_j D_j$ and $|s_\infty - s_0||\omega^*| \le \sup_j D_j \le \min\{\mu/4, \nu/2\}$ with the bound of Step S3.22: (b). Passing to the limit in $\nu(s_j) \ge \nu - D_j$ (continuity of $\nu(\cdot)$) gives $\nu(s_\infty) \ge \nu/2 > 0$: (c).

**Step S3.27: Margins.** If $|z - K_\infty(\theta)| < \mu/2$, then $|z - K(\theta)| < 3\mu/4$, so $z \in D$; $|DK_\infty| = \lim|DK_j| \le \sigma_K - \mu/2$ by (iii): (d).

**Step S3.28: Invariance.** For $\theta \in T_{\rho_\infty}$, $K_\infty(\theta) \in D$ where $X_H$ is continuous, so $E_j(\theta) = X_H(K_j(\theta)) - DK_j(\theta)s_j\omega^* \to E_{H, K_\infty, s_\infty\omega^*}(\theta)$; and $|E_j| \le \varepsilon_j \le 2^{-2aj}e \to 0$. □

**Lemma 3.16** (Passage of the energy average to the limit)**.** *In the setting of Proposition 3.15,* $\bar{E}_H(K_\infty) = E_0$, *and* $\bar{E}_H(K_j) \to \bar{E}_H(K_\infty)$.

*Proof.* **Step S3.29: Uniform convergence of $H \circ K_j$ on $\mathbb{R}^n$.** For $\theta \in \mathbb{R}^n$ both $K_j(\theta)$ and $K_\infty(\theta)$ lie in $B_\mu(K(\theta)) \subset D$ (distance to $K(\theta)$ at most $\mu/4$), a convex subset of $D$. By [K:bp:p1:increments-open] ($F = H$, $|DH| \le M$): $|H(K_j) - H(K_\infty)| \le M|K_j - K_\infty| \to 0$ uniformly on $\mathbb{R}^n$.

**Step S3.30: Integrals.** $H \circ K_\infty$ is continuous on $\mathbb{R}^n$, 1-periodic (K primary, H periodic), so integrable on $[0,1]^n$ ([K:bp:p1:ext-integral]); $|\bar{E}_H(K_j) - \bar{E}_H(K_\infty)| \le \sup_{\mathbb{R}^n} |H(K_j) - H(K_\infty)| \to 0$ (volume one). Since $|\bar{E}_H(K_j) - E_0| \le \varepsilon_j \to 0$ (B7), $\bar{E}_H(K_\infty) = E_0$. □

## 3.6. **Proof of I-G1.**

**Theorem 3.17** (I-G1 with explicit constants)**.** *Let the data of Theorem 1.14 be fixed $(n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma}, \sigma_\omega, \nu, \omega^*)$. With Definition 3.3 put*

$$C_1 := \rho^{4\tau} \max\{\bar{\gamma}^4,\ \hat{\mathfrak{c}}_* \delta_0^{-2a}\}, \qquad C_2 = C_3 := 2\hat{C}_\Delta\, \rho^{2\tau} \delta_0^{-a}.$$

*Then $C_1, C_2, C_3 > 0$ depend only on $(n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma}, \nu)$ (in particular they are fixed before $D, H, K, s, \gamma, e, E_0$, and do not use $\omega^*, \sigma_\omega$), and the conclusions (O1)–(O6) of Theorem 1.14 hold for all $D, H, K, s, \gamma, e, E_0$ satisfying (I1)–(I13).*
*Frozen Lean declaration:* `KAM.Iso.isoenergetic_kam`
*Source: HL19 Theorem 3.6 and Rem. 3.7, pp. 18–19; proof pp. 45–53; Table 5, p. 59, (6.45)–(6.51).*

*Proof.* **Step S3.31: The constants.** All constants of Definition 3.3 are finite, positive, and $\delta_0 \in (0,1]$, $\rho > 0$; so $C_1 \ge \rho^{4\tau}\bar{\gamma}^4 > 0$ and $C_2 > 0$. Their dependence is read off the definitions; $\rho^{4\tau}, \rho^{2\tau}$ are real powers of a positive number.

**Step S3.32: Standing data.** (I1)–(I4) and $E_0 \in \mathbb{R}$, $\gamma > 0$, $\omega^* \in \mathcal{D}_{\gamma,\tau}$ (I10) are Assumption 3.1.

**Step S3.33: Size of the errors.** From (I13) and $C_1 \ge \rho^{4\tau}\bar{\gamma}^4$: $e < \gamma^4/\bar{\gamma}^4 \le 1$ ($\gamma \le \bar{\gamma}$, I10); and $\hat{\mathfrak{c}}_* e/(\gamma^4 \delta_0^{2a}) < 1$ from $C_1 \ge \rho^{4\tau}\hat{\mathfrak{c}}_* \delta_0^{-2a}$.

**Step S3.34: Dictionary for the hypotheses.** (I5): (B1). (I6): $B_\mu(K) \subset D$, $|DK| \le \sigma_K - \mu$. (I7): $\det G \ne 0$, $|B| \le \sigma_B - \mu$ (Gram invertibility is a hypothesis, so $N$, $T$ are genuine, Remark 1.13(T2)). (I8): $\det\langle \mathsf{T}_H \rangle \ne 0$, $|\langle \mathsf{T}_H \rangle^{-1}| \le \sigma_T - \mu$; the average is defined by Lemma 2.5. (I9): $s > 1$ and $\nu(s) \ge \nu$. (I11): (B6). (I12): (B7). $s > 1$ gives $s\omega^* \in \mathcal{D}_{\gamma,\tau}$ (Lemma 3.2).

**Step S3.35: Iteration and limit.** Propositions 3.14, 3.15 apply and give $K_\infty$, $s_\infty$.

**Step S3.36: (O1), (O4)–(O6).** (O1) is Proposition 3.15(a); (O4) is (d). (O5): $|K_\infty - K| \le 2\hat{C}_\Delta e/(\gamma^2 \delta_0^a) = C_2 e/(\gamma^2 \rho^{2\tau})$ on $T_{\rho_\infty}$. (O6): $|s_\infty \omega^* - s\omega^*| \le 2\hat{C}_\Delta e/(\gamma^2 \delta_0^a) = C_3 e/(\gamma^2 \rho^{2\tau})$ (b). Also $1 < s_\infty < \sigma_\omega$ by (c).

**Step S3.37: (O2).** Proposition 3.15(e).

**Step S3.38: (O3).** $K_\infty$ is holomorphic, primary, real on reals on $T_{\rho_\infty}$ with $K_\infty(T_{\rho_\infty}) \subset D$ (by (d)), so Assumption 2.1 holds with $r = \rho_\infty$, $\omega = s_\infty \omega^* \in \mathcal{D}_{\gamma,\tau}$; $E_{H,K_\infty,s_\infty\omega^*} = 0$ there (O2) and $\bar{E}_H(K_\infty) = E_0$ (Lemma 3.16). Lemma 2.10(b) gives $H(K_\infty(\theta)) = E_0$ for all $\theta \in T_{\rho_\infty}$ (Remark 2.11); no frame or Gram determinant of $K_\infty$ is used. □

*Remark* 3.18 (Register of constants). $\hat{C}_S, \hat{C}_{\xi N}, \hat{C}_g, \hat{C}_\xi, \hat{C}_{\Delta K}, \hat{C}_\Delta$: Lemma 3.7 (HL19 $C_{\xi_0^N}, C_{\xi^\omega}, C_\omega, C_{\Delta\omega}$, (6.29)–(6.31)); $C_P, \hat{C}_\partial, \hat{C}_{\mathrm{lin}}, \hat{C}_{E1}, \hat{C}_{E2}, \hat{C}_E$: Lemma 3.8 ($C_{\mathrm{lin}}, C^\omega_{\mathrm{lin}}, C_{L\Delta K}, C_E, C_{E^\omega}, C_{E_c}$, (6.34)–(6.39)); $C_Y, C_{\Delta\mathsf{T}}$: Lemma 3.10 ($C_{\Delta T_c}$, (6.43)); $\hat{\mathfrak{c}}_{\mathrm{step}}$: Proposition 3.11; $\hat{\mathfrak{c}}_{\mathrm{sum}}, \hat{\mathfrak{c}}_*$: Proposition 3.14 ((6.46)–(6.49)); $C_1, C_2, C_3$: Theorem 3.17 ((6.45), (6.50), (6.51)).

*Remark* 3.19 (Deviations from HL19 and from track K). (W1) As in track K: open strips with pointwise bounds, the exponent $\varkappa = \tau + n$, $\gamma$-independent step constants ($\gamma \le |\omega| \le \Theta$ from [K:`bp:p2:frequency`]), uniform margin $\mu$, loss $4\delta_s$ per step; HL19's $\nu$ of (4.40)/(6.28) is replaced by $e \le 1$, $\delta \le 1$; the frozen $\nu$ is HL19's $\operatorname{dist}(\omega, \partial\Theta)$ (relative boundary: the two endpoints, review finding I01). (W2) $|\omega| \le \Theta$ comes from [K:`bp:p2:frequency`] at every iterate, not from $|\omega| < \sigma_\omega|\omega^*|$ (HL19 (6.30)); the latter is unused, which is why $C_i$ do not depend on $\omega^*, \sigma_\omega$. (W3) The first-order term $L\Delta\omega$ and the term $-\binom{0}{\Omega_K\omega}\xi^\omega$ are carried explicitly (Remark 3.9); HL19 p. 49 print $E_{\text{lin}}$ with the signs of their $\mathfrak{L}_\omega = -\partial_\omega$. (W4) Conclusions (O5), (O6) read $\le$ (not $<$), (O3) is pointwise (Part 2); (O6) is in the form $C_3 e/(\gamma^2\rho^{2\tau})$ of the frozen statement.

*Remark* 3.20 (Candidate errata and doubts). (E4) HL19 (6.51), p. 53: "$|\omega_\infty - \omega| < C_3\|E_c\|_\rho/(\gamma\rho^\tau)$", with $C_3 = a_3^\tau C_{\Delta\omega}/(1 - a_1^{-3\tau})$ (Table 5), then "thus obtaining the estimates in (3.16)", where (3.16) has $\gamma^2\rho^{2\tau}$. The exponents of $\gamma$ and $\rho$ differ, so (6.51) does not literally give (3.16). The proof here obtains the form of (3.16) directly ($d_j$ has $\gamma^{-2}\delta_j^{-a}$). (E5) Candidate source typos: HL19 pp. 49–51, (6.33), (6.36), (6.44) print $\|E\|_\rho$ where the combined error $\|E_c\|_\rho$ is needed (a pure energy discrepancy with $E = 0$ can require nonzero corrections); p. 52 prints "when $s \to 0$" for $s \to \infty$; the eight-term maximum is (6.16), not (6.15). The proof here uses the combined error $\varepsilon$ throughout. No doubt about the frozen I-G1 arose: it is proved as stated; $\nu$ is used only through $\hat{\mathfrak{c}}_*$.

## 4. Applying I-G1: local energy charts and uniform constants

This part turns the isoenergetic a-posteriori theorem (Theorem 1.14, proved in Part 3 as Theorem 3.17) into the family of invariant tori that the confinement proof of I-HEAD (Part 5) consumes. The data are those of the frozen I-HEAD (Theorem 1.16) with $n = 2$; $H = h + f$ is the perturbed Hamiltonian, and the approximate torus is the flat torus $K(\theta) = (\theta, I)$. Four things are proved.

(1) *Local energy charts* (§4.2). The nondegeneracy $\det \mathsf{B}_h \neq 0$ forces $\nabla h \neq 0$ and makes $I \mapsto (h(I), \text{direction of } \nabla h(I))$ a local diffeomorphism (Lemma 4.7). A finite cover by such charts (Lemma 4.9) is the only place where compactness of $\overline{B'}$ enters.

(2) *Two Diophantine directions* (§4.3). In each chart there are Diophantine vectors $\omega_\pm^*$ in two open sectors on either side of the direction of $\nabla h$; this uses the measure theorem G3 of track K with the fixed exponent $\tau = 2 > n - 1$ (finding I05 of the Part-0 review; no density is claimed at $\tau = 1$). For every energy $E$ in the window of the chart the two points $I_\pm(E)$ on the curve $\{h = E\}$ with $\nabla h(I_\pm) \parallel \omega_\pm^*$ exist (Lemma 4.13); no implicit function theorem with a parameter is needed.

(3) *Verification of (I1)–(I13)* for $H = h + f$ and $K(\theta) = (\theta, I_\pm)$, $e = C\varepsilon$ (§4.4), using Part 2: $\mathsf{T}_H = \mathsf{B}_h + O(\varepsilon)$ (Lemmas 2.6, 2.8), and the Cauchy estimates and the Neumann series of track K.

(4) *Uniform constants* (§4.5). All constants of Theorem 1.14 are chosen once, per ray, before $f, \varepsilon, E, I_0$; $\varepsilon_1$ is the minimum over the finitely many rays and the other smallness conditions. The output is Proposition 4.19, whose conclusion (T1)–(T7) is what Part 5 needs.

Source: The local charts, the finite cover and the choice of constants are ours (ledger D-I-CONFINEMENT). Input from the sources: the isoenergetic nondegeneracy is HL19 (3.13)–(3.14) and Rem. 3.7, p. 19, and the bordered determinant of Chi05 p. 17 (Kepler case); both are about the extended torsion, not about charts.
Source gap: Neither HL19 nor Chi05 passes from $\det \mathsf{B}_h \neq 0$ to the existence of invariant tori on both sides of a given point of an energy surface; Lemmas 4.4–4.13 supply the bridge.

### 4.1. Standing data, constants, external facts.

**Standing assumption 4.1** (Data of Part 4)**.** The data of Theorem 1.16 with $n = 2$: $B \subset \mathbb{R}^2$ open, bounded, nonempty; $s_0, r_0 > 0$; $h$ holomorphic on $B_{r_0}$ with $\operatorname{Im} h(I) = 0$ and $\det \mathsf{B}_h(I) \neq 0$ for every

real $I \in B_{r_0}$; $B' \subset \mathbb{R}^2$ open with $\overline{B'}$ compact, $\overline{B'} \subset B$; $\eta > 0$. We assume $B' \neq \emptyset$ (otherwise the conclusion of I-HEAD is vacuous and no patch is needed). The exponent is fixed:

$$\tau := 2 \quad (> n - 1 = 1).$$

The perturbation is $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$, $\varepsilon \in \mathbb{R}$ as in I-HEAD (holomorphic on $T_{s_0} \times B_{r_0}$, 1-periodic in $\theta$ there, real at real $\theta$ and real $I \in B_{r_0}$, $0 \le \varepsilon$, $|f| \le \varepsilon$ on $T_{s_0} \times B_{r_0}$), and $H := h + f$, i.e. $H(\theta, I) = h(I) + f(\theta, I)$. Write $\iota : \mathbb{R}^2 \to \mathbb{C}^2$ for the inclusion and $U := \iota^{-1}(B_{r_0}) \subset \mathbb{R}^2$, open, $B \subset U$ ([K:`bp:p4:freq-map`](i): $\iota B \subset B_{r_0}$, since $r_0 > 0$).

The setting of track K's Part 4 ([K:`bp:p4:data`]) is exactly this one (with $n = 2$, $B$ open bounded, $h$ holomorphic on $B_{r_0}$, $f$ as above), apart from its Kolmogorov and boundary-layer hypotheses, which are not used here. We therefore take its shrunken domains and constants verbatim; they are listed so that the proofs below can quote them.

**Definition 4.2** (Constants of the data)**.** *(a) From* [K:`bp:p4:data`]. $s := \min\{s_0, 1\}$, $\delta_* := \frac{1}{4}\min\{s, r_0\}$, and

$$\mu := \tfrac{1}{2}\delta_*, \qquad \rho := \tfrac{s}{4}, \qquad \rho_\infty := \tfrac{s}{8}, \qquad \mathfrak{D} := T_{s/2} \times B_{r_0/2}, \qquad \mathfrak{D}' := T_{3s/4} \times B_{3r_0/4}.$$

$M_h := \sup\{|h(I)| : I \in B_{3r_0/4}\} < \infty$ ([K:`bp:p4:h-bounded`]); $c(m, j)$ are the constants of [K:`bp:p1:cauchy-open`], $\bar{c} := \max\{1, c(2, j), c(4, j) : j = 1, 2, 3\}$, $F_j := \bar{c}\,\delta_*^{-j}$ $(j = 1, 2, 3)$, and $M := \bar{c}(M_h + 1)\delta_*^{-3}$. Note $\delta_* \le \frac{1}{4}$, $\mu \le \frac{1}{8}\min\{s, r_0\}$, $\rho_\infty < \rho$, $F_1 \le F_2$.
*(b) The compact action set.* $d_B := \inf\{\|I - J\| : I \in \overline{B'},\ J \in \mathbb{R}^2 \setminus B\}$ if $B \neq \mathbb{R}^2$ (always, since $B$ is bounded); $d_B > 0$ because $\overline{B'}$ is compact and contained in the open set $B$. Put

$$\eta_1 := \min\{\eta,\ d_B/2\}, \qquad \mathcal{S} := \{I \in \mathbb{R}^2 : \operatorname{dist}(I, \overline{B'}) \le d_B/2\}.$$

*(c) Gradient and bordered inverse on $\mathcal{S}$.*

$$m_h := \min_{I \in \mathcal{S}} \|\nabla h(I)\|, \qquad M_g := \max_{I \in \mathcal{S}} \|\nabla h(I)\|, \qquad S_h := \max_{I \in \mathcal{S}} |\mathsf{B}_h(I)^{-1}| \quad \text{(row-sum norm)}.$$

*(d) Data for I-G1.*

$$\nu := \tfrac{1}{4}m_h, \qquad \sigma_\omega := \tfrac{4M_g}{m_h} + 2, \qquad \sigma_K := 1 + \mu, \qquad \sigma_B := 1 + \mu, \qquad \sigma_T := 2S_h + \mu,$$

$$\varepsilon_T := \frac{1}{4S_h F_2}, \qquad e := F_1\,\varepsilon \quad \text{(the error bound of I-G1, } C_e := F_1\text{)}.$$

All of these depend only on $(B, s_0, r_0, h, B', \eta)$ (in fact $\eta$ enters only $\eta_1$), and not on $f$, $\varepsilon$ or any point.

**Lemma 4.3** ($\mathcal{S}$ and the constants are well defined)**.** *$\mathcal{S}$ is compact, $\overline{B'} \subset \mathcal{S} \subset B$, and $0 < m_h \le M_g < \infty$, $0 < S_h < \infty$.*

*Proof.* **Step S4.1: Compactness and inclusion.** $I \mapsto \operatorname{dist}(I, \overline{B'})$ is continuous, so $\mathcal{S}$ is closed; it is bounded since $\overline{B'}$ is. If $I \notin B$ then $\|I - J\| \ge d_B$ for the nearest $J \in \overline{B'}$ (definition of $d_B$), so $\operatorname{dist}(I, \overline{B'}) \ge d_B > d_B/2$; hence $\mathcal{S} \subset B$.

**Step S4.2: The gradient is nonzero.** By Lemma 4.4(a) below (which does not use this lemma), $\nabla h(I) \neq 0$ for every $I \in U \supset \mathcal{S}$. $\nabla h$ is continuous on $U$ ([K:`bp:p4:freq-map`](iii): it is differentiable), so $\|\nabla h\|$ attains a positive minimum and a finite maximum on the compact set $\mathcal{S}$.

**Step S4.3: The bordered inverse.** The entries of $\mathsf{B}_h(I)$ are continuous on $U$ ([K:`bp:p4:freq-map`](iii) and the continuity of the second derivatives of the holomorphic $h$, [K:`bp:p1:holomorphic-regularity`]), and $\det \mathsf{B}_h \neq 0$ on $U$ (hypothesis, $\iota U \subset B_{r_0}$). By Cramer's rule ([K:`bp:p1:ext-cramer`]), $\mathsf{B}_h^{-1} = \operatorname{adj} \mathsf{B}_h / \det \mathsf{B}_h$ has continuous entries on $U$, which are real ([K:`bp:p1:real-inverse`]). The row-sum norm is continuous, hence bounded on $\mathcal{S}$. □

External facts used in this part. In addition to [K:bp:p1:ext-cramer], [K:bp:p1:ext-integral] and the external facts of Part 2:

(Y1) *Inverse function theorem.* If $\Omega \subset \mathbb{R}^m$ is open, $F : \Omega \to \mathbb{R}^m$ is $C^1$ and $DF(c)$ is invertible, then there is an open $W \ni c$, $W \subset \Omega$, such that $F|_W$ is a homeomorphism onto an open set $F(W)$ with $C^1$ inverse. (Mathlib: `HasStrictFDerivAt.toOpenPartialHomeomorph`.) The restriction of such an $F|_W$ to an open $W' \subset W$ has the same property.

(Y2) *Compactness.* A continuous function on a compact set attains its extrema; a compact subset of $\mathbb{R}^2$ covered by open sets is covered by finitely many of them (Heine–Borel).

(Y3) *Lebesgue measure on $\mathbb{R}^2$.* It is monotone and countably subadditive on outer measures; a nonempty open set has positive measure; a bounded set has finite measure.

4.2. **Local energy charts.** The bordered Hessian is, in dimension two, the curvature of the energy curve times a positive factor; we use only its determinant. For $g = (g_1, g_2) \in \mathbb{R}^2$ put $g^\perp := (-g_2, g_1)$ and $\det(g, w) := g_1 w_2 - g_2 w_1 = g^\perp \cdot w$, where $\cdot$ is the Euclidean dot product on $\mathbb{R}^2$.

**Lemma 4.4** (The bordered determinant)**.** *Let $I \in U$ and $g := \nabla h(I)$, $\nabla^2 h(I) := D(\nabla h)(I)$ (real symmetric,* [K:bp:p4:freq-map]*(iii)). Then*

$$\det \mathsf{B}_h(I) = -\, g^\perp \cdot \nabla^2 h(I)\, g^\perp.$$

*Consequently (a) $\nabla h(I) \neq 0$ whenever $\det \mathsf{B}_h(I) \neq 0$ (in particular on all of $U$); (b) if $g \neq 0$, then $\det \mathsf{B}_h(I) \neq 0$ iff $\det\big(g, \nabla^2 h(I) g^\perp\big) \neq 0$, and $\det(g, \nabla^2 h(I) g^\perp) = g^\perp \cdot \nabla^2 h(I) g^\perp = -\det \mathsf{B}_h(I)$.*

*Proof.* **Step S4.4: Expansion.** Write $h_a = \partial_a h(I)$, $h_{ab} = \partial_a \partial_b h(I)$ (real, $h_{12} = h_{21}$). By Definition 1.7 and Lemma 2.6 (Step S2.10), $\mathsf{B}_h = \begin{pmatrix} h_{11} & h_{12} & h_1 \\ h_{12} & h_{22} & h_2 \\ h_1 & h_2 & 0 \end{pmatrix}$. Expanding along the last row, $\det \mathsf{B}_h = h_1(h_{12}h_2 - h_1 h_{22}) - h_2(h_{11}h_2 - h_1 h_{12}) = -(h_{22}h_1^2 - 2h_{12}h_1h_2 + h_{11}h_2^2)$. With $g^\perp = (-h_2, h_1)$, $g^\perp \cdot \nabla^2 h\, g^\perp = h_{11}h_2^2 - 2h_{12}h_1h_2 + h_{22}h_1^2$.

**Step S4.5: Consequences.** (a) If $g = 0$ the right-hand side is 0. (b) is the identity $\det(g, w) = g^\perp \cdot w$ with $w = \nabla^2 h\, g^\perp$. □

**Definition 4.5** (Direction coordinates at a point)**.** Let $c \in \overline{B'}$ and $g_c := \nabla h(c) \neq 0$, $q_c := g_c \cdot g_c > 0$. For $w \in \mathbb{R}^2$ put

$$a_c(w) := \frac{g_c \cdot w}{q_c}, \qquad b_c(w) := \frac{\det(g_c, w)}{q_c}, \qquad \text{so that } w = a_c(w)\, g_c + b_c(w)\, g_c^\perp.$$

(check: first component $\big(g_1(g \cdot w) - g_2 \det(g, w)\big)/q = w_1$, similarly the second). On $\{w : a_c(w) > 0\}$ define the *slope* $\sigma_c(w) := b_c(w)/a_c(w) = \det(g_c, w)/(g_c \cdot w)$, and on

$$\mathcal{U}_c := \{I \in U :\ a_c(\nabla h(I)) > 0\} \quad (\text{open},\ c \in \mathcal{U}_c,\ \text{since } a_c(g_c) = 1)$$

the *chart map*

$$m_c(I) := \sigma_c(\nabla h(I)), \qquad \Phi_c(I) := \big(h(I),\, m_c(I)\big) \in \mathbb{R}^2, \qquad I \in \mathcal{U}_c.$$

Since $\nabla h$ is $C^1$ on $U$ ([K:bp:p4:freq-map]), $\Phi_c$ is $C^1$ on $\mathcal{U}_c$; $\Phi_c(c) = (h(c), 0)$.

*Remark* 4.6 (Slopes encode rays). If $w, w'$ have $a_c(w), a_c(w') > 0$, then $\sigma_c(w) = \sigma_c(w')$ iff $w' = \lambda w$ for some $\lambda > 0$, and then $\lambda = a_c(w')/a_c(w)$ (from $w = a(g + \sigma g^\perp)$). Hence for $I \in \mathcal{U}_c$ and a vector $\omega^*$ with $a_c(\omega^*) > 0$: $m_c(I) = \sigma_c(\omega^*)$ iff $\nabla h(I) = s\,\omega^*$ with $s = a_c(\nabla h(I))/a_c(\omega^*) > 0$, and then, for the max norm, $s = \|\nabla h(I)\|/\|\omega^*\|$.

**Lemma 4.7** (Local chart)**.** *For every $c \in \overline{B'}$ the derivative $D\Phi_c(c)$ is invertible, with $\det D\Phi_c(c) = -\det \mathsf{B}_h(c)/q_c \neq 0$. Hence there is an open $W \subset \mathcal{U}_c$ with $c \in W$ such that $\Phi_c|_W : W \to V := \Phi_c(W)$ is a homeomorphism onto an open set with $C^1$ inverse, and $W$ may be taken inside any prescribed open neighbourhood of $c$.*

*Source: ours (the isoenergetic nondegeneracy as a chart); the determinant identity is the one of Chi05 p. 17 in the Kepler case, HL19 Rem. 3.7, p. 19.*

*Proof.* **Step S4.6: Derivative of $h$ and of $m_c$.** $Dh(c)u = g_c \cdot u$ ([K:bp:p4:freq-map](ii)). The slope map $\sigma_c$ has numerator $\det(g_c, \cdot)$ vanishing at $g_c$ and denominator $g_c \cdot w = q_c$ at $w = g_c$, so $D\sigma_c(g_c)[\delta w] = \det(g_c, \delta w)/q_c$. By the chain rule and $D(\nabla h)(c) = \nabla^2 h(c)$, $Dm_c(c)u = \det(g_c, \nabla^2 h(c)u)/q_c = g_c^\perp \cdot \nabla^2 h(c)u/q_c = (\nabla^2 h(c) g_c^\perp) \cdot u/q_c$ by the symmetry of $\nabla^2 h(c)$.

**Step S4.7: Determinant.** $D\Phi_c(c)$ has the rows $g_c^\top$ and $(\nabla^2 h(c) g_c^\perp)^\top/q_c$, so $\det D\Phi_c(c) = \det(g_c, \nabla^2 h(c) g_c^\perp)/q_c = -\det \mathsf{B}_h(c)/q_c$ by Lemma 4.4(b). It is nonzero because $c \in \overline{B'} \subset B \subset U$ and $\iota U \subset B_{r_0}$.

**Step S4.8: Inverse function theorem.** $\mathcal{U}_c$ is open, $\Phi_c$ is $C^1$ on it, and (Y1) gives $W$; intersecting with a prescribed open neighbourhood of $c$ keeps the conclusion. □

**Definition 4.8** (Patch)**.** A *patch* is a tuple $\mathfrak{p} = (c, W, \delta_E, \delta_m, \rho_c)$ with $c \in \overline{B'}$, $W$ as in Lemma 4.7, $\delta_E, \delta_m, \rho_c > 0$, such that, with $V = \Phi_c(W)$,

(P1) $W \subset \{I \in \mathbb{R}^2 : \|I - c\| < \eta_1/4\}$ (so $W \subset \mathcal{S}$);
(P2) the box $Q := [h(c) - \delta_E,\, h(c) + \delta_E] \times [-\delta_m, \delta_m]$ lies in $V$;
(P3) $\rho_c \le \eta_1/4$, the ball $\{\|I - c\| < \rho_c\}$ lies in $W$, and on it $|h(I) - h(c)| < \delta_E/2$ and $|m_c(I)| < \delta_m/4$.

We write $\Phi_{\mathfrak{p}}^{-1} : V \to W$ for the inverse.

**Lemma 4.9** (Finite cover)**.** *There are finitely many patches $\mathfrak{p}_j = (c_j, W_j, \delta_{E,j}, \delta_{m,j}, \rho_j)$, $j \in \mathcal{P}$, with $\overline{B'} \subset \bigcup_j \{\|I - c_j\| < \rho_j\}$. The family depends only on $(B, s_0, r_0, h, B', \eta)$.*
*Source: ours.*

*Proof.* **Step S4.9: A patch at every $c \in \overline{B'}$.** Fix $c$. Lemma 4.7 with the neighbourhood $\{\|I - c\| < \eta_1/4\}$ gives $W$ with (P1); $V$ is open and contains $\Phi_c(c) = (h(c), 0)$, so (P2) holds for $\delta_E, \delta_m$ small. $W$ is open and $h$, $m_c$ are continuous on $\mathcal{U}_c$ with $m_c(c) = 0$, so (P3) holds for $\rho_c$ small.

**Step S4.10: Compactness.** The open balls $\{\|I - c\| < \rho_c\}$, $c \in \overline{B'}$, cover the compact set $\overline{B'}$; take a finite subcover ((Y2)). □

### 4.3. Two Diophantine directions and the two points on each energy curve.

**Lemma 4.10** (Diophantine vectors in an open set)**.** *Let $\tau = 2$ and let $O \subset \mathbb{R}^2$ be nonempty, open and bounded. Then there are $\gamma > 0$ and $\omega \in O \cap \mathcal{D}_{\gamma,2}$.*
*Source: standard; measure input is G3 of track K, Pos01 §1d, p. 5, via* [K:bp:p0:g3]*. At the endpoint $\tau = n - 1 = 1$ G3 is not available and this argument gives nothing (Part-0 review, I05): we do not use $\tau = 1$.*
*Frozen Lean declaration:* `KAM.volume_not_diophantine_le`

*Proof.* **Step S4.11: Positive measure.** $0 < \operatorname{meas} O < \infty$ ((Y3)). [K:bp:p0:g3] with $n = 2$, $\tau = 2 > 1$ and $\Omega = O$ gives $C \ge 0$ with $\operatorname{meas}\{\omega \in O : \omega \notin \mathcal{D}_{\alpha,2}\} \le C\alpha$ for all $\alpha > 0$.

**Step S4.12: Choice of $\gamma$.** Take $\gamma := \operatorname{meas} O/(2(C+1))$. Then $\operatorname{meas}\{\omega \in O : \omega \notin \mathcal{D}_{\gamma,2}\} \le C\gamma < \operatorname{meas} O$, so by subadditivity $O \cap \mathcal{D}_{\gamma,2} \neq \emptyset$. □

**Lemma 4.11** (Rays in a sector)**.** *Let $c \in \overline{B'}$, $J \subset \mathbb{R}$ a nonempty open bounded interval. Then there are $\gamma > 0$ and $\omega^* \in \mathbb{R}^2$ with*

$$\omega^* \in \mathcal{D}_{\gamma,2}, \qquad a_c(\omega^*) > 0, \qquad \sigma_c(\omega^*) \in J, \qquad \tfrac{1}{4} m_h < \|\omega^*\| < \tfrac{1}{2} m_h.$$

*Proof.* **Step S4.13: An open nonempty set.** $O := \{\omega \in \mathbb{R}^2 : a_c(\omega) > 0,\ \sigma_c(\omega) \in J,\ \frac{1}{4}m_h < \|\omega\| < \frac{1}{2}m_h\}$ is open (preimage of open sets under continuous maps on the open set $\{a_c > 0\}$) and bounded. For $\varsigma \in J$ and $t > 0$ put $\omega_t = t(g_c + \varsigma g_c^{\perp})$. Then $a_c(\omega_t) = t$, $\sigma_c(\omega_t) = \varsigma$, and $\|\omega_t\| = t\|g_c + \varsigma g_c^{\perp}\|$ with $\|g_c + \varsigma g_c^{\perp}\| > 0$ (its Euclidean length is $\sqrt{q_c(1+\varsigma^2)}$). Choosing $t = \frac{3}{8}m_h/\|g_c + \varsigma g_c^{\perp}\|$ gives $\omega_t \in O$.

**Step S4.14: Diophantine.** Lemma 4.10 applied to $O$. □

**Definition 4.12** (Rays of a patch)**.** For each patch $\mathfrak{p} = (c, W, \delta_E, \delta_m, \rho_c)$ of Lemma 4.9 choose, by Lemma 4.11 with $J_+ = (\delta_m/2, \delta_m)$ and $J_- = (-\delta_m, -\delta_m/2)$, vectors $\omega^*_{\mathfrak{p},\pm}$ and constants $\gamma_{\mathfrak{p},\pm} > 0$ with $\omega^*_{\mathfrak{p},\pm} \in \mathcal{D}_{\gamma_{\mathfrak{p},\pm},2}$, $a_c(\omega^*_{\mathfrak{p},\pm}) > 0$, $\frac{1}{4}m_h < \|\omega^*_{\mathfrak{p},\pm}\| < \frac{1}{2}m_h$, and put

$$m_{\mathfrak{p},\pm} := \sigma_c(\omega^*_{\mathfrak{p},\pm}), \qquad \text{so } m_{\mathfrak{p},+} \in (\tfrac{\delta_m}{2}, \delta_m),\quad m_{\mathfrak{p},-} \in (-\delta_m, -\tfrac{\delta_m}{2}).$$

There are finitely many rays ($2|\mathcal{P}|$); the choice is made once and depends only on $(B, s_0, r_0, h, B', \eta)$.

**Lemma 4.13** (The two points on the energy curve)**.** *Let $\mathfrak{p}$ be a patch, $E \in \mathbb{R}$ with $|E - h(c)| \le \delta_E$, and write $\omega^*_{\pm}, m_{\pm}$ for the data of Definition 4.12. Then $(E, m_{\pm}) \in Q \subset V$, so*

$$I_{\pm}(E) := \Phi_{\mathfrak{p}}^{-1}(E, m_{\pm}) \in W \quad (\textit{real points of } \mathbb{R}^2)$$

*are defined, and*

- *(a) $h(I_{\pm}) = E$, $m_c(I_{\pm}) = m_{\pm}$, and $\nabla h(I_{\pm}) = s^{\circ}_{\pm}\,\omega^*_{\pm}$ with $s^{\circ}_{\pm} = \|\nabla h(I_{\pm})\|/\|\omega^*_{\pm}\| \in \big(2,\ 4M_g/m_h\big)$;*
- *(b) $\|I_{\pm} - c\| < \eta_1/4$, $I_{\pm} \in \mathcal{S} \subset B$, and $I_- \neq I_+$; for every $I_0$ with $\|I_0 - c\| < \rho_c$: $\|I_{\pm} - I_0\| < \eta_1/2$;*
- *(c) (ordering) $m_c(I_-) \le -\delta_m/2$, $m_c(I_+) \ge \delta_m/2$, whereas $|m_c(I_0)| < \delta_m/4$ for $\|I_0 - c\| < \rho_c$.*

*Moreover $\Phi_{\mathfrak{p}}$ maps $\{I \in W :\ h(I) = E,\ |m_c(I)| \le \delta_m\}$ homeomorphically onto $\{E\} \times [-\delta_m, \delta_m]$, so that in the chart the energy curve is the segment $m \mapsto (E, m)$ and $I_-$, $I_+$ are the points with $m = m_{\pm}$ ("$I_-$, $I_0$, $I_+$ in this order along the curve").*

*Source: ours.*

*Proof.* **Step S4.15: Existence.** $|m_{\pm}| < \delta_m$ and $|E - h(c)| \le \delta_E$, so $(E, m_{\pm}) \in Q \subset V = \Phi_c(W)$ by (P2).

**Step S4.16: Parallel gradient.** $\Phi_c(I_{\pm}) = (E, m_{\pm})$ gives $h(I_{\pm}) = E$ and $m_c(I_{\pm}) = \sigma_c(\omega^*_{\pm})$, with $a_c(\nabla h(I_{\pm})) > 0$ since $I_{\pm} \in W \subset \mathcal{U}_c$ and $a_c(\omega^*_{\pm}) > 0$. Remark 4.6 gives $\nabla h(I_{\pm}) = s^{\circ}_{\pm}\omega^*_{\pm}$ with $s^{\circ}_{\pm} = \|\nabla h(I_{\pm})\|/\|\omega^*_{\pm}\|$. By (P1), $I_{\pm} \in \mathcal{S}$ (since $\eta_1/4 \le d_B/8$), so $m_h \le \|\nabla h(I_{\pm})\| \le M_g$, while $\frac{1}{4}m_h < \|\omega^*_{\pm}\| < \frac{1}{2}m_h$; hence $2 < s^{\circ}_{\pm} < 4M_g/m_h$.

**Step S4.17: Position.** $I_{\pm} \in W$ and (P1) give $\|I_{\pm} - c\| < \eta_1/4$; with $\|I_0 - c\| < \rho_c \le \eta_1/4$ ((P3)), $\|I_{\pm} - I_0\| < \eta_1/2$. $I_- \neq I_+$ since $m_c(I_-) < 0 < m_c(I_+)$. Ordering: (c) is $m_{\pm}$ vs. (P3).

**Step S4.18: Energy segment.** $\Phi_{\mathfrak{p}} : W \to V$ is a homeomorphism, and $\{E\} \times [-\delta_m, \delta_m] \subset Q \subset V$, whose preimage is exactly $\{I \in W : h(I) = E, |m_c(I)| \le \delta_m\}$. □

## 4.4. The approximate torus and the hypotheses of I-G1.

**Lemma 4.14** (The hypotheses (I1)–(I4) for $H = h + f$)**.** *Let Assumption 4.1 hold with $\varepsilon \le 1$ and $D := \mathfrak{D} = T_{s/2} \times B_{r_0/2}$. Then (I1)–(I4) of Theorem 1.14 hold for $(D, H)$ with the constant $M$ of Definition 4.2. Moreover for $z \in \mathfrak{D}$ and $j = 1, 2, 3$: $\|D^j f(z)\| \le F_j\varepsilon$, in particular $\|X_f(z)\| \le F_1\varepsilon$ and $\|D^2 f(z)\| \le F_2\varepsilon$.*

*Source: ours, quoting* [K:`bp:p4:h1`] *and* [K:`bp:p4:derivative-bounds`].

*Proof.* **Step S4.19: (I1)–(I3).** Steps [K:`step:p4:h1-open`] and [K:`step:p4:h1-holo`] of [K:`bp:p4:h1`]: $D$ is open and invariant under $\theta \mapsto \theta + k$; $H$ is holomorphic on $D \subset T_{s_0} \times B_{r_0}$,

1-periodic in $\theta$, and real at real phase points of $D$ (for real $\theta$ and real $I \in B_{r_0}$: $h(I)$ and $f(\theta, I)$ are real by the hypotheses of I-HEAD).

**Step S4.20: (I4).** By [K:`bp:p4:derivative-bounds`](a), (c), (d) (Step [K:`step:p4:db-cauchy`]) with $\varepsilon \le 1$: $\|DH\|, \|X_H\|, \|DX_H\|, \|D^2 X_H\| \le M$ on $D$. The same step gives $\|D^j H\| \le \bar{c}(M_h + \varepsilon)\delta_*^{-j} \le M$ ($\delta_* \le 1$) for $j = 2$, which is the bound on $\|D^2 H\|$ in (I4). The bounds $\|D^j f\| \le F_j \varepsilon$ are [K:`bp:p4:derivative-bounds`](c). □

**Lemma 4.15** (The flat torus of $H = h + f$)**.** *Let Assumption 4.1 hold with $\varepsilon \le 1$, $D = \mathfrak{D}$. Let $I \in B$ (a real point), $\omega^* \in \mathbb{R}^2$, $s' \in \mathbb{R}$ with $\nabla h(I) = s'\omega^*$, $\omega := s'\omega^*$, and $K(\theta) := (\theta, I)$ for $\theta \in \mathbb{C}^2$. Then for $\theta \in T_{s/2}$:*

*(a) $K$ is holomorphic, primary and real on reals; $K(T_{s/2}) \subset D$; $DK(\theta)v = (v, 0)$, $\|DK(\theta)\| = 1$; $G_K \equiv I_2$, $|G_K^{-1}| = 1$; $N_{\cdot j} \equiv (0, e_j)$. For $\theta \in T_\rho$: $B_\mu(K(\theta)) \subset D$.*

*(b) $E_{H,K,\omega}(\theta) = X_f(K(\theta)) = \big((\partial_{I_a} f)_a, (-\partial_{\theta_a} f)_a\big)(\theta, I)$ and $\|E_{H,K,\omega}(\theta)\| \le F_1\varepsilon$.*

*(c) $\mathsf{T}_H(\theta) = \mathsf{B}_h(I) + R(\theta)$ with $R(\theta) = \begin{pmatrix} (\partial_{I_i}\partial_{I_j} f)_{ij} & 0 \\ (\partial_{I_b} f)_b^\top & 0 \end{pmatrix}(\theta, I)$, $|R(\theta)| \le 2F_2\varepsilon$, hence $|\langle \mathsf{T}_H \rangle - \mathsf{B}_h(I)| \le 2F_2\varepsilon$.*

*(d) If $I \in \mathcal{S}$ and $\varepsilon \le \varepsilon_T$, then $\det\langle \mathsf{T}_H \rangle \ne 0$ and $|\langle \mathsf{T}_H \rangle^{-1}| \le 2S_h = \sigma_T - \mu$.*

*(e) $|\bar{E}_H(K) - h(I)| \le \varepsilon$; hence $|\bar{E}_H(K) - E_0| \le \varepsilon$ whenever $h(I) = E_0$.*

*Source: ours; Part 2, Lemmas 2.6 and 2.8 for the frame and the last row; HL19 Rem. 3.7, p. 19.*

*Proof.* **Step S4.21: The torus and its frame.** Steps S2.5 and S2.6 of Lemma 2.6 apply verbatim ($K$ and the frame do not involve the Hamiltonian): $K$ is affine, primary, real on reals ($I$ real), $DK = (\mathrm{id}, 0)$, $G_K = I_2$, $N_{\cdot j} = (0, e_j)$ constant. $\|DK(\theta)v\| = \|(v, 0)\| = \|v\|$. For $\theta \in T_{s/2}$ and $I \in B$, $K(\theta) \in T_{s/2} \times B \subset D$. If $\theta \in T_\rho$ and $\|(\theta', I') - K(\theta)\| < \mu$, then $|\mathrm{Im}\,\theta_i'| < s/4 + \mu < s/2$ ($\mu \le s/8$) and $\|I' - I\| < \mu \le r_0/8$, so $I' \in B_{r_0/2}$: $B_\mu(K(\theta)) \subset D$.

**Step S4.22: The error.** $X_H(\theta, I) = \big((\partial_{I_a} h + \partial_{I_a} f)_a, (-\partial_{\theta_a} f)_a\big)$ and $(\partial_{I_a} h(I))_a = \nabla h(I) = \omega$ (Lemma 2.6, Step S2.7); with $DK\omega = (\omega, 0)$ we get $E = X_f(K(\theta))$, and $\|X_f\| \le F_1\varepsilon$ on $D$ (Lemma 4.14).

**Step S4.23: Torsion and border.** $DX_H(z)(0, e_j) = \big((\partial_{I_a}\partial_{I_j} H)_a, (-\partial_{\theta_a}\partial_{I_j} H)_a\big)$, $\Omega_0(x, y) = (-y, x)$, $D_\theta N_{\cdot j} = 0$. So $T_{ij} = (0, e_i) \cdot \Omega_0 DX_H N_{\cdot j} = \partial_{I_i}\partial_{I_j} H = \partial_i\partial_j h(I) + \partial_{I_i}\partial_{I_j} f(\theta, I)$ (as in Step S2.8). The last column is $\omega = \nabla h(I)$. By Lemma 2.8(a), the last row is $Y = \omega^\top - (\eta^L)^\top$ with $\eta_b^L = -N_{\cdot b}^\top \Omega_0 E = -\partial_{I_b} f$; i.e. $Y_b = \partial_b h(I) + \partial_{I_b} f(\theta, I)$ (equal to the direct value $DH(K)N_{\cdot b}$). The upper-left block of $\mathsf{B}_h(I)$ is $(\partial_a\partial_b h(I))$ and its border is $\nabla h(I)$ (Step S2.10), which gives (c)'s identity. For the size: row $i \le 2$ of $R$ has absolute sum $\sum_j |D^2 f[e_i, e_j]| \le 2\|D^2 f\| \le 2F_2\varepsilon$, the last row $\sum_b |\partial_{I_b} f| \le 2F_1\varepsilon \le 2F_2\varepsilon$ (Lemma 4.14). The average over $[0, 1]^2 \subset \mathbb{R}^2 \subset T_{s/2}$ is integrable (Lemma 2.5(a)) and the norm of an average is at most the supremum ([K:`bp:p1:ext-integral`]); $\langle \mathsf{B}_h \rangle = \mathsf{B}_h$ since the cube has volume one.

**Step S4.24: Invertibility.** For $I \in \mathcal{S}$, $|\mathsf{B}_h(I)^{-1}| \le S_h$ and $|\mathsf{B}_h^{-1}||\langle R \rangle| \le S_h \cdot 2F_2\varepsilon \le \frac{1}{2}$ when $\varepsilon \le \varepsilon_T$. [K:`bp:p1:neumann`] gives invertibility of $\mathsf{B}_h(I) + \langle R \rangle = \langle \mathsf{T}_H \rangle$ and $|\langle \mathsf{T}_H \rangle^{-1}| \le 2S_h$ (a real matrix, [K:`bp:p1:real-inverse`]); $2S_h = \sigma_T - \mu$.

**Step S4.25: Energy average.** $H(K(\theta)) = h(I) + f(\theta, I)$ and for $\theta \in [0, 1]^2$, $(\theta, I) \in T_{s_0} \times B_{r_0}$ so $|f(\theta, I)| \le \varepsilon$; hence $|\bar{E}_H(K) - h(I)| = |\int f(\theta, I)\, d\theta| \le \varepsilon$. □

**Lemma 4.16** (One application of I-G1)**.** *Let $\mathfrak{p}$ be a patch with ray $\omega^* = \omega^*_{\mathfrak{p},\pm}$, $\gamma = \gamma_{\mathfrak{p},\pm}$. Apply Theorem 1.14 with $n = 2$, $\tau = 2$, $\rho = s/4$, $\rho_\infty = s/8$ and the constants $M, \sigma_K, \sigma_B, \sigma_T, \mu, \sigma_\omega, \nu$ of Definition 4.2, $\bar{\gamma} := \gamma$, $\omega^*$ as given. Its constants $C_1, C_2, C_3 > 0$ (denoted $C_i^{\mathfrak{p},\pm}$) depend only on*

*$(B, s_0, r_0, h, B', \eta)$ and the ray. Then, for every $f, \varepsilon$ of Assumption 4.1 with*

$$\varepsilon \le 1, \qquad \varepsilon \le \varepsilon_T, \qquad \varepsilon < \varepsilon^{\mathfrak{p},\pm} := \frac{\gamma^4 \rho^8}{C_1^{\mathfrak{p},\pm} F_1},$$

*every $E \in \mathbb{R}$ with $|E - h(c)| \le \delta_E$, and $I := I_\pm(E)$, $s' := s_\pm^\circ$ of Lemma 4.13, the hypotheses (I1)–(I13) of Theorem 1.14 hold for*

$$D = \mathfrak{D}, \quad H = h + f, \quad K(\theta) = (\theta, I), \quad s = s_\pm^\circ, \quad \gamma, \quad e = F_1\varepsilon, \quad E_0 = E.$$

*Consequently there are $K_\infty$ and $s_\infty \in (1, \sigma_\omega)$ with (O1)–(O6), in particular $\|K_\infty - K\| \le C_2 F_1 \varepsilon/(\gamma^2\rho^4)$ on $T_{\rho_\infty}$, $\|(s_\infty - s_\pm^\circ)\omega^*\| \le C_3 F_1 \varepsilon/(\gamma^2\rho^4)$.*

*Proof.* **Step S4.26: (I1)–(I4).** Lemma 4.14 ($\varepsilon \le 1$).

**Step S4.27: (I5)–(I7).** Lemma 4.15(a) with $I \in B$ (Lemma 4.13(b)): $K$ is holomorphic, primary, real on reals on $T_\rho$; $B_\mu(K(\theta)) \subset D$ and $\|DK\| = 1 = \sigma_K - \mu$; $\det G_K = 1$, $|G_K^{-1}| = 1 = \sigma_B - \mu$.

**Step S4.28: (I8).** $\nabla h(I) = s_\pm^\circ \omega^*$ (Lemma 4.13(a)), $I \in \mathcal{S}$; Lemma 4.15(d) with $\omega = s_\pm^\circ \omega^*$ gives $\det\langle \mathsf{T}_H \rangle \ne 0$, $|\langle \mathsf{T}_H \rangle^{-1}| \le \sigma_T - \mu$ ($\varepsilon \le \varepsilon_T$).

**Step S4.29: (I9).** $1 < 2 < s < 4M_g/m_h < \sigma_\omega$. $(s-1)\|\omega^*\| > \|\omega^*\| > \frac{1}{4} m_h = \nu$ and $(\sigma_\omega - s)\|\omega^*\| > 2\|\omega^*\| > \nu$.

**Step S4.30: (I10)–(I12).** $0 < \gamma \le \bar\gamma = \gamma$, $\omega^* \in \mathcal{D}_{\gamma,2}$, $e = F_1\varepsilon \ge 0$ (Definition 4.12). (I11) is Lemma 4.15(b). (I12): $h(I) = E = E_0$, Lemma 4.15(e), and $\varepsilon \le F_1\varepsilon = e$ since $F_1 \ge 1$.

**Step S4.31: (I13).** $C_1 e/(\gamma^4\rho^{4\tau}) = C_1 F_1 \varepsilon/(\gamma^4\rho^8) < 1$ by $\varepsilon < \varepsilon^{\mathfrak{p},\pm}$ (and trivially if $\varepsilon = 0$).

**Step S4.32: Conclusion.** Theorem 1.14, with $C_2 e/(\gamma^2\rho^{2\tau}) = C_2 F_1 \varepsilon/(\gamma^2\rho^4)$. □

*Remark* 4.17 (Constants of Part 3 make the order of quantifiers harmless). The frozen Theorem 1.14 chooses $C_1, C_2, C_3$ after $\omega^*$ and $\bar\gamma$, which is why the application is per ray: finitely many rays, finitely many applications, $\varepsilon_1$ below is a minimum over a finite set. Theorem 3.17 shows that its constants do not depend on $\omega^*$ and $\sigma_\omega$ (only on $\bar\gamma$ through $C_1$); this is not needed.

### 4.5. Uniform application: the output of Part 4.

**Definition 4.18** (Uniform constants)**.** With the patches $\mathfrak{p}_j$ ($j \in \mathcal{P}$) of Lemma 4.9, their rays (Definition 4.12) and the constants $C_i^{j,\pm}$ of Lemma 4.16, put

$$C_K := \max_{j\in\mathcal{P},\pm} \frac{F_1 \max\{C_2^{j,\pm}, C_3^{j,\pm}\}}{(\gamma_{j,\pm})^2 \rho^4},$$

$$\varepsilon_1 := \min\Big\{1,\ \varepsilon_T,\ \min_j \tfrac{1}{2}\delta_{E,j},\ \min_{j,\pm} \varepsilon^{j,\pm},\ \frac{\eta_1}{2C_K}\Big\} \in (0,1].$$

*Order of choice:* $(B, s_0, r_0, h, B', \eta) \to \mathfrak{D}, \mu, \delta_*, M, F_j, \mathcal{S}, m_h, M_g, S_h, \nu, \sigma$'s $\to$ patches $\to$ rays, $\gamma_{j,\pm} \to C_i^{j,\pm} \to C_K, \varepsilon_1$. Neither depends on $f$, $\varepsilon$, $E$, $I_0$.

**Proposition 4.19** (Invariant tori on both sides: output of Part 4)**.** *Let Assumption 4.1 hold (the data of I-HEAD with $n = 2$, $B' \ne \emptyset$, $\eta > 0$, $\tau = 2$). Let $\varepsilon_1, C_K$ be as in Definition 4.18, $\rho_\infty = s/8$, $\mathcal{P}$, $\mathfrak{p}_j$ as above. Then for every $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ and $\varepsilon$ with the hypotheses of I-HEAD and $\varepsilon < \varepsilon_1$, every $I_0 \in \overline{B'}$ and every $E \in \mathbb{R}$ with $|E - h(I_0)| \le \varepsilon$ there are $j \in \mathcal{P}$ with $\|I_0 - c_j\| < \rho_j$ and, for each sign $\pm$, a map $K_\pm : \mathbb{C}^2 \to \mathbb{C}^2 \times \mathbb{C}^2$, numbers $s_\pm \in (1, \sigma_\omega)$ and a real point $I_\pm \in \mathbb{R}^2$, such that, with $H = h + f$, $\omega_\pm^* = \omega_{j,\pm}^*$ and $\omega_\pm := s_\pm \omega_\pm^*$:*

- *(T1)* (torus) *$K_\pm$ is holomorphic on $T_{\rho_\infty}$, primary on $T_{\rho_\infty}$ and real on reals; its restriction to $\mathbb{R}^2$ is real-analytic with values in $\mathbb{R}^2 \times \mathbb{R}^2$.*
- *(T2)* (invariance, Diophantine frequency) *$E_{H,K_\pm,\omega_\pm}(\theta) = 0$ for all $\theta \in T_{\rho_\infty}$; $\omega_\pm^* \in \mathcal{D}_{\gamma_{j,\pm},2}$, so $\omega_\pm$ is nonresonant; $\omega_\pm \in \mathbb{R}^2 \setminus \{0\}$.*

*(T3)* (energy) $H(K_\pm(\theta)) = E$ *for all* $\theta \in T_{\rho_\infty}$ *(pointwise); in particular* $\mathcal{H}_{h,f}(K_\pm(\theta)) = E$ *for real* $\theta$*.*

*(T4)* (closeness to the flat torus; graph inputs) $\|K_\pm(\theta) - (\theta, I_\pm)\| \le C_K\varepsilon$ *on* $T_{\rho_\infty}$*, and, writing* $K_\pm(\theta) = (\theta + u_\pm(\theta), I_\pm + v_\pm(\theta))$ *with* $u_\pm, v_\pm : T_{\rho_\infty} \to \mathbb{C}^2$ *holomorphic and* 1*-periodic,* $\|u_\pm\|_{\rho_\infty}, \|v_\pm\|_{\rho_\infty} \le C_K\varepsilon$ *and* $\|Du_\pm\|_{\rho_\infty/2}, \|Dv_\pm\|_{\rho_\infty/2} \le 4C_K\varepsilon/\rho_\infty$ *(operator norms;* $\rho_\infty = s/8$*).*

*(T5)* (the base points and the frequencies) $h(I_\pm) = E$*,* $\nabla h(I_\pm) = s^\circ_\pm \omega^*_\pm$ *with* $s^\circ_\pm \in (2, 4M_g/m_h)$*,* $\|\omega_\pm - \nabla h(I_\pm)\| = |s_\pm - s^\circ_\pm|\|\omega^*_\pm\| \le C_K\varepsilon$*; and* $\nabla h(I_\pm)$*,* $\omega_\pm$ *are positive multiples of* $\omega^*_\pm$*.*

*(T6)* (position) $I_\pm \in B$*,* $\|I_\pm - I_0\| < \eta_1/2 \le \eta/2$*, and for every real* $\theta$*:* $\|K_\pm(\theta)_2 - I_0\| < \eta$ *and* $K_\pm(\theta)_2 \in \mathcal{S} \subset B$ *(the action component stays in* $B$ *and within* $\eta$ *of* $I_0$*). Moreover* $B_{\mu/2}(K_\pm(\theta)) \subset \mathfrak{D}$ *and* $\|DK_\pm(\theta)\| \le \sigma_K - \mu/2$ *on* $T_{\rho_\infty}$*.*

*(T7)* (ordering in the energy chart) *With the chart* $\Phi_j = (h, m_{c_j})$ *of* $\mathfrak{p}_j$*:* $I_\pm = \Phi_j^{-1}(E, m_{j,\pm})$*,* $m_{c_j}(I_-) \le -\delta_{m,j}/2$*,* $m_{c_j}(I_+) \ge \delta_{m,j}/2$ *and* $|m_{c_j}(I_0)| < \delta_{m,j}/4$*, so that, on the arc* $\Phi_j^{-1}(\{E\} \times [-\delta_{m,j}, \delta_{m,j}])$ *of the energy curve* $\{h = E\}$ *(the segment* $m \mapsto (E, m)$ *in the chart), the slope coordinate of* $I_-$ *is below* $-\delta_{m,j}/2$ *and that of* $I_+$ *above* $\delta_{m,j}/2$*, while* $I_0$ *has slope coordinate in* $(-\delta_{m,j}/4, \delta_{m,j}/4)$*.*

*The numbers* $\varepsilon_1$*,* $C_K$*,* $\rho_\infty$*,* $\mathcal{P}$*,* $\sigma_\omega$*,* $M_g$ *depend only on* $(B, s_0, r_0, h, B', \eta)$*.*

*Source: ours (D-I-CONFINEMENT); the isoenergetic theorem used is HL19 Thm 3.6 and Rem. 3.7, pp. 18–19, in the form of Theorem 1.14.*

*Proof.* **Step S4.33: Choice of the patch.** Since $I_0 \in \overline{B'}$ and the balls cover (Lemma 4.9), there is $j$ with $\|I_0 - c_j\| < \rho_j$. Write $c = c_j$, $\delta_E = \delta_{E,j}$ etc. By (P3), $|h(I_0) - h(c)| < \delta_E/2$, hence $|E - h(c)| \le \varepsilon + \delta_E/2 \le \delta_E$ because $\varepsilon < \varepsilon_1 \le \delta_E/2$.

**Step S4.34: The base points.** Lemma 4.13 gives $I_\pm := I_\pm(E)$, $s^\circ_\pm$: this is (T5) (first part), (T7) and $I_\pm \in B$, $\|I_\pm - I_0\| < \eta_1/2$.

**Step S4.35: Application of I-G1.** $\varepsilon < \varepsilon_1 \le \min\{1, \varepsilon_T, \varepsilon^{j,\pm}\}$, so Lemma 4.16 gives $K_\pm := K_\infty$ and $s_\pm := s_\infty \in (1, \sigma_\omega)$ with (O1)–(O6) of Theorem 1.14 for $e = F_1\varepsilon$, $E_0 = E$. (T1): (O1), and the real-analyticity by [K:`bp:p1:real-restriction`]. (T2): (O2); $\omega^*_\pm \in \mathcal{D}_{\gamma,2}$ is a property of the ray. (T3): (O3), and $\mathcal{H}_{h,f} = \operatorname{Re} H = H$ at real phase points of $D$ whose value is real (Definition 1.4). (T5): (O6) with $C_3F_1\varepsilon/(\gamma^2\rho^4) \le C_K\varepsilon$; the multiples are positive since $s_\pm > 1$ and $s^\circ_\pm > 2$.

**Step S4.36: Closeness and derivatives.** (O5): $\|K_\pm - K\| \le C_2F_1\varepsilon/(\gamma^2\rho^4) \le C_K\varepsilon$ on $T_{\rho_\infty}$, $K = (\theta, I_\pm)$. $u_\pm, v_\pm$ are the components of $K_\pm - K$, holomorphic on $T_{\rho_\infty}$, 1-periodic because both $K_\pm$ and $K$ are primary: $K_\pm(\theta + k) - K(\theta + k) = K_\pm(\theta) + (k, 0) - K(\theta) - (k, 0)$. The max norm gives $\|u\|, \|v\| \le C_K\varepsilon$. [K:`bp:p1:cauchy-strip`] with $n = 2$ and $\delta = \rho_\infty/2$ gives $\|Du\|_{\rho_\infty/2}, \|Dv\|_{\rho_\infty/2} \le 2 \cdot \frac{2}{\rho_\infty} C_K\varepsilon = 4C_K\varepsilon/\rho_\infty$ (applied to the pair $(u, v)$ as a vector-valued function bounded by $C_K\varepsilon$).

**Step S4.37: Position.** For real $\theta$, $\|K_\pm(\theta)_2 - I_\pm\| \le C_K\varepsilon < \eta_1/2$ since $\varepsilon < \eta_1/(2C_K)$. With $\|I_\pm - I_0\| < \eta_1/2$ this gives $\|K_\pm(\theta)_2 - I_0\| < \eta_1 \le \eta$. Since $I_0 \in \overline{B'}$, $\operatorname{dist}(K_\pm(\theta)_2, \overline{B'}) < \eta_1 \le d_B/2$, so $K_\pm(\theta)_2 \in \mathcal{S} \subset B$ (Lemma 4.3). The remaining claims of (T6) are (O4) of Theorem 1.14 and the shape of $\sigma_K$: $\sigma_K - \mu/2$ and the ball $B_{\mu/2}$ are those of (O4), with $D = \mathfrak{D}$. □

*Remark* 4.20 (What Part 5 does with this). Given $z_0 = (\theta_0, I_0)$ with $I_0 \in B'$, its energy is $E := \mathcal{H}_{h,f}(z_0) = h(I_0) + \operatorname{Re} f(\theta_0, I_0)$, so $|E - h(I_0)| \le \varepsilon$ and Proposition 4.19 applies; take $\varepsilon_0 \le \varepsilon_1$. The two tori are invariant under the flow of $\mathcal{H}_{h,f}$ ($E_{H,K,\omega} = 0$ and $\mathcal{H} = H$ at real points), lie on the energy surface $\{\mathcal{H}_{h,f} = E\}$ of $z_0$, are graphs over $\mathbb{T}^2$ for $\varepsilon$ small by (T4) (Part 5, p5.2), and sit on the two sides $m_c \le -\delta_m/2$, $m_c \ge \delta_m/2$ of $z_0$ in the energy chart (T7) (Part 5, p5.1, p5.3); the confinement $\|z(t)_2 - I_0\| < \eta$ is (T6) and the compact enclosure in $\mathcal{S} \subset B$ gives global existence. No positive measure of tori is used: only the existence of the $2|\mathcal{P}|$ Diophantine rays.

*Remark* 4.21 (Deviations, doubts). (1) $\tau = 2$, not $\tau \geq 1$ (review I05); (2) the hypothesis of I-HEAD that $\det \mathsf{B}_h \neq 0$ on all of the real slice of $B_{r_0}$ is used only on $\mathcal{S} \subset B$ (and for $\nabla h \neq 0$, Lemma 4.4); (3) $\mathcal{H}_{h,f}$ is the real part of $H$, and $\mathcal{H}_{h,f} = H$ because $H$ is real at real points of $\mathfrak{D}$ (I3); (4) the finite cover is not a Lebesgue-number argument: each patch has its own $\varepsilon^{j,\pm}$ and $\varepsilon_1$ is the minimum; (5) no frozen statement looks doubtful for this part. The only matrix identity we use beyond Part 2 is Lemma 4.4; the nonsingular-derivative argument of track N ([N:`bpN:p2:qc-kernel`]) is not needed.

## 5. Separation by invariant tori; proof of I-HEAD

This part proves Theorem 1.16 from Proposition 4.19 (Part 4) and the shared ODE layer of track N ([N:`bpN:p3:gen-uniqueness`], [N:`bpN:p3:gen-barrier`], [N:`bpN:p3:gen-continuation-periodic`], [N:`bpN:p3:ham-energy`], [N:`bpN:p3:ham-periodic`], [N:`bpN:p3:ham-reduction`]).

Idea. A solution with initial point $z_0 = (\theta_0, I_0)$ stays on the energy level $\{\mathcal{H}_{h,f} = E\}$, $E = \mathcal{H}_{h,f}(z_0)$. Part 4 supplies two invariant tori $\mathcal{T}_\pm$ of exact energy $E$, one on each side of $I_0$ along the energy curve of the unperturbed problem. We do not invoke a Jordan-type separation theorem. Instead, near $I_0$ we use the coordinate $m_c$ of the energy chart (Definition 4.5) as a continuous function, and we describe each torus as a graph $I = Y_\pm(\varphi)$ over *all* real angles $\varphi \in \mathbb{R}^2$ (Lemma 5.4). The set $\mathcal{V} = \{(\varphi, I) : \ m_c(Y_-(\varphi)) < m_c(I) < m_c(Y_+(\varphi)),\ I \in \Omega\}$ is open and contains $z_0$. On the energy level, a point of the boundary of $\mathcal{V}$ has $m_c(I) = m_c(Y_\pm(\varphi))$ and the same energy as $(\varphi, Y_\pm(\varphi))$; the energy equation is uniquely solvable for the energy coordinate $e = h(I)$ because $f$ is small (Lemma 5.6), so $I = Y_\pm(\varphi)$ and the boundary point lies on a torus. A solution that starts in $\mathcal{V}$ therefore cannot leave it (the barrier lemma [N:`bpN:p3:gen-barrier`]); $\mathcal{V}$ projects into a compact subset of $B$ at distance $< \eta$ from $I_0$ (Proposition 5.8), and the continuation theorem gives global existence. The constant $\varepsilon_0$ is the minimum of $\varepsilon_1$ of Part 4 and finitely many patch constants (Definition 5.3).

Source: ours (D-I-CONFINEMENT). CC03 Theorem 1.1, p. 7 (read on the page image) asserts, for one numerical Hamiltonian, "a $\phi^t$-invariant region $\mathcal{J}$, whose boundary is given by two KAM tori"; it is a precedent for the mechanism only and is not used as an input. Arnold's original paper (Arn63b) is **[unverified]** for the corresponding statement (docs/tracks/I/sources.md); nothing below depends on it.

Source gap: No pinned source proves, for a general real-analytic $h$ with $\det \mathsf{B}_h \neq 0$, the separation of the energy level by two invariant tori or the resulting all-time confinement; Propositions 5.8 and the proof of Theorem 1.16 below supply this bridge.

### 5.1. The real Hamiltonian and external facts.

Standing data: Assumption 4.1 ($n = 2$, $\tau = 2$, $B' \neq \emptyset$), the constants of Definition 4.2 and the patches, rays and constants of Part 4. Write $H_{\mathbb{R}} := \mathcal{H}_{h,f} : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ (Definition 1.4), $U = \iota^{-1}(B_{r_0}) \supset B$ (open), $W_B := \mathbb{R}^2 \times B$ (open in $\mathbb{R}^2 \times \mathbb{R}^2$), $P = \mathbb{R}^2 \times \mathbb{R}^2$ with the max norm, and $X_{\mathbb{R}} := X_{H_{\mathbb{R}}}$.

**Lemma 5.1** (Regularity and ODE reduction)**.** *For $f, \varepsilon$ as in Theorem 1.16:*

- *(a) $H_{\mathbb{R}}(\theta, I) = h(\iota I) + f(\iota\theta, \iota I)$ on $\mathbb{R}^2 \times U$ (the real part is the value), and $H_{\mathbb{R}}$ is $C^\infty$ on the open set $\mathbb{R}^2 \times U \supset W_B$; $H_{\mathbb{R}}(\theta + k, I) = H_{\mathbb{R}}(\theta, I)$ for $k \in \mathbb{Z}^2$.*
- *(b) At every real $z \in \mathbb{R}^2 \times U$, $X_{\mathbb{R}}(z) = X_H(z)$, where $H = h + f$ is the holomorphic function on $T_{s_0} \times B_{r_0}$ and $X_H$ its complex Hamiltonian field.*
- *(c) $X_{\mathbb{R}}$ is $C^1$ on $W_B$ and $X_{\mathbb{R}}(\theta + k, I) = X_{\mathbb{R}}(\theta, I)$ on $W_B$. For $J \subset \mathbb{R}$ and $z : \mathbb{R} \to P$: `IsHamiltonianSolutionOn` $H_{\mathbb{R}}\, B\, J\, z \iff$ `IsSolIn` $X_{\mathbb{R}}\, W_B\, J\, z$.*
- *(d) For real $(\varphi, I) \in \mathbb{R}^2 \times B$ and $v \in \mathbb{R}^2$: $|\partial_t f(\varphi, I + tv)|_{t=0}| \leq F_1 \varepsilon \|v\|$ (real derivative), provided $\varepsilon \leq 1$.*
- *(e) $|H_{\mathbb{R}}(\theta, I) - h(\iota I)| \leq \varepsilon$ on $\mathbb{R}^2 \times U$.*

*Source: standard;* [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`], [N:`bpN:p3:ham-reduction`].

*Proof.* **Step S5.1: Values.** For $(\theta, I) \in \mathbb{R}^2 \times U$: $\iota\theta \in T_{s_0}$ ($s_0 > 0$) and $\iota I \in B_{r_0}$. $h(\iota I)$ and $f(\iota\theta, \iota I)$ are real by hypothesis, so $H_{\mathbb{R}} = h \circ \iota + f \circ (\iota, \iota)$, and $|f| \le \varepsilon$ gives (e).

**Step S5.2: Smoothness and periodicity.** $H = h + f$ is holomorphic on the open set $T_{s_0} \times B_{r_0}$, hence $C^\infty$ ([K:`bp:p1:holomorphic-regularity`]); $H_{\mathbb{R}}$ is its composition with the $\mathbb{R}$-linear inclusion, hence $C^\infty$ on $\mathbb{R}^2 \times U$. Periodicity is that of $f$.

**Step S5.3: The fields.** (b) is [K:`bp:p1:hamiltonian-real`] ($H$ real at the real points of its domain, $H_{\mathbb{R}} = \operatorname{Re} H$). (c): [N:`bpN:p3:ham-reduction`] with the open set $U' := B$, $W = \mathbb{R}^2 \times U$ and $H_{\mathbb{R}}$ $C^2$ on $W$; periodicity of $X_{\mathbb{R}}$ is [N:`bpN:p3:ham-periodic`].

**Step S5.4: Derivative bound.** $\mathbb{R}^2 \times B \subset \mathfrak{D} = T_{s/2} \times B_{r_0/2}$ ($B \subset B_{r_0/2}$ since $\|I - I\| = 0 < r_0/2$), so Lemma 4.14 gives $\|Df(z)\| \le F_1\varepsilon$. By the chain rule the real directional derivative is $Df(\varphi, I)(0, v)$ of modulus $\le F_1\varepsilon\|v\|$. □

External facts used in this part.

(Z1) Banach's fixed point theorem: [K:`bp:p1:ext-banach`]; mean value inequality [K:`bp:p1:ext-mvt`].

(Z2) *Topology.* A continuous image of a compact set is compact, hence closed; a compact set and a disjoint closed set have positive distance; a continuous real function on a compact set attains its maximum; the closure of a subset of a closed set lies in it; for $A$ open, $\partial A = \overline{A} \setminus A$. (Mathlib: `IsCompact.exists_forall_ge`, `IsCompact.isClosed`, `IsCompact.exists_infDist_eq_dist`, `closure_minimal`.)

(Z3) Inverse function theorem in the form used in Part 4 ((Y1)): $\Phi_c|_W$ is a homeomorphism onto the open set $V$ with $C^1$ inverse.

No Jordan curve/surface theorem and no theorem on connected components is used.

5.2. **Patch constants.** Let $\mathfrak{p} = (c, W, \delta_E, \delta_m, \rho_c)$ be a patch of Lemma 4.9, $\Phi = \Phi_c$, $V = \Phi(W)$, $Q = [E_c - \delta_E, E_c + \delta_E] \times [-\delta_m, \delta_m] \subset V$ with $E_c := h(c)$, $O := (E_c - \delta_E, E_c + \delta_E) \times (-\delta_m, \delta_m)$, $\Omega := \Phi^{-1}(O) \subset W$ (open), $\mathcal{K} := \Phi^{-1}(Q) \subset W$. Let $m_\pm = m_{\mathfrak{p},\pm}$ be the slopes of its rays (Definition 4.12).

**Definition 5.2** (Patch constants)**.** (a) $\mathcal{K}$ is compact (image of compact $Q$ under the continuous $\Phi^{-1}$); $r_{\mathfrak{p}} := \min\{1, \frac{1}{2}\operatorname{dist}(\mathcal{K}, \mathbb{R}^2 \setminus W)\} > 0$ ($W$ is bounded, by (P1), so the complement is nonempty and closed); $\mathcal{N} := \{I : \operatorname{dist}(I, \mathcal{K}) \le r_{\mathfrak{p}}\}$, compact, $\mathcal{N} \subset W$.

(b) $L_m := \max\{1, \max_{I \in \mathcal{N}} \|Dm_c(I)\|\}$ ($m_c$ is $C^1$ on $\mathcal{U}_c \supset W$).

(c) $L_\Phi := \max\{1, \max_{(e,m) \in Q} \|D(\Phi^{-1})(e, m)\|\}$.

(d) $\kappa_{\mathfrak{p}} := \min\{\delta_m/4,\ \delta_m - m_+,\ \delta_m + m_-\} > 0$ (Definition 4.12: $m_+ \in (\delta_m/2, \delta_m)$, $m_- \in (-\delta_m, -\delta_m/2)$).

They depend only on $\mathfrak{p}$ and its rays, hence only on $(B, s_0, r_0, h, B', \eta)$.

**Definition 5.3** (The constant $\varepsilon_0$)**.** With $\varepsilon_1, C_K, F_1, \rho_\infty$ from Part 4:

$$\varepsilon_0 := \min\Big\{\varepsilon_1,\ \min_{j \in \mathcal{P}}\ \min\Big\{\tfrac{\delta_{E,j}}{4},\ \tfrac{\rho_\infty}{8C_K},\ \tfrac{r_j}{C_K},\ \tfrac{\kappa_j}{L_{m,j}C_K},\ \tfrac{1}{2F_1 L_{\Phi,j}}\Big\}\Big\} > 0.$$

Order of choice: all data of Part 4 → patch constants → $\varepsilon_0$; independent of $f, \varepsilon, z_0$. In the sequel $\varepsilon < \varepsilon_0$ (so every listed bound holds with $\varepsilon$, strictly).

5.3. **The tori are graphs over all angles, and are invariant.** Fix $f, \varepsilon$ as in Theorem 1.16 with $\varepsilon < \varepsilon_0$, $I_0 \in \overline{B'}$, $E$ with $|E - h(I_0)| \le \varepsilon$, and the output of Proposition 4.19 ($j$, $K_\pm$, $I_\pm$, $\omega_\pm$). Let $k_\pm : \mathbb{R}^2 \to \mathbb{R}^2 \times \mathbb{R}^2$ be the real restriction of $K_\pm$, $k_\pm(\theta) = (\theta + u_\pm(\theta), I_\pm + v_\pm(\theta))$, $u_\pm, v_\pm : \mathbb{R}^2 \to \mathbb{R}^2$ real-analytic ((T1), [K:`bp:p1:real-restriction`]), and $\mathcal{T}_\pm := k_\pm(\mathbb{R}^2)$.

**Lemma 5.4** (Graph property)**.** *For each sign, with $\Psi := \theta \mapsto \theta + u_\pm(\theta)$ and $\|\cdot\|$ the max norm:*

*(a) $\Psi : \mathbb{R}^2 \to \mathbb{R}^2$ is a bijection, $\|\Psi\theta - \Psi\theta'\| \ge \frac{1}{2}\|\theta - \theta'\|$, and $\Psi^{-1}$ is 2-Lipschitz (in particular continuous).*

*(b)* $Y_\pm := I_\pm + v_\pm \circ \Psi^{-1} : \mathbb{R}^2 \to \mathbb{R}^2$ *is continuous,* $\|Y_\pm(\varphi) - I_\pm\| \le C_K\varepsilon$*, and* $\mathcal{T}_\pm = \{(\varphi, Y_\pm(\varphi)) : \varphi \in \mathbb{R}^2\}$.
*(c)* $H_{\mathbb{R}}(\varphi, Y_\pm(\varphi)) = E$ *for all* $\varphi$*, and* $Y_\pm(\varphi) \in \mathcal{N}$*,* $|h(Y_\pm(\varphi)) - E| \le \varepsilon$.

*Proof.* **Step S5.5: Contraction.** By (T4), $\|Du_\pm\|_{\rho_\infty/2} \le 4C_K\varepsilon/\rho_\infty \le \frac{1}{2}$ ($\varepsilon < \rho_\infty/(8C_K)$) and $\mathbb{R}^2 \subset T_{\rho_\infty/2}$; the real derivative has norm at most the complex one ([K:bp:p1:real-restriction]). By [K:bp:p1:ext-mvt] on the convex set $\mathbb{R}^2$, $u_\pm$ is $\frac{1}{2}$-Lipschitz.

**Step S5.6: Bijectivity.** For $\varphi \in \mathbb{R}^2$ the map $\theta \mapsto \varphi - u_\pm(\theta)$ is a $\frac{1}{2}$-contraction of the complete space $\mathbb{R}^2$; its unique fixed point $\theta$ is the unique solution of $\Psi\theta = \varphi$ ((Z1)). Moreover $\|\Psi\theta - \Psi\theta'\| \ge \|\theta - \theta'\| - \frac{1}{2}\|\theta - \theta'\|$, which is the lower bound and the Lipschitz bound of $\Psi^{-1}$.

**Step S5.7: Graph.** $k_\pm(\theta) = (\Psi\theta, I_\pm + v_\pm(\theta))$; substituting $\theta = \Psi^{-1}\varphi$ gives (b); $\|v_\pm\| \le C_K\varepsilon$ is (T4).

**Step S5.8: Energy and position.** $H_{\mathbb{R}}(k_\pm(\theta)) = \operatorname{Re} H(K_\pm(\theta)) = E$ by (T3). $I_\pm = \Phi^{-1}(E, m_\pm) \in \mathcal{K}$ since $(E, m_\pm) \in Q$ (Lemma 4.13), and $C_K\varepsilon < r_j$, so $Y_\pm(\varphi) \in \mathcal{N}$. Finally $h(Y) = H_{\mathbb{R}}(\varphi, Y) - f = E - f$ and $|f| \le \varepsilon$ (Lemma 5.1(e)). □

**Lemma 5.5** (Invariance)**.** $\mathcal{T}_\pm \subset W_B$ *and* $\mathcal{T}_+ \cup \mathcal{T}_-$ *is invariant for* $(X_{\mathbb{R}}, W_B)$ *(Definition* [N:bpN:p3:gen-invariant]*): if* $J'$ *is an interval,* $y$ *satisfies* `IsSolIn` $X_{\mathbb{R}}\, W_B\, J'\, y$ *and* $y(t_1) \in \mathcal{T}_\pm$ *for one* $t_1 \in J'$*, then* $y(t) \in \mathcal{T}_\pm$ *for all* $t \in J'$.

*Proof.* **Step S5.9: Inclusion.** $k_\pm(\theta)_2 = Y_\pm(\Psi\theta) \in \mathcal{N} \subset W \subset B$ by Lemma 5.4(c) and (P1) (this uses the smallness in $\varepsilon_0$, not T6 alone).

**Step S5.10: The torus solution.** If $y(t_1) = k_\pm(\theta_1)$, put $w(t) := k_\pm(\theta_1 + (t - t_1)\omega_\pm)$. By [K:bp:p1:real-restriction], $w$ is differentiable with $w'(t) = DK_\pm(\theta)\omega_\pm|_{\theta=\theta_1+(t-t_1)\omega_\pm} = X_H(K_\pm(\theta))$ ((T2), $E_{H,K,\omega} = 0$ on $T_{\rho_\infty} \supset \mathbb{R}^2$) $= X_{\mathbb{R}}(w(t))$ (Lemma 5.1(b)). With Step S5.9, `IsSolIn` $X_{\mathbb{R}}\, W_B\, \mathbb{R}\, w$.

**Step S5.11: Uniqueness.** $w(t_1) = y(t_1)$; [N:bpN:p3:gen-uniqueness] (with $F = X_{\mathbb{R}}$ $C^1$ on $W_B$, Lemma 5.1(c)) on $J'$ gives $y = w$ on $J'$, and $w(J') \subset \mathcal{T}_\pm$. The union of two invariant sets is invariant. □

## 5.4. **The energy coordinate.**

**Lemma 5.6** (Energy equation is uniquely solvable in $e = h(I)$)**.** *Let* $\varepsilon \le \min\{1, 1/(2F_1L_\Phi)\}$*,* $\varphi \in \mathbb{R}^2$*,* $(e_1, m), (e_2, m) \in Q$ *and* $I_i = \Phi^{-1}(e_i, m)$*. If* $H_{\mathbb{R}}(\varphi, I_1) = H_{\mathbb{R}}(\varphi, I_2)$ *then* $I_1 = I_2$*. (Equivalently: on the level* $\{H_{\mathbb{R}} = E\}$ *the points over* $(\varphi, m)$ *with chart energy in the window are at most one — the energy level is a graph over* $(\varphi, m_c)$*, where uniqueness is the part used.)*

*Proof.* **Step S5.12: Reduction to one variable.** Put $g(e) = e + f(\varphi, \Phi^{-1}(e, m))$ on $[E_c - \delta_E, E_c + \delta_E]$; since $h(\Phi^{-1}(e, m)) = e$ (first component of $\Phi$) and Lemma 5.1(a), $H_{\mathbb{R}}(\varphi, \Phi^{-1}(e, m)) = g(e)$ (these points lie in $\mathcal{K} \subset W \subset B$).

**Step S5.13: Contraction.** $e \mapsto \Phi^{-1}(e, m)$ is $C^1$ with derivative of norm $\le L_\Phi$ (the segment $\{(e, m)\}$ lies in the convex box $Q$). By Lemma 5.1(d) and the chain rule, $|\frac{d}{de} f(\varphi, \Phi^{-1}(e, m))| \le F_1\varepsilon L_\Phi \le \frac{1}{2}$; so by [K:bp:p1:ext-mvt], $|e_1 - e_2| = |g(e_1) - g(e_2) - (f_1 - f_2)|$ with $g(e_1) = g(e_2)$ gives $|e_1 - e_2| = |f(\varphi, I_1) - f(\varphi, I_2)| \le \frac{1}{2}|e_1 - e_2|$, hence $e_1 = e_2$ and $I_1 = I_2$. □

5.5. **Separation.** Keep the notation of §5.3 ($j$ is the patch with $\|I_0 - c_j\| < \rho_j$; we write $c, \Phi, Q, O, \Omega, \mathcal{K}, \mathcal{N}, \delta_E, \delta_m, \kappa$ for its data) and put

$$\mu_\pm(\varphi) := m_c(Y_\pm(\varphi)), \qquad G := \{z \in P : H_{\mathbb{R}}(z) = E\},$$
$$\mathcal{V} := \{(\varphi, I) \in \mathbb{R}^2 \times \Omega : \ \mu_-(\varphi) < m_c(I) < \mu_+(\varphi)\}.$$

**Lemma 5.7** (Position of the graphs in the chart)**.** *$\mu_\pm$ is continuous on $\mathbb{R}^2$, $|\mu_\pm(\varphi) - m_\pm| \le L_m C_K \varepsilon < \kappa \le \delta_m/4$. Hence $\mu_-(\varphi) < -\delta_m/4$, $\mu_+(\varphi) > \delta_m/4$ and $\mu_\pm(\varphi) \in (-\delta_m, \delta_m)$ for all $\varphi$.*

*Proof.* **Step S5.14: Continuity.** $Y_\pm$ is continuous with values in $\mathcal{N} \subset W \subset \mathcal{U}_c$, where $m_c$ is continuous (Lemma 5.4).

**Step S5.15: Lipschitz comparison.** $I_\pm \in \mathcal{K}$ and $\|Y_\pm(\varphi) - I_\pm\| \le C_K\varepsilon \le r_j$, so the segment between them lies in $\mathcal{N}$, where $\|Dm_c\| \le L_m$ ([K:bp:p1:ext-mvt]); $m_c(I_\pm) = m_\pm$ (Lemma 4.13). The bound $L_m C_K \varepsilon < \kappa$ is $\varepsilon < \varepsilon_0$.

**Step S5.16: Ordering.** $m_+ \ge \delta_m/2$ and $m_- \le -\delta_m/2$ (patch rays) give $\mu_+ > \delta_m/2 - \delta_m/4$, $\mu_- < -\delta_m/4$; and $\mu_+ < m_+ + \kappa \le \delta_m$, $\mu_- > m_- - \kappa \ge -\delta_m$. □

**Proposition 5.8** (Separation by two invariant tori)**.** *Let $z_0 = (\theta_0, I_0)$ with $I_0 \in B'$, $E := H_{\mathbb{R}}(z_0)$, and $\varepsilon < \varepsilon_0$ as above (so Proposition 4.19 applies with this $E$). Then:*

*(a) $\mathcal{V}$ is open in $P$, $z_0 \in \mathcal{V}$, $z_0 \notin \mathcal{T}_+ \cup \mathcal{T}_-$.*
*(b) $\overline{\mathcal{V}} \subset \mathbb{R}^2 \times \mathcal{K}$ and $\partial\mathcal{V} \cap G \subset \mathcal{T}_+ \cup \mathcal{T}_-$.*
*(c) (Separation) If $J$ is an interval with $0 \in J$ and `IsSolIn` $X_{\mathbb{R}}\, W_B\, J\, z$ with $z(0) = z_0$, then $z(t) \in \mathcal{V}$ for all $t \in J$.*

*Proof.* **Step S5.17: Openness.** $\Omega$ is open, $m_c$ is continuous on $\Omega \subset \mathcal{U}_c$, and $\mu_\pm$ are continuous (Lemma 5.7); $\mathcal{V}$ is the intersection of $\mathbb{R}^2 \times \Omega$ with two open strict-inequality sets.

**Step S5.18: The initial point.** Since $\|I_0 - c\| < \rho_c$: $|h(I_0) - E_c| < \delta_E/2$, $|m_c(I_0)| < \delta_m/4$ ((P3)), so $\Phi(I_0) \in O$, $I_0 \in \Omega$, and $\mu_-(\theta_0) < -\delta_m/4 < m_c(I_0) < \delta_m/4 < \mu_+(\theta_0)$ (Lemma 5.7): $z_0 \in \mathcal{V}$. If $z_0 \in \mathcal{T}_\pm$, Lemma 5.4(b) gives $I_0 = Y_\pm(\theta_0)$ and $m_c(I_0) = \mu_\pm(\theta_0)$, contradicting the strict inequalities.

**Step S5.19: Closure.** $\overline{\Omega} \subset \mathcal{K}$, because $\Omega \subset \mathcal{K}$ and $\mathcal{K}$ is compact, hence closed ((Z2)); thus $\overline{\mathcal{V}} \subset \mathbb{R}^2 \times \mathcal{K}$.

**Step S5.20: Energy window.** By $|E - h(I_0)| \le \varepsilon$ (Lemma 5.1(e)) and $|h(I_0) - E_c| < \delta_E/2$: $|E - E_c| < \delta_E/2 + \varepsilon$. If $(\varphi, I) \in G$ with $I \in \mathcal{K}$, then $h(I) = E - f(\varphi, I)$ so $|h(I) - E_c| < \delta_E/2 + 2\varepsilon \le \delta_E$ ($\varepsilon < \delta_E/4$).

**Step S5.21: Boundary on the level.** Let $p = (\varphi, I) \in \partial\mathcal{V} \cap G$, $p = \lim p_k$, $p_k = (\varphi_k, I_k) \in \mathcal{V}$. By Step S5.19, $I \in \mathcal{K} \subset W$; as $m_c$ and $\mu_\pm$ are continuous, $\mu_-(\varphi) \le m_c(I) \le \mu_+(\varphi)$. Suppose $\mu_-(\varphi) < m_c(I) < \mu_+(\varphi)$. Then $|m_c(I)| < \delta_m$ (Lemma 5.7) and $|h(I) - E_c| < \delta_E$ (Step S5.20), i.e. $\Phi(I) \in O$, $I \in \Omega$, $p \in \mathcal{V}$, contradicting $p \in \partial\mathcal{V} = \overline{\mathcal{V}} \setminus \mathcal{V}$. So $m_c(I) = \mu_s(\varphi)$ for one sign $s \in \{\pm\}$. Put $Y = Y_s(\varphi) \in \mathcal{N} \subset W$; by Lemma 5.4(c), $H_{\mathbb{R}}(\varphi, Y) = E = H_{\mathbb{R}}(\varphi, I)$, and $\Phi(Y) = (h(Y), \mu_s(\varphi))$ with $|h(Y) - E| \le \varepsilon$, so $(h(Y), \mu_s(\varphi)) \in Q$ (Lemma 5.7, Step S5.20); also $\Phi(I) \in Q$. Both have the same second coordinate $\mu_s(\varphi)$, and $Y = \Phi^{-1}\Phi(Y)$, $I = \Phi^{-1}\Phi(I)$ ($\Phi$ is injective on $W$). Lemma 5.6 gives $I = Y_s(\varphi)$, i.e. $p \in \mathcal{T}_s$ (Lemma 5.4(b)). This is (b).

**Step S5.22: Barrier.** $z(J) \subset G$ by [N:bpN:p3:ham-energy] (Lemma 5.1(c): the predicate gives differentiability of $H_{\mathbb{R}}$ and the derivative $X_{\mathbb{R}}$). Apply [N:bpN:p3:gen-barrier] with $F = X_{\mathbb{R}}$, $W = W_B$, $V = \mathcal{V}$, $S = \mathcal{T}_+ \cup \mathcal{T}_-$ (invariant, Lemma 5.5), $G$ as above, $\partial\mathcal{V} \cap G \subset S$ by (b), $z(0) \notin S$ by (a): $z(t) \in \mathcal{V}$ for all $t \in J$. □

*Remark* 5.9 (On the topology). The statement used is exactly (c): a continuous-function description of the region between the tori, namely $\mathcal{V} = \{\Lambda > 0\} \cap (\mathbb{R}^2 \times \Omega)$ with $\Lambda(\varphi, I) = (m_c(I) - \mu_-(\varphi))(\mu_+(\varphi) - m_c(I))$, together with the boundary identification (b). What replaces the Jordan theorem is Lemma 5.4 (the tori are global graphs over $\mathbb{R}^2$, by a contraction argument) and Lemma 5.6. Passing to $\mathbb{T}^2$ is not needed: the lifted angles are unbounded but only the action components are confined. The full energy level in $\mathbb{R}^2 \times B$ may contain further components (other points with $h \approx E$); they are irrelevant, since the solution never leaves $\mathcal{V}$.

## 5.6. **Confinement, global existence, and the proof of I-HEAD.**

**Proposition 5.10** (Confinement and global existence)**.** *In the setting of Proposition 5.8, let $z$ be the maximal solution of $\dot{z} = X_{\mathbb{R}}(z)$ in $W_B$ with $z(0) = z_0$ (*[N:bpN:p3:gen-maximal]*), with maximal interval $(T_-, T_+)$. Then $T_- = -\infty$, $T_+ = +\infty$, and for all $t \in \mathbb{R}$: $z(t) \in \mathcal{V}$, $z(t)_2 \in \mathcal{K} \subset \mathcal{S} \subset B$ and $\|z(t)_2 - I_0\| < \eta/2 < \eta$.*

*Proof.* **Step S5.23: Confinement on the maximal interval.** Proposition 5.8(c) with $J = (T_-, T_+)$: $z(t) \in \mathcal{V}$, hence $z(t)_2 \in \Omega \subset W$. By (P1), $\|z(t)_2 - c\| < \eta_1/4$, and $\|I_0 - c\| < \rho_c \le \eta_1/4$: $\|z(t)_2 - I_0\| < \eta_1/2 \le \eta/2$.

**Step S5.24: Continuation.** $\mathcal{K}$ is a compact subset of $B$, $X_{\mathbb{R}}$ is $C^1$ and $\mathbb{Z}^2$-periodic in the angles on $\mathbb{R}^2 \times B$ (Lemma 5.1(c)), and $z(t)_2 \in \mathcal{K}$ on $[0, T_+)$ and $(T_-, 0]$; [N:bpN:p3:gen-continuation-periodic] (the angles $z(t)_1$ may be unbounded) gives $T_\pm = \pm\infty$. □

*Proof of Theorem 1.16.* **Step S5.25: The constant.** If $B' = \emptyset$ the conclusion is vacuous and $\varepsilon_0 = 1$ works. Otherwise let $\varepsilon_0$ be that of Definition 5.3; it depends only on $(B, s_0, r_0, h, B', \eta)$ — all fixed before $f, \varepsilon, z_0$, matching the frozen quantifier order (Remark 1.17(a)); nothing depends on $f, \varepsilon, z_0$.

**Step S5.26: Fixing $f, \varepsilon, z_0$.** Let $f, \varepsilon$ satisfy the hypotheses with $0 \le \varepsilon < \varepsilon_0$ ($\varepsilon = 0$ allowed: then $f \equiv 0$ on the domain and all objects above exist as stated, with $u_\pm = v_\pm = 0$), and $z_0 = (\theta_0, I_0)$ with $I_0 \in B'$ (all $\theta_0 \in \mathbb{R}^2$). Put $E = H_{\mathbb{R}}(z_0)$; then $|E - h(I_0)| \le \varepsilon$ (Lemma 5.1(e)) and $I_0 \in \overline{B'}$, so Proposition 4.19 applies.

**Step S5.27: The solution.** Let $z$ be the maximal solution through $z_0$ in $W_B$; by Proposition 5.10 it is defined on $\mathbb{R}$, so $\texttt{IsSolIn}\, X_{\mathbb{R}}\, W_B\, \mathbb{R}\, z$, which by Lemma 5.1(c) is $\texttt{IsHamiltonianSolutionOn}\, H_{\mathbb{R}}\, B\, \mathrm{univ}\, z$, i.e. $\mathrm{Sol}(\mathcal{H}_{h,f}, B, \mathbb{R}, z)$; $z(0) = z_0$; $\|z(t)_2 - I_0\| < \eta$ for all $t$ (Proposition 5.10).

**Step S5.28: Uniqueness among all solutions.** Let $a < 0 < b$, $y(0) = z_0$ and $\mathrm{Sol}(\mathcal{H}_{h,f}, B, (a, b), y)$. By Lemma 5.1(c), $\texttt{IsSolIn}\, X_{\mathbb{R}}\, W_B\, (a, b)\, y$; [N:bpN:p3:gen-uniqueness] on the interval $(a, b) \ni 0$ gives $y = z$ on $(a, b)$. No smallness, confinement in $\mathcal{V}$ or in $\Omega$ is assumed of $y$: only that its actions stay in $B$, as in the definition of Sol. □

*Remark* 5.11 (What Part 4 does not provide, and where it is proved here). Following the review of Part 4 (findings I3-01, I3-02): (T7) orders the *base points* $I_\pm$ in the slope coordinate only, and $I_0$ need not lie on $\{h = E\}$ (only $|E - h(I_0)| \le \varepsilon$); we never place $I_0$ on that curve — the initial point is handled by the strict inequalities $|m_c(I_0)| < \delta_m/4$ and $|h(I_0) - E_c| < \delta_E/2$ (Step S5.18); the point $\tilde{I}_0 = \Phi^{-1}(E, m_c(I_0))$ is not needed. The *deformed* tori are controlled here: that their actions lie in the chart domain ($Y_\pm(\varphi) \in \mathcal{N} \subset W$) and that their slope coordinates satisfy $\mu_- < -\delta_m/4$, $\mu_+ > \delta_m/4$, $\mu_\pm \in (-\delta_m, \delta_m)$ is Lemma 5.7, proved with the explicit smallness $\varepsilon < \min\{r_j/C_K, \kappa_j/(L_{m,j} C_K)\}$ added to $\varepsilon_0$ (Definition 5.3). The *whole* enclosed region is bounded: $\mathcal{V} \subset \mathbb{R}^2 \times \Omega$ and $\Omega \subset W \subset \{\|I - c\| < \eta_1/4\}$ (Proposition 5.10); confinement is not deduced from (T6), and the remark 4.20 is superseded on this point. No strengthening of Part 4 is required.

*Remark* 5.12 (Guards, dependence, doubts). (1) Lean totality: Sol asserts differentiability of $H_{\mathbb{R}}$ at $z(t)$ and a two-sided derivative; Lemma 5.1 shows that $H_{\mathbb{R}}$ is $C^\infty$ on $\mathbb{R}^2 \times U \supset W_B$ so the clause is automatic ([N:bpN:p3:ham-reduction]); derivatives of $f, h$ are genuine on the open domains (Remark 1.13). The values of $m_c, \Phi^{-1}$ are used only on $W \subset \mathcal{U}_c$, where $a_c(\nabla h) > 0$ guards the division in $\sigma_c$. (2) Hypothesis use: $\det \mathsf{B}_h \neq 0$ is used only through Part 4 on $\mathcal{S}$; the part used of the energy level is the bounded region $\mathcal{V}$. (3) The case where $\overline{B'}$ meets several patches is handled pointwise: the patch is chosen from $I_0$, and only finitely many constants are involved. (4) The tori $\mathcal{T}_\pm$ are needed only at the energy of $z_0$ (the point of the pointwise energy identity (T3)); no foliation of a neighbourhood is used. (5) The frozen statement I-HEAD looked correct as written; no frozen statement is doubtful for this part. (6) Forward/backward references: all cited labels exist (Part 4, [N:bpN:p3:*] as in the label index of track N Part 3, [K:bp:p1:*]).

## 6. Kepler bordered determinant and period-one conversion

This part proves the frozen I-KEPLER (Theorem 1.18) completely, and supplies the bridge from the Kepler Hamiltonian in the conventional variables ($2\pi$-periodic angles, actions $(L, G)$) to the frozen period-one conventions of I-G1 and I-HEAD: the transformed Hamiltonian is again of Kepler type, with the constants $\kappa' = \kappa(2\pi)^2$, $\Omega' = \Omega/(2\pi)$ (finding I06 of the Part-0 review: $\kappa, \Omega$ are kept general), and its bordered Hessian `borderedHessian` is exactly the frozen matrix `keplerBorder` of the converted constants. Part 7 uses the result for the nondegeneracy hypothesis of I-HEAD; track P cites Lemma 6.7 for the action normalisation.

### 6.1. The determinant.

**Lemma 6.1** (Determinant of the Kepler bordered matrix)**.** *For all $\kappa, \Omega \in \mathbb{R}$ and $L > 0$ (no sign condition on $\kappa, \Omega$),*

$$\det \mathsf{K}_{\kappa,\Omega}(L) = \frac{3\kappa\Omega^2}{L^4}.$$

*Source: Chi05 p. 17, display after (3.43), for $\kappa = \Omega = 1$ ($\det = 3/L^4$); the general case is the same computation (ours).*

*Proof.* **Step S6.1: Expansion.** Write $\mathsf{K} = (k_{ij})$ with $k_{11} = -3\kappa/L^4$, $k_{12} = 0$, $k_{13} = \kappa/L^3$, $k_{21} = 0$, $k_{22} = 0$, $k_{23} = -\Omega$, $k_{31} = \kappa/L^3$, $k_{32} = -\Omega$, $k_{33} = 0$ (Definition 1.12; the denominators are nonzero because $L > 0$, the carrier of `keplerBorder` being $\{L : \mathbb{R} \mid 0 < L\}$). The Leibniz (Sarrus) expansion of a $3 \times 3$ determinant (`Matrix.det_fin_three`) is

$$\det \mathsf{K} = k_{11}k_{22}k_{33} + k_{12}k_{23}k_{31} + k_{13}k_{21}k_{32} - k_{13}k_{22}k_{31} - k_{12}k_{21}k_{33} - k_{11}k_{23}k_{32}.$$

**Step S6.2: Evaluation.** Since $k_{12} = k_{21} = k_{22} = k_{33} = 0$, the first five products vanish (each contains one of these factors), and $-k_{11}k_{23}k_{32} = -\big(-3\kappa/L^4\big)(-\Omega)(-\Omega) = 3\kappa\Omega^2/L^4$. (Equivalently: expand along the first column or second row; the cofactor of the $(1,3)$-entry has the vanishing minor $\det\left(\begin{smallmatrix} 0 & 0 \\ \kappa/L^3 & -\Omega \end{smallmatrix}\right) = 0$, so only the $(1,1)$-entry contributes, with minor $\det\left(\begin{smallmatrix} 0 & -\Omega \\ -\Omega & 0 \end{smallmatrix}\right) = -\Omega^2$.) □

**Theorem 6.2** (I-KEPLER, proof)**.** *For $\kappa, \Omega > 0$ and $L > 0$: $\det \mathsf{K}_{\kappa,\Omega}(L) = 3\kappa\Omega^2/L^4$ and $\det \mathsf{K}_{\kappa,\Omega}(L) > 0$. This is Theorem 1.18, with the same quantifiers: $\kappa, \Omega$ first, then $L$ in the subtype; no invertibility of any Hessian is assumed.*
*Frozen Lean declaration: `KAM.Iso.kepler_bordered_det`*
*Source: Chi05 p. 17; general $\kappa, \Omega$ ours.*

*Proof.* **Step S6.3: Value.** Lemma 6.1.

**Step S6.4: Positivity.** $\kappa > 0$, $\Omega^2 > 0$ (as $\Omega \neq 0$) and $L^4 > 0$ (as $L > 0$), so the quotient of the positive numerator $3\kappa\Omega^2$ by $L^4$ is positive (`div_pos`, `pow_pos`). □

*Remark* 6.3 (Lean totality). In the statement of Theorem 6.2 the only divisions are by powers of $L = L.\mathrm{val} > 0$; there is no `fderiv`, inverse or square root. The hypotheses $\kappa, \Omega > 0$ are used only for positivity; the determinant formula holds for all real $\kappa, \Omega$.

6.2. **The Kepler Hamiltonian as a function on $\mathbb{C}^2$; its bordered Hessian.**

**Definition 6.4** (The Kepler–rotation Hamiltonian)**.** For $\kappa, \Omega \in \mathbb{R}$ let $h_{\kappa,\Omega} : \mathbb{C}^2 \to \mathbb{C}$,

$$h_{\kappa,\Omega}(I) = -\frac{\kappa}{2I_1^2} - \Omega I_2 \qquad (I_1 \neq 0),$$

(Lean division: the value at $I_1 = 0$ is junk and is never used; $h_{\kappa,\Omega}$ is holomorphic on $\{I_1 \neq 0\}$, which is open). At real $I$ with $I_1 \neq 0$ its value is real. In the conventional variables $(L, G)$ the Kepler Hamiltonian with the rotating-frame term (Chi05 (3.43) at $\varepsilon = 0$, $\kappa = \Omega = 1$) is $h_{1,1}(L, G) = -1/(2L^2) - G$.
Source: Chi05 (3.43), p. 16 ($\kappa = \Omega = 1$); ours for general constants.

**Lemma 6.5** (Frequency map and bordered Hessian of $h_{\kappa,\Omega}$)**.** *Let $\kappa, \Omega \in \mathbb{R}$ and $I \in \mathbb{R}^2$ with $I_1 > 0$. Then*

$$\mathit{frequencyMap}\ h_{\kappa,\Omega}\ I = \Big(\frac{\kappa}{I_1^3}, -\Omega\Big), \qquad D\big(\mathit{frequencyMap}\ h_{\kappa,\Omega}\big)(I) = \begin{pmatrix} -3\kappa/I_1^4 & 0 \\ 0 & 0 \end{pmatrix}$$

*(the real Fréchet derivative, $a$-th component applied to $e_b$ being the $(a, b)$ entry), and with the reindexing $\epsilon : \mathrm{Fin}\,3 \simeq \mathrm{Fin}\,2 \oplus \mathrm{Unit}$, $0, 1 \mapsto \mathrm{inl}\,0, \mathrm{inl}\,1$, $2 \mapsto \mathrm{inr}$,*

$$\mathit{borderedHessian}\ h_{\kappa,\Omega}\ I = \mathrm{reindex}_\epsilon\ \mathsf{K}_{\kappa,\Omega}(I_1),$$

*and in particular*

$$\det \mathit{borderedHessian}\ h_{\kappa,\Omega}\ I = \frac{3\kappa\Omega^2}{I_1^4}.$$

*Frozen Lean declaration: `KAM.Iso.borderedHessian`, `KAM.frequencyMap`, `KAM.Iso.keplerBorder`*
*Source: Chi05 p. 17 ($\kappa = \Omega = 1$); ours.*

*Proof.* **Step S6.5: Real part near a real point.** Let $V = \{J \in \mathbb{R}^2 : J_1 > 0\}$, open. For $J \in V$ the complex number $h_{\kappa,\Omega}(\iota J)$ is real and equals $\phi(J) := -\kappa/(2J_1^2) - \Omega J_2$ ($\iota : \mathbb{R}^2 \to \mathbb{C}^2$ the inclusion, `Complex.ofReal` applied entrywise), so $J \mapsto \mathrm{Re}\, h_{\kappa,\Omega}(\iota J)$ coincides with the smooth function $\phi$ on the neighbourhood $V$ of $I$.

**Step S6.6: Gradient.** Derivatives at a point depend only on the germ (`Filter.EventuallyEq.fderiv_eq`); so $\texttt{frequencyMap}\, h\, J_0 = \nabla\phi(J_0)$ for $J_0 \in V$, with $\partial_1\phi = \kappa/J_1^3$, $\partial_2\phi = -\Omega$ (power rule). Thus on the open set $V$ the map $\texttt{frequencyMap}\, h$ coincides with $g(J) = (\kappa J_1^{-3}, -\Omega)$.

**Step S6.7: Derivative of the frequency map.** Again by germs, $D(\texttt{frequencyMap}\, h)(I) = Dg(I)$, whose first row is $(-3\kappa I_1^{-4}, 0)$ and whose second row is 0.

**Step S6.8: Entries of the bordered matrix.** Insert into Definition 1.7: entries $(a, b)$ with $a, b \in \{0, 1\}$ are those of $Dg(I)$; the border entries $(a, \mathrm{inr})$ and $(\mathrm{inr}, b)$ are $g(I)_a$ and $g(I)_b$, i.e. $\kappa/I_1^3, -\Omega$; the corner is 0. Compare entrywise with Definition 1.12 at $L = I_1$: the matrices agree (the matrix $\mathsf{K}$ is symmetric, the bordered Hessian was not assumed to be).

**Step S6.9: Determinant.** The determinant is invariant under reindexing by a bijection of the index set (`Matrix.det_reindex_self`), and Lemma 6.1 gives the value. □

6.3. **Conversion to period-one angles.**

**Definition 6.6** (Period-one conversion map)**.** $\mathsf{S}:\mathbb{R}^2\times\mathbb{R}^2\to\mathbb{R}^2\times\mathbb{R}^2$, $\mathsf{S}(\theta,I)=\big(2\pi\theta,\ I/(2\pi)\big)$, so that $(\ell,g)=2\pi\theta$ and $(L,G)=I/(2\pi)$, i.e. $\theta=(\ell,g)/(2\pi)$ and $I=2\pi(L,G)$; the same formula on $\mathbb{C}^2\times\mathbb{C}^2$. It is a linear isomorphism, and it maps $\mathbb{Z}^2$-translations of $\theta$ to $2\pi\mathbb{Z}^2$-translations of $(\ell,g)$. Source: ours; frozen convention of [K:bp:p0:conventions].

**Lemma 6.7** (Conversion)**.** *Let $\kappa,\Omega\in\mathbb{R}$, and $\kappa'=\kappa(2\pi)^2$, $\Omega'=\Omega/(2\pi)$.*

*(a)* Symplectic. $\mathsf{S}^*\big(d\ell_1\wedge dL_1+d\ell_2\wedge dL_2\big)=d\theta_1\wedge dI_1+d\theta_2\wedge dI_2$, *i.e. with* $\Omega_0=\begin{pmatrix}0&\mathbf{1}\\-\mathbf{1}&0\end{pmatrix}$ *(the matrix of the standard form in the order angles, actions):* $D\mathsf{S}^\top\Omega_0 D\mathsf{S}=\Omega_0$ *and* $D\mathsf{S}\,\Omega_0\,D\mathsf{S}^\top=\Omega_0$. *Consequently, for $H$ differentiable, $D\mathsf{S}\,X_{H\circ\mathsf{S}}=X_H\circ\mathsf{S}$ (with $X_H=(\partial_I H,-\partial_\theta H)$), and a curve $w$ solves $\dot w=X_{H\circ\mathsf{S}}(w)$ iff $\mathsf{S}w$ solves $\dot y=X_H(y)$.*

*(b)* Hamiltonian. *For all $I\in\mathbb{C}^2$ with $I_1\neq0$: $h_{\kappa,\Omega}\big(I/(2\pi)\big)=h_{\kappa',\Omega'}(I)$. Thus $\tilde h:=h_{\kappa,\Omega}\circ(I\mapsto I/2\pi)$ is the Kepler–rotation Hamiltonian of the converted constants, holomorphic on $\{I_1\neq0\}$, real at real $I$. For $\kappa=\Omega=1$: $\kappa'=(2\pi)^2>0$, $\Omega'=1/(2\pi)>0$.*

*(c)* Frequencies. *For real $I$ with $I_1>0$, $\nabla\tilde h(I)=\big(\kappa'/I_1^3,-\Omega'\big)=\frac{1}{2\pi}\nabla h_{\kappa,\Omega}\big(I/2\pi\big)$: the frequency per unit of the period-one angle is the conventional angular velocity divided by $2\pi$.*

*(d)* Bordered Hessian. *For real $I$ with $I_1>0$, with $\mathsf{D}=\operatorname{diag}\big(\frac{1}{2\pi},\frac{1}{2\pi},1\big)$,*

$$\mathsf{K}_{\kappa',\Omega'}(I_1)=\mathsf{D}\,\mathsf{K}_{\kappa,\Omega}\big(I_1/2\pi\big)\,\mathsf{D},\qquad \texttt{borderedHessian}\,\tilde h\,I=\operatorname{reindex}_\epsilon\mathsf{K}_{\kappa',\Omega'}(I_1),$$

$$\det\texttt{borderedHessian}\,\tilde h\,I=\frac{3\kappa'\Omega'^2}{I_1^4}=\frac{3\kappa\Omega^2}{I_1^4}.$$

*In particular for $\kappa=\Omega=1$: $\det\texttt{borderedHessian}\,\tilde h\,I=3/I_1^4>0$.*

*Source: ours; the conversion by $2\pi$ is forced by the period-one convention of* [K:bp:p0:conventions] *(Chi05 uses $2\pi$-periodic angles). Finding I06 of the Part-0 review.*

*Proof.* **Step S6.10: Symplectic.** $D\mathsf{S}=\operatorname{diag}(2\pi,2\pi,\frac{1}{2\pi},\frac{1}{2\pi})$ in the order $(\theta_1,\theta_2,I_1,I_2)$. A block computation gives $D\mathsf{S}^\top\Omega_0 D\mathsf{S}=\begin{pmatrix}0&2\pi\cdot\frac{1}{2\pi}\mathbf{1}\\-\frac{1}{2\pi}\cdot2\pi\mathbf{1}&0\end{pmatrix}=\Omega_0$, and likewise for $D\mathsf{S}\Omega_0 D\mathsf{S}^\top$ (the matrix is diagonal, hence equal to its transpose). Since $X_H=\Omega_0\nabla H$ and $\nabla(H\circ\mathsf{S})=D\mathsf{S}^\top\nabla H\circ\mathsf{S}$: $D\mathsf{S}\,X_{H\circ\mathsf{S}}=D\mathsf{S}\,\Omega_0 D\mathsf{S}^\top\nabla H\circ\mathsf{S}=\Omega_0\nabla H\circ\mathsf{S}$. For a curve, the chain rule gives $\frac{d}{dt}(\mathsf{S}w)=D\mathsf{S}\dot w$.

**Step S6.11: Hamiltonian.** For $I_1\neq0$: $-\kappa/(2(I_1/2\pi)^2)=-\kappa(2\pi)^2/(2I_1^2)$ and $\Omega\,I_2/(2\pi)=\Omega' I_2$ (field arithmetic).

**Step S6.12: Frequencies and bordered Hessian.** Part (c) is Lemma 6.5 applied to $(\kappa',\Omega')$ together with (b) and the chain rule for $I\mapsto I/2\pi$. For the matrix identity: $(\mathsf{DKD})_{ij}=\mathsf{D}_{ii}\mathsf{K}_{ij}\mathsf{D}_{jj}$ with $L=I_1/2\pi$: $(1,1)$: $-3\kappa/(L^4(2\pi)^2)=-3\kappa(2\pi)^2/I_1^4$; $(1,3)$: $\kappa/(L^3\,2\pi)=\kappa(2\pi)^2/I_1^3$; $(2,3)$: $-\Omega/(2\pi)=-\Omega'$; the other entries are 0. These are the entries of $\mathsf{K}_{\kappa',\Omega'}(I_1)$. The determinant: $3\kappa'\Omega'^2/I_1^4=3\kappa(2\pi)^2\Omega^2(2\pi)^{-2}/I_1^4=3\kappa\Omega^2/I_1^4$; also $\det(\mathsf{DKD})=(2\pi)^{-4}\cdot3\kappa\Omega^2(2\pi)^4/I_1^4$, the same value. □

**Corollary 6.8** (Nondegeneracy in period-one variables)**.** *With $\kappa=\Omega=1$, $\tilde h=h_{(2\pi)^2,1/(2\pi)}$ is holomorphic on $\{I\in\mathbb{C}^2:I_1\neq0\}$, real at real $I$, and $\det\texttt{borderedHessian}\,\tilde h\,I=3/I_1^4>0$ for every real $I$ with $I_1>0$. If $B\subset\{I\in\mathbb{R}^2:I_1>0\}$ and $0<r_0\le\frac12\inf_B I_1$, then $\tilde h$ is holomorphic on $B_{r_0}$, every real $I\in B_{r_0}$ has $I_1>0$, and the nondegeneracy hypothesis of Theorem 1.16 ($\det\texttt{borderedHessian}\,\tilde h\,I\neq0$ for every real $I\in B_{r_0}$) holds.*

*Proof.* **Step S6.13: Domain.** If $I\in B_{r_0}$ then $\|I-J\|<r_0$ for some $J\in B$, so $|\operatorname{Re}I_1-J_1|<r_0\le\frac12 J_1$ and $\operatorname{Re}I_1>\frac12 J_1>0$; in particular $I_1\neq0$, and $I_1>0$ for real $I$.

**Step S6.14: Conclusion.** Lemma 6.7(b),(d) and Lemma 6.5. □

*Remark* 6.9 (What is used where). Theorem 6.2 is the whole of I-KEPLER (no consumer needs more). Corollary 6.8 is the nondegeneracy hypothesis `hiso` of I-HEAD for the Kepler part of the restricted problem (Part 7, Proposition 7.21); Lemma 6.7(a) is the symplectic bookkeeping for the passage between the Delaunay chart of Part 7 (conventional $2\pi$ variables) and I-HEAD (period one). Track P cites Lemma 6.7 for the action normalisation. No external (Mathlib-level) fact beyond field arithmetic, the determinant of a $3\times 3$ matrix, the power rule and the germ locality of the Fréchet derivative is used.

## 7. Delaunay coordinates and the exact restricted Hamiltonian

This part owns, for tracks I and P, the *Kepler problem fundamentals*: the planar Kepler Hamiltonian, its energy and angular-momentum identities, Kepler's equation $u-e\sin u=\ell$ solved analytically on a complex neighbourhood of all $|e|\le e_1<1$ (so including $e=0$), the orbit maps (position and momentum as functions of the elements) and the mean values. On top of these sit the track-I specific statements: the Delaunay chart away from $e=0$ and $e=1$ as a symplectic diffeomorphism, the exact barycentric restricted Hamiltonian in Delaunay variables $H=-\frac{1}{2L^2}-G+\mu F_\mu$ with $F_\mu$ holomorphic and uniformly bounded on a common complex neighbourhood for $|\mu|\le\mu_2$, the transfer of solutions to the physical problem, and the nondegeneracy $\mathsf{B}_h=\mathsf{K}$ via Part 6. Track P cites [I:`bpI:p7:kepler-eq`], [I:`bpI:p7:orbit-maps-e`], [I:`bpI:p7:orbit-identities`], [I:`bpI:p7:means`], [I:`bpI:p7:scaling`].

Source: Chi05 §2, (2.23)–(2.26), pp. 8–9, and (2.29)–(2.32), pp. 11–12 (Delaunay variables, Kepler's equation, true anomaly, radius); §3, (3.33), (3.34), (3.37), pp. 12–13 (barycentre, unit separation, total mass one); the Delaunay Hamiltonian $-1/(2L^2)-G$ is Chi05 (3.43) at $\varepsilon=0$, p. 16.

Source gap: Chi05 states the inversion of Kepler's equation and the analyticity of $\hat f_0,\tilde u,\tilde f$ only "for $|e|$ small enough" (p. 11) and then uses it on the whole range $e<1/4$. Proposition 7.3 supplies the full argument (all $|e|<1$, with a complex neighbourhood), and Part 7 does not use Chi05's truncated, heliocentric model (3.38)–(3.43): $F_\mu$ below is exact and barycentric. Chi05 does not prove that the Delaunay map is a symplectic bijection onto the Cartesian region; it is derived by generating functions (pp. 8–9). Proposition 7.9 and Theorem 7.10 give a direct verification.

### 7.1. Notation and external facts.

Notation. $|\cdot|_2$ is the Euclidean norm on $\mathbb{R}^2$ and $\mathbb{C}^2$ ($|z|_2^2=|z_0|^2+|z_1|^2$), with $|z|_\infty\le|z|_2\le\sqrt{2}|z|_\infty$; it is used for planar geometry only, the ambient norms of the frozen statements being max norms (Part 1). For $z\in\mathbb{C}^2$ put $\vartheta(z)=z_0^2+z_1^2\in\mathbb{C}$ (so $\mathrm{rad}(q)=\sqrt{\vartheta(q)}$ for real $q$), $\mathbf{e}=(1,0)$ (the first unit vector of the plane; the letter $e$ without bold is always an eccentricity). $\mathsf{R}(g)=\left(\begin{smallmatrix}\cos g & -\sin g\\ \sin g & \cos g\end{smallmatrix}\right)$. For $q,p\in\mathbb{R}^2$: $q\cdot p=q_0p_0+q_1p_1$, angular momentum $\Gamma(q,p)=q_0p_1-q_1p_0$, Kepler energy $E(q,p)=\frac12|p|_2^2-1/|q|_2$ ($q\neq0$). The open cone of Delaunay actions is

$$\mathcal{C}_0=\{(L,G)\in\mathbb{R}^2:0<G<L\},\qquad e(L,G)=\sqrt{1-G^2/L^2}\in(0,1)\quad\text{on }\mathcal{C}_0.$$

$\Omega_0=\left(\begin{smallmatrix}0 & \mathbf{1}\\ -\mathbf{1} & 0\end{smallmatrix}\right)$ is the matrix of the standard form $\sum dq_i\wedge dp_i$ in the order (positions, momenta); $X_H=\Omega_0\nabla H=(\partial_pH,-\partial_qH)$. $D(c,r)$ is the open disc in $\mathbb{C}$.

External facts used in this part.

(X1) *Holomorphic functions of several variables.* Sums, products, compositions and quotients (with nonvanishing denominator) of holomorphic functions are holomorphic; $\exp,\sin,\cos$ are entire; the principal power $z\mapsto z^{-1/2}=\exp(-\frac12\operatorname{Log}z)$ is holomorphic on $\{\operatorname{Re}z>0\}$, with $|z^{-1/2}|=|z|^{-1/2}\le(\operatorname{Re}z)^{-1/2}$ and $z^{-1/2}=1/\sqrt{z}$ for $z>0$; $\sqrt{\cdot}$ is holomorphic on $\{\operatorname{Re}z>0\}$. (`DifferentiableOn`, `Complex.cpow`.)

(X2) *Weierstrass.* A uniform limit of holomorphic functions on an open set of $\mathbb{C}^m$ is holomorphic. (`TendstoLocallyUniformlyOn.differentiableOn`.)

(X3) *Parameter integrals.* If $\Phi$ is holomorphic on an open set $U \subset \mathbb{C}^m \times \mathbb{C}$ with $(q, t\mu) \in U$ for all $t \in [0,1]$, then $(q,\mu) \mapsto \int_0^1 \Phi(q, t\mu)\, dt$ is holomorphic on that set (differentiation under the integral, dominated convergence).

(X4) *Schwarz lemma (maximum modulus).* If $g$ is holomorphic on a neighbourhood of $\bar{D}(0,\varrho)$ and $g(0) = 0$, then $|g(\mu)| \le \frac{|\mu|}{\varrho} \max_{|\zeta|=\varrho} |g(\zeta)|$ for $|\mu| \le \varrho$.

(X5) *Inverse function theorem (real-analytic).* A real-analytic map between open subsets of $\mathbb{R}^m$ with invertible differential at every point is a local real-analytic diffeomorphism; a bijective such map is a real-analytic diffeomorphism.
(`HasStrictFDerivAt.toOpenPartialHomeomorph`; Mathlib's analytic version.)

(X6) *Exterior calculus on open subsets of* $\mathbb{R}^m$. $d^2 = 0$; $d$ commutes with pull-back by smooth maps; a form is determined by its values on all tangent vectors. If $F$ is a smooth map with $F^*(\sum_i p_i\, dq_i) = \sum_i P_i\, dQ_i + dS$ for a smooth $S$, then $DF^\top \Omega_0 DF = \Omega_0$ whenever $\sum dP_i \wedge dQ_i$ stands for the matrix $-\Omega_0$ in the order (positions $Q$, momenta $P$).

(X7) *Change of variables in* $\mathbb{R}$. If $u : [a,b] \to \mathbb{R}$ is $C^1$, increasing, then $\int_a^b \phi(u(\ell))u'(\ell)\, d\ell = \int_{u(a)}^{u(b)} \phi$.

(X8) *Banach's fixed point theorem* ([K:`bp:p1:ext-banach`]) and the *mean value inequality* ([K:`bp:p1:ext-mvt`]).

7.2. **Hamilton's equations of the restricted problem.** Let $\mu \in \mathbb{R}$, $W_\mu = \{z = (q,p) \in \mathbb{R}^2 \times \mathbb{R}^2 : \texttt{CollisionFree}\ \mu\ z\} = \{\mathrm{rad}(q - a_1(\mu)) > 0,\ \mathrm{rad}(q - a_2(\mu)) > 0\} = \{q \ne a_1(\mu), q \ne a_2(\mu)\}$, open (Definition 1.9); write $r_j = \mathrm{rad}(q - a_j(\mu))$. Both $\mathcal{H}_\mu^{\mathrm{R3}}$ and $V_\mu$ are given on $W_\mu$ by the formulas of Definition 1.10.

**Lemma 7.1** (Hamilton's equations)**.** *$\mathcal{H}_\mu^{\mathrm{R3}}$ is $C^\infty$ (indeed real-analytic) on $W_\mu$ and $V_\mu = X_{\mathcal{H}_\mu^{\mathrm{R3}}} = (\partial_p \mathcal{H}_\mu^{\mathrm{R3}}, -\partial_q \mathcal{H}_\mu^{\mathrm{R3}})$ at every point of $W_\mu$. Consequently* $\mathrm{RSol}_\mu(J,z)$ *holds iff for every $t \in J$ one has $z(t) \in W_\mu$ and $z$ is differentiable at $t$ with $z'(t) = X_{\mathcal{H}_\mu^{\mathrm{R3}}}(z(t))$; and the shared ODE layer applies to it (*[N:`bpN:p3:newton-reduction`]*: `IsRestrictedSolutionOn` $\iff$ `IsSolIn` for the field $V_\mu$ on $W_\mu$).*
*Frozen Lean declaration: `KAM.Iso.restrictedHamiltonian`, `KAM.Iso.restrictedVectorField`, `KAM.Iso.IsRestrictedSolutionOn`*
*Source: Chi05 (1.2), p. 2, in the rotating frame of (3.34), (3.37), pp. 12–13; the check is ours (it is not displayed in Chi05).*

*Proof.* **Step S7.1: Smoothness.** $r_j^2 = \vartheta(q - a_j)$ is a polynomial, positive on $W_\mu$, so $r_j = \sqrt{\cdot}$ and $1/r_j$ are $C^\infty$ (real-analytic) there; the remaining terms are polynomial in $(q,p)$.

**Step S7.2: Derivatives.** With $\partial_{q_i}(1/r_j) = -(q - a_j)_i / r_j^3$:

$$\partial_{p_0}\mathcal{H} = p_0 + q_1, \qquad \partial_{p_1}\mathcal{H} = p_1 - q_0,$$
$$\partial_{q_0}\mathcal{H} = -p_1 + \tfrac{(1-\mu)(q-a_1)_0}{r_1^3} + \tfrac{\mu(q-a_2)_0}{r_2^3}, \qquad \partial_{q_1}\mathcal{H} = p_0 + \tfrac{(1-\mu)(q-a_1)_1}{r_1^3} + \tfrac{\mu(q-a_2)_1}{r_2^3}.$$

Hence $(\partial_p\mathcal{H}, -\partial_q\mathcal{H}) = \big((p_0 + q_1,\, p_1 - q_0),\ (p_1, -p_0) - \frac{1-\mu}{r_1^3}(q - a_1) - \frac{\mu}{r_2^3}(q - a_2)\big) = V_\mu$, which is the display of Definition 1.10.

**Step S7.3: Solution predicates.** The first clause of $\mathrm{RSol}_\mu$ (collision-freeness at each $t$) is membership $z(t) \in W_\mu$, the second is differentiability with the value $V_\mu = X_{\mathcal{H}}$. The reduction to `IsSolIn` is [N:`bpN:p3:newton-reduction`]. □

**Lemma 7.2** (Newtonian meaning of $V_\mu$)**.** *Let $\mathcal{J} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so $\mathsf{R}(t) = e^{t\mathcal{J}}$. Let $x : I \to \mathbb{R}^2$ ($I$ an interval) be a twice differentiable curve with $x(t) \ne \mathsf{R}(t)a_j(\mu)$ for $j = 1,2$, and put $q(t) = \mathsf{R}(-t)x(t)$,*

$p(t) = \mathsf{R}(-t)\dot{x}(t)$. *Then $x$ solves Newton's equations of the massless body in the field of two primaries of masses $1-\mu$, $\mu$ revolving on the unit circle about their barycentre at angular velocity 1,*

$$\ddot{x} = -(1-\mu)\frac{x-\mathsf{R}(t)a_1}{|x-\mathsf{R}(t)a_1|_2^3} - \mu\frac{x-\mathsf{R}(t)a_2}{|x-\mathsf{R}(t)a_2|_2^3},$$

*iff $(q,p)$ solves $\dot{z} = V_\mu(z)$. (Not used by the frozen statements, which are formulated for $V_\mu$; it identifies $V_\mu$ with Chi05 (1.2), (3.34)–(3.37).)*
*Source: Chi05 (1.2) with (3.34), (3.37), pp. 2, 12–13; ours.*

*Proof.* **Step S7.4: Rotation.** $\frac{d}{dt}\mathsf{R}(-t) = -\mathcal{J}\mathsf{R}(-t) = -\mathsf{R}(-t)\mathcal{J}$, so $\dot{q} = \mathsf{R}(-t)\dot{x} - \mathcal{J}q = p - \mathcal{J}q = (p_0+q_1, p_1-q_0)$ since $\mathcal{J}q = (-q_1, q_0)$; this is the first slot of $V_\mu$.

**Step S7.5: Force.** $\dot{p} = \mathsf{R}(-t)\ddot{x} - \mathcal{J}p$ and $-\mathcal{J}p = (p_1, -p_0)$. By orthogonality of $\mathsf{R}(-t)$ and $\mathsf{R}(-t)\mathsf{R}(t)a_j = a_j$: $\mathsf{R}(-t)\frac{x-\mathsf{R}(t)a_j}{|x-\mathsf{R}(t)a_j|^3} = \frac{q-a_j}{|q-a_j|^3}$. So Newton's equation holds iff $\dot{p} = (p_1, -p_0) - \frac{1-\mu}{r_1^3}(q-a_1) - \frac{\mu}{r_2^3}(q-a_2)$, the second slot of $V_\mu$. The positions $\mathsf{R}(t)a_j$ are the barycentric positions of masses $1-\mu$ at $-\mu$ and $\mu$ at $1-\mu$ (distance 1, barycentre 0) rotating at angular velocity 1, as Kepler's third law gives for total mass one (Chi05 (3.37)). □

## 7.3. **Kepler's equation on a complex neighbourhood.**

**Proposition 7.3** (Kepler's equation, analytic solution)**.** *Let $0 < e_1 < 1$ and put*

$$e_2 = \tfrac{1+e_1}{2}, \qquad \vartheta_* = \tfrac{1+e_2}{2}, \qquad y_0 = \operatorname{arcosh}\frac{\vartheta_*}{e_2} > 0, \qquad \delta_K = (1-\vartheta_*)\,y_0 > 0,$$

$$\mathcal{P} = \mathcal{P}(e_1) = \{(\ell, e) \in \mathbb{C}^2 :\ |\operatorname{Im}\ell| < \delta_K,\ |e| < e_2\},$$

*an open set containing $\{(\ell,e)\in\mathbb{R}^2 : |e|\le e_1\}$ (the constants depend only on $e_1$). There is a unique holomorphic $u : \mathcal{P} \to \mathbb{C}$ with $|\operatorname{Im} u| \le y_0$ and*

$$u(\ell,e) - e\sin u(\ell,e) = \ell \qquad ((\ell,e)\in\mathcal{P}).$$

*It has the following properties.*

- *(a) $|\sin u|, |\cos u| \le \cosh y_0 = \vartheta_*/e_2$, $|e\cos u| \le \vartheta_* < 1$ and so $|1 - e\cos u| \ge 1 - \vartheta_* > 0$ on $\mathcal{P}$ (the quantity $\partial_u(u - e\sin u) = 1 - e\cos u$ never vanishes).*
- *(b) $\partial_\ell u = \dfrac{1}{1-e\cos u}$, $\partial_e u = \dfrac{\sin u}{1-e\cos u}$; $u(\ell,0) = \ell$; $u(\ell+2\pi, e) = u(\ell,e) + 2\pi$; hence $\sin u, \cos u, (1-e\cos u)^{-1}$ are holomorphic on $\mathcal{P}$ and $2\pi$-periodic in $\ell$.*
- *(c) $u$ is real at real $(\ell,e)\in\mathcal{P}$. For every real $\ell$ and $e\in(-1,1)$ the equation $u - e\sin u = \ell$ has exactly one real solution $u_\mathbb{R}(\ell,e)$; $u_\mathbb{R}$ is real-analytic on $\mathbb{R}\times(-1,1)$, $u_\mathbb{R}(\cdot,e)$ is an increasing diffeomorphism of $\mathbb{R}$ with $u_\mathbb{R}(0,e) = 0$, $u_\mathbb{R}(2\pi,e) = 2\pi$, and $u = u_\mathbb{R}$ on the real points of every $\mathcal{P}(e_1)$.*

*Source: Chi05 p. 11 (the inversion $u_0(\lambda,e) = \lambda + e\tilde{u}(\lambda,e)$ for "$|e|$ small enough", no radius); full argument ours.*
*Source gap: Chi05 gives no domain of analyticity in $e$ and none in $\lambda$; the uniform statement above (all $|e|<1$, a strip in $\ell$, with the periodic structure) is what the use on $e \le 1/4$ (Chi05 §3) and track P's use near $e = 0$ require.*

*Proof.* **Step S7.6: Constants.** $0 < e_1 < e_2 < \vartheta_* < 1$, and $\vartheta_*/e_2 > 1$, so $y_0 > 0$ and $e_2\cosh y_0 = \vartheta_*$. Also $\sinh y \le y\cosh y$ for $y \ge 0$ ($\tanh y \le y$), hence $e_2\sinh y_0 \le \vartheta_* y_0$.

**Step S7.7: Trigonometric bounds.** For $u = x + iy$: $|\sin u|^2 = \sin^2 x + \sinh^2 y$, $|\cos u|^2 = \cos^2 x + \sinh^2 y$, both $\le \cosh^2 y$; $\operatorname{Im}\sin u = \cos x \sinh y$, so $|\operatorname{Im}\sin u| \le \sinh|y|$.

**Step S7.8: A contraction.** Let $S = \{u\in\mathbb{C} : |\operatorname{Im} u| \le y_0\}$ (closed, convex, complete) and, for $(\ell,e)\in\mathcal{P}$, $\Phi_{\ell,e}(u) = \ell + e\sin u$. For $u\in S$: $|\operatorname{Im}\Phi_{\ell,e}(u)| \le |\operatorname{Im}\ell| + e_2\sinh y_0 < \delta_K + \vartheta_* y_0 = y_0$, so

$\Phi_{\ell,e}(S) \subset S$; and $|\Phi'_{\ell,e}(u)| = |e\cos u| \le e_2 \cosh y_0 = \vartheta_* < 1$, so by the mean value inequality (X8) on the convex $S$, $\Phi_{\ell,e}$ is a $\vartheta_*$-contraction of $S$. Banach gives a unique fixed point $u(\ell,e) \in S$, i.e. a unique solution of Kepler's equation in $S$.

**Step S7.9: Holomorphy.** Put $u_0(\ell,e) = \ell \in S$ ($|\mathrm{Im}\,\ell| < \delta_K < y_0$) and $u_{m+1} = \Phi_{\ell,e}(u_m)$: each $u_m$ is holomorphic on $\mathcal{P}$ with values in $S$ ((X1)), and $|u_{m+1} - u_m| \le \vartheta_*^m |u_1 - u_0| \le \vartheta_*^{m+1}$ (as $|u_1 - u_0| = |e\sin\ell| \le \vartheta_*$), so $|u_m - u| \le \vartheta_*^{m+1}/(1-\vartheta_*)$ uniformly on $\mathcal{P}$. By (X2), $u$ is holomorphic.

**Step S7.10: (a).** The bounds follow from $u \in S$, Step S7.7 and $e_2 \cosh y_0 = \vartheta_*$.

**Step S7.11: (b).** Differentiate $u - e\sin u = \ell$ (a holomorphic identity): $(1 - e\cos u)\partial_\ell u = 1$, $(1-e\cos u)\partial_e u = \sin u$; divide using (a). $u(\ell,0) = \ell$ because $\ell \in S$ solves the equation. Periodicity: $u(\ell,e) + 2\pi \in S$ solves the equation at $(\ell + 2\pi, e)$, so equals the unique solution there.

**Step S7.12: (c).** For real $e$, $|e| < 1$, the map $\phi(u) = \ell + e\sin u$ is a $|e|$-contraction of $\mathbb{R}$, so it has exactly one real fixed point $u_{\mathbb{R}}(\ell,e)$. For real $(\ell,e) \in \mathcal{P}$ this fixed point lies in $S$, so it equals $u(\ell,e)$ by uniqueness in $S$ (Step S7.8). Every $e \in (-1,1)$ lies in some $\mathcal{P}(e_1)$ ($e_1 \in (|e|,1)$), hence $u_{\mathbb{R}}$ is the restriction of a holomorphic function, so real-analytic on $\mathbb{R}\times(-1,1)$ ([K:`bp:p1:holomorphic-analytic`]). $\partial_\ell u_{\mathbb{R}} = 1/(1 - e\cos u_{\mathbb{R}}) > 0$; $|u_{\mathbb{R}} - \ell| \le |e| < 1$ gives $u_{\mathbb{R}}(\ell,e) \to \pm\infty$ as $\ell \to \pm\infty$, so $u_{\mathbb{R}}(\cdot,e)$ is an increasing diffeomorphism of $\mathbb{R}$. $u_{\mathbb{R}}(0,e) = 0$ since $0 - e\sin 0 = 0$ and by uniqueness; $u_{\mathbb{R}}(2\pi,e) = 2\pi$ similarly. □

*Remark* 7.4 (Alternative route). The holomorphic implicit function theorem at a real solution $(\ell_0, e_0, u_0)$, using $\partial_u(u - e\sin u) = 1 - e_0\cos u_0 \ge 1 - e_1 > 0$, gives local holomorphic solutions that agree with the real one on a connected real-symmetric overlap, hence glue; the contraction argument above makes the size of the neighbourhood explicit and independent of the point, which is what the uniform bounds below need.

### 7.4. The planar Kepler problem: elements, orbit maps, identities.

**Definition 7.5** (Kepler region and elements)**.** The Kepler Hamiltonian (strength one) is $E(q,p) = \frac{1}{2}|p|_2^2 - 1/|q|_2$ on $\{q \ne 0\}$. The *bound prograde non-circular region* is

$$\mathcal{C} = \{(q,p) \in \mathbb{R}^2 \times \mathbb{R}^2 :\ q \ne 0,\ E(q,p) < 0,\ \Gamma(q,p) > 0,\ \Gamma(q,p)^2 < -1/(2E(q,p))\},$$

open (strict inequalities between continuous functions on the open set $\{q \ne 0\}$). On $\mathcal{C}$ the *elements* are

$$L = \frac{1}{\sqrt{-2E}}, \quad G = \Gamma, \quad a = L^2 = \frac{1}{-2E}, \quad e = \sqrt{1 - G^2/L^2} = \sqrt{1 + 2E\Gamma^2},$$

with $L > G > 0$ and $e \in (0,1)$, so that $E = -1/(2a)$ and $\Gamma^2 = a(1-e^2)$.
Source: Chi05 (2.24)–(2.26), p. 9 ($\mu = M = 1$): $h_{\mathrm{Kep}} = -1/(2\Lambda^2)$, $e = \sqrt{1 - (\Gamma/\Lambda)^2}$, $a = \Lambda^2$, $\Gamma > 0$, momentum space the cone $0 < \Gamma < \Lambda$ (Remark 2.3).

**Definition 7.6** (Orbit maps)**.** Let $\mathsf{Q}_{\mathrm{el}}$ be the following map of $(\ell, g, L, e, k)$ ($k$ stands for $G/L$): with $u = u(\ell,e)$ of Proposition 7.3, $c = \cos u$, $s = \sin u$, $\mathsf{D} = 1 - ec$,

$$X = L^2(c - e), \quad Y = L^2 k\, s, \quad P_X = -\frac{s}{L\,\mathsf{D}}, \quad P_Y = \frac{k\,c}{L\,\mathsf{D}}, \qquad q = \mathsf{R}(g)\begin{pmatrix} X \\ Y \end{pmatrix}, \quad p = \mathsf{R}(g)\begin{pmatrix} P_X \\ P_Y \end{pmatrix}.$$

The *Delaunay orbit map* is $\mathsf{Q}(\ell, g, L, G) = \mathsf{Q}_{\mathrm{el}}(\ell, g, L, e(L,G), G/L)$, where $e(L,G) = \sqrt{1 - G^2/L^2}$. On the real domain $\mathcal{D}_{\mathbb{R}} = \mathbb{R}^2 \times \mathcal{C}_0$ ($\ell, g \in \mathbb{R}$, $(L,G) \in \mathcal{C}_0$) all quantities are real and $\mathsf{D} \ge 1 - e > 0$ ($u = u_{\mathbb{R}}$). Here $q = \mathsf{R}(g)(X,Y)$ is the position on the Kepler ellipse of semi-major axis $a = L^2$, eccentricity $e$, perihelion direction $g$, at eccentric anomaly $u$ and mean anomaly $\ell$ ($X = a(\cos u - e)$, $Y = a\sqrt{1-e^2}\sin u$, Chi05 p. 12, (A.45), $r = a(1 - e\cos u)$), and $p$ is the velocity ($\dot\ell = L^{-3}$, $\dot u = L^{-3}/\mathsf{D}$).
Source: Chi05 (2.29)–(2.32), pp. 11–12 and (A.44)–(A.45), p. 17; the explicit velocity formula is ours.

**Proposition 7.7** (Orbit maps near $e = 0$)**.** *Let $0 < e_1 < 1$ and $\mathcal{P} = \mathcal{P}(e_1)$ as in Proposition 7.3. The map $\mathsf{Q}_{\mathrm{el}}$ is holomorphic on $\{(\ell, g, L, e, k) : (\ell, e) \in \mathcal{P},\ g \in \mathbb{C},\ L \in \mathbb{C} \setminus \{0\},\ k \in \mathbb{C}\}$ (note $\mathsf{D} \neq 0$ there), $2\pi$-periodic in $\ell$ and in $g$, and real on real points with $|e| < 1$; in particular it is holomorphic across $e = 0$ (only $e$, not $\sqrt{1 - G^2/L^2}$, enters), e.g. with $k = \sqrt{1 - e^2}$ (principal branch, holomorphic for $|e| < 1$).*

*Source: ours (the consequence of Proposition 7.3 that track P's Poincaré chart uses).*

*Proof.* **Step S7.13: Holomorphy.** $u, c, s$ are holomorphic and $\mathsf{D} = 1 - e\cos u$ satisfies $|\mathsf{D}| \geq 1 - \vartheta_* > 0$ (Proposition 7.3(a)); the formulas for $X, Y, P_X, P_Y$ are polynomial in $(c, s, e, k, L, L^{-1}, \mathsf{D}^{-1})$, and $\mathsf{R}(g)$ is entire ((X1)). $\mathrm{Re}(1 - e^2) \geq 1 - |e|^2 > 0$, so $k = \sqrt{1 - e^2}$ is holomorphic.

**Step S7.14: Periodicity, reality.** Proposition 7.3(b),(c) and $2\pi$-periodicity of $\mathsf{R}$. □

**Proposition 7.8** (Identities for the orbit maps)**.** *Let $(\ell, g, L, G) \in \mathcal{D}_{\mathbb{R}}$, $e = e(L, G)$, $(q, p) = \mathsf{Q}(\ell, g, L, G)$, $r = |q|_2$, and $u, c, s, \mathsf{D}$ as above. Then*

*(a) $r = L^2\mathsf{D} = L^2(1 - e\cos u)$ and $L^2(1 - e) \leq r \leq L^2(1 + e)$; $|p|_2^2 = \dfrac{1 + e\cos u}{L^2\mathsf{D}}$;*

*(b) $\Gamma(q, p) = G$ and $q \cdot p = L\, e \sin u$;*

*(c) $E(q, p) = -\dfrac{1}{2L^2}$; hence $\mathsf{Q}(\mathcal{D}_{\mathbb{R}}) \subset \mathcal{C}$ and the elements of Definition 7.5 of $\mathsf{Q}(\ell, g, L, G)$ are $L, G, a = L^2, e$;*

*(d) $(Le\sin u)^2 = -G^2 + 2r - r^2/L^2$ and $L^2 e\cos u = L^2 - r$.*

*Source: Chi05 (2.24)–(2.26), (2.32), pp. 9, 12; computations ours.*

*Proof.* Use $G = Lk$, $k^2 = 1 - e^2$, $c^2 + s^2 = 1$ (real case, $k = G/L > 0$).

**Step S7.15: Radius.** $\mathsf{R}(g)$ is orthogonal, so $r^2 = X^2 + Y^2 = L^4\big[(c - e)^2 + (1 - e^2)s^2\big] = L^4\big[1 - 2ec + e^2c^2\big] = L^4\mathsf{D}^2$ and $r = L^2\mathsf{D} > 0$ since $\mathsf{D} \geq 1 - e$. Likewise $|p|_2^2 = P_X^2 + P_Y^2 = \frac{s^2 + k^2c^2}{L^2\mathsf{D}^2} = \frac{1 - e^2c^2}{L^2\mathsf{D}^2} = \frac{1 + ec}{L^2\mathsf{D}}$.

**Step S7.16: Angular momentum, radial momentum.** Rotations preserve $q_0p_1 - q_1p_0$ and $q \cdot p$, so $\Gamma = XP_Y - YP_X = \frac{L^2(c - e)kc}{L\mathsf{D}} + \frac{L^2ks \cdot s}{L\mathsf{D}} = \frac{Lk(c^2 - ec + s^2)}{\mathsf{D}} = Lk = G$, and $q \cdot p = XP_X + YP_Y = \frac{-L(c - e)s + Lk^2sc}{\mathsf{D}} = \frac{Ls\, e(1 - ec)}{\mathsf{D}} = Le\, s$.

**Step S7.17: Energy.** $E = \frac{1}{2}|p|^2 - \frac{1}{r} = \frac{(1 + ec)/2 - 1}{L^2\mathsf{D}} = -\frac{1 - ec}{2L^2\mathsf{D}} = -\frac{1}{2L^2}$. As $G > 0$ and $G^2 = L^2(1 - e^2) < L^2 = -1/(2E)$, $\mathsf{Q}(\ell, g, L, G) \in \mathcal{C}$, and $L = 1/\sqrt{-2E}$, $\Gamma = G$ give the elements.

**Step S7.18: (d).** $L^2ec = L^2 - r$ is Step S7.15. Then $(Les)^2 = L^2e^2 - L^2e^2c^2 = L^2e^2 - (L^2 - r)^2/L^2 = (L^2 - G^2) - L^2 + 2r - r^2/L^2$, using $L^2e^2 = L^2 - G^2$. □

**Proposition 7.9** (The Delaunay map is symplectic)**.** *On $\mathcal{D}_{\mathbb{R}}$ (open in $\mathbb{R}^4$), with $F = 2Le\sin u$:*

$$\mathsf{Q}^*\Big(\sum_{i=0}^{1} p_i\, dq_i\Big) = L\, d\ell + G\, dg + dF, \qquad \textit{hence} \qquad D\mathsf{Q}^{\top}\Omega_0 D\mathsf{Q} = \Omega_0 = D\mathsf{Q}\,\Omega_0\, D\mathsf{Q}^{\top},$$

*i.e. $\mathsf{Q}^*(dq_0 \wedge dp_0 + dq_1 \wedge dp_1) = d\ell \wedge dL + dg \wedge dG$ in the order $(\ell, g; L, G)$, and $D\mathsf{Q}(w)$ is invertible at every point. Consequently, for every $C^1$ function $H$ on an open subset of $\mathbb{R}^4$ on which $\mathsf{Q}$ is defined,*

$$D\mathsf{Q}\; X_{H\circ\mathsf{Q}} = X_H \circ \mathsf{Q}, \qquad \textit{i.e.} \qquad \dot w = X_{H\circ\mathsf{Q}}(w) \iff \tfrac{d}{dt}\mathsf{Q}(w) = X_H(\mathsf{Q}(w)) \quad (\textit{for } w \in C^1).$$

*Source: Chi05 pp. 8–9 obtains the same canonicity by generating functions $S_0, S_1, S_2$ (not reproduced); the direct verification below is ours.*

*Proof.* **Step S7.19: Reduction of the angle $g$.** Write $q = \mathsf{R}(g)\xi$, $\xi = (X, Y)$, $\pi = (P_X, P_Y)$, $p = \mathsf{R}(g)\pi$. Then $dq = \mathsf{R}(g)d\xi + \mathsf{R}(g)J\xi\, dg$ ($\frac{d}{dg}\mathsf{R} = \mathsf{R}J$), so $p \cdot dq = \pi \cdot d\xi + (\pi \cdot J\xi)\, dg = \pi \cdot d\xi + (XP_Y - YP_X)\, dg = \pi \cdot d\xi + G\, dg$ by Proposition 7.8(b).

**Step S7.20: Coordinates $(u, L, e)$.** On the real domain use $(u, L, e)$ in place of $(\ell, L, G)$: $\ell = u - e\sin u$, $G = Lk$, $k = \sqrt{1-e^2}$; the change of variables $(\ell, L, G) \mapsto (u, L, e)$ is a diffeomorphism ($\partial_\ell u > 0$, $e \in (0,1) \leftrightarrow G/L \in (0,1)$), so identities between 1-forms may be checked in the new coordinates. With $X = L^2(c-e)$, $Y = L^2 ks$:

$$dX = 2L(c-e)\,dL - L^2 s\,du - L^2\,de, \qquad dY = 2Lks\,dL - \frac{L^2 es}{k}\,de + L^2 kc\,du,$$

and $P_X = -s/(L\mathsf{D})$, $P_Y = kc/(L\mathsf{D})$. Collecting coefficients:

$$\begin{aligned} dL &: \tfrac{2}{\mathsf{D}}\big[-s(c-e) + k^2 cs\big] = \tfrac{2s}{\mathsf{D}}\,e(1-ec) = 2es, \\ de &: \tfrac{Ls}{\mathsf{D}} - \tfrac{Lecs}{\mathsf{D}} = Ls, \\ du &: \tfrac{Ls^2}{\mathsf{D}} + \tfrac{Lk^2c^2}{\mathsf{D}} = \tfrac{L(1-e^2c^2)}{\mathsf{D}} = L(1+ec). \end{aligned}$$

So $\pi \cdot d\xi = L(1+ec)\,du + 2es\,dL + Ls\,de$.

**Step S7.21: Comparison with $L\,d\ell$.** $d\ell = (1-ec)\,du - s\,de$, so $\pi\cdot d\xi - L\,d\ell = 2Lec\,du + 2es\,dL + 2Ls\,de = 2\,d(Les) = dF$. With Step S7.19: $\mathsf{Q}^*(\sum p_i dq_i) = L\,d\ell + G\,dg + dF$.

**Step S7.22: Exterior derivative.** Apply $d$ ((X6)): $\mathsf{Q}^*(\sum dp_i \wedge dq_i) = dL\wedge d\ell + dG \wedge dg$, i.e. $\mathsf{Q}^* \sum dq_i \wedge dp_i = d\ell \wedge dL + dg \wedge dG$, which in matrices is $D\mathsf{Q}^\top \Omega_0 D\mathsf{Q} = \Omega_0$. Taking determinants, $\det D\mathsf{Q} \neq 0$. Since $M^\top \Omega_0 M = \Omega_0$ with $M$ invertible gives $M\Omega_0 M^\top = \Omega_0$ ($M^{-1}\Omega_0 M^{-\top} = \Omega_0$ is obtained by inverting both sides and using $\Omega_0^{-1} = -\Omega_0$, then replace $M$ by $M^{-1}$ in the first identity), the second identity follows.

**Step S7.23: Hamiltonian fields.** $\nabla(H\circ\mathsf{Q}) = D\mathsf{Q}^\top \nabla H \circ \mathsf{Q}$, so $D\mathsf{Q}\, X_{H\circ\mathsf{Q}} = D\mathsf{Q}\,\Omega_0 D\mathsf{Q}^\top \nabla H \circ \mathsf{Q} = \Omega_0 \nabla H \circ \mathsf{Q} = X_H \circ \mathsf{Q}$; the statement on curves is the chain rule. □

**Theorem 7.10** (Delaunay chart)**.** *$\mathsf{Q}$ is $2\pi$-periodic in $\ell$ and in $g$ and induces a bijection from $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \mathcal{C}_0$ onto $\mathcal{C}$; $\mathsf{Q} : \mathcal{D}_{\mathbb{R}} \to \mathcal{C}$ is a surjective real-analytic local diffeomorphism, and the induced map on the quotient is a real-analytic diffeomorphism onto the open set $\mathcal{C}$, symplectic by Proposition 7.9. Moreover $E\circ\mathsf{Q} = -\frac{1}{2L^2}$, $\Gamma\circ\mathsf{Q} = G$ (so $\mathcal{H}^{\mathrm{K}}\circ\mathsf{Q} = -\frac{1}{2L^2} - G$, Lemma 7.12). The two singular loci of the Delaunay variables, $e = 0$ ($G = L$) and $e = 1$ ($G = 0$), are excluded from $\mathcal{C}_0$.*

*Source: Chi05 pp. 8–9 and Remark 2.3, p. 9 (cone $0 < \Gamma < \Lambda$); the bijectivity proof is ours.*

*Proof.* **Step S7.24: Regularity.** $e(L,G)$ is real-analytic on $\mathcal{C}_0$ ($1 - G^2/L^2 > 0$), $u_{\mathbb{R}}$ is real-analytic (Proposition 7.3(c)); so $\mathsf{Q}$ is real-analytic on $\mathcal{D}_{\mathbb{R}}$; the invertibility of $D\mathsf{Q}$ is Proposition 7.9; $\mathsf{Q}(\mathcal{D}_{\mathbb{R}}) \subset \mathcal{C}$ is Proposition 7.8(c).

**Step S7.25: Injectivity modulo $2\pi$.** Let $\mathsf{Q}(\ell, g, L, G) = (q, p)$. Proposition 7.8 gives $L = 1/\sqrt{-2E(q,p)}$, $G = \Gamma(q,p)$, hence $e = e(L,G)$, and $r = |q|_2$. Then $\cos u = (1 - r/(L^2))/e$ and $\sin u = q\cdot p/(Le)$ are determined, so $u$ modulo $2\pi$, so $\ell = u - e\sin u$ modulo $2\pi$ (since $u \mapsto u - e\sin u$ commutes with $u \mapsto u + 2\pi$); the vector $(X, Y)$ is then determined and nonzero ($|(X,Y)|_2 = r > 0$), so $g$ modulo $2\pi$ is determined by $\mathsf{R}(g)(X,Y) = q$.

**Step S7.26: Surjectivity: the elements.** Let $(q,p) \in \mathcal{C}$ and define $L, G, e$ as in Definition 7.5; then $0 < G < L$, i.e. $(L,G) \in \mathcal{C}_0$, $e = e(L,G)$. Put $r = |q|_2 > 0$. From Lagrange's identity $r^2|p|_2^2 = (q\cdot p)^2 + \Gamma^2$ and $\frac{1}{2}|p|^2 = E + 1/r = -\frac{1}{2L^2} + \frac{1}{r}$:

$$(*) \qquad (q\cdot p)^2 = 2r - \frac{r^2}{L^2} - G^2.$$

**Step S7.27: Surjectivity: the anomaly.** Set $c := (1 - r/L^2)/e$ and $s := q\cdot p/(Le)$. Then $e^2c^2L^2 + (q\cdot p)^2 = (L^2 - r)^2/L^2 + 2r - r^2/L^2 - G^2 = L^2 - G^2 = L^2e^2$ by $(*)$, so $c^2 + s^2 = 1$ (divide by $L^2e^2$). Choose $u \in \mathbb{R}$ with $(\cos u, \sin u) = (c, s)$, put $\ell = u - e\sin u$ (so $u = u_{\mathbb{R}}(\ell, e)$ by uniqueness

in Proposition 7.3(c)), $X = L^2(c-e)$, $Y = LGs$, so $X^2+Y^2 = L^4\mathsf{D}^2 = r^2$ ($\mathsf{D} = 1-ec = r/L^2$). Choose $g$ with $\mathsf{R}(g)(X,Y) = q$ (possible as $|(X,Y)| = |q|$).

**Step S7.28: Surjectivity: the momentum.** Let $(q,p') = \mathsf{Q}(\ell,g,L,G)$. By Proposition 7.8(b),(c): $E(q,p') = E(q,p)$, $\Gamma(q,p') = G = \Gamma(q,p)$, $q\cdot p' = Les = q\cdot p$. A vector $v\in\mathbb{R}^2$ is determined by $q\cdot v$ and $q_0v_1 - q_1v_0$ when $q\neq 0$ ($q\neq 0$: the $2\times 2$ system has determinant $-|q|_2^2$), so $p' = p$.

**Step S7.29: Quotient diffeomorphism.** Periodicity of $\mathsf{Q}$ in $\ell, g$ follows from Proposition 7.3(b) and the $2\pi$-periodicity of $\mathsf{R}$; the quotient map is a bijective local diffeomorphism, hence a real-analytic diffeomorphism ((X5)). □

**Lemma 7.11** (The preimage of the near-circular region)**.** *For $a_-, a_+, e_-, e_+ \in \mathbb{R}$ the set $\mathcal{N} = \mathcal{N}(a_-,a_+,e_-,e_+)$ of Definition 1.11 satisfies*

$$\begin{aligned}\mathcal{N} &= \{(q,p)\in\mathcal{C} :\ a_- < a(q,p) < a_+,\ e_- < e(q,p) < e_+\}\\ &= \mathsf{Q}\big(\{(\ell,g,L,G)\in\mathcal{D}_{\mathbb{R}} :\ a_- < L^2 < a_+,\ e_- < e(L,G) < e_+\}\big),\end{aligned}$$

*with the elements of Definition 7.5; the witnesses $r, a, e$ of* `NearCircular` *are $|q|_2$, $a(q,p)$, $e(q,p)$ and are unique. In particular $\mathcal{N}$ is open, and nonempty iff there is $(L,G)\in\mathcal{C}_0$ with $a_- < L^2 < a_+$, $e_- < e(L,G) < e_+$ (e.g. for $0 < a_- < a_+$ and $0\le e_- < e_+ < 1$ with $(e_-,e_+)\cap(0,1)\neq\emptyset$).*
*Frozen Lean declaration:* `KAM.Iso.NearCircular`
*Source: Chi05 (2.24)–(2.26), p. 9; ours.*

*Proof.* **Step S7.30: $\subseteq$.** Let $(q,p)\in\mathcal{N}$ with witnesses $r,a,e$. $r>0$, $r^2 = |q|_2^2$ gives $q\neq 0$, $r = |q|_2$. $E = \frac12|p|^2 - 1/r = -1/(2a) < 0$, so $a = -1/(2E)$ is forced; $\Gamma > 0$; $\Gamma^2 = a(1-e^2) < a = -1/(2E)$ since $e > 0$: thus $(q,p)\in\mathcal{C}$ and $a = a(q,p)$; $e>0$ and $e^2 = 1-\Gamma^2/a$ force $e = e(q,p)$. The bounds are those of the definition.

**Step S7.31: $\supseteq$.** If $(q,p)\in\mathcal{C}$ with the element bounds, take $r = |q|_2$, $a = a(q,p) > 0$, $e = e(q,p)\in(0,1)$: the equations $E = -1/(2a)$ and $\Gamma^2 = a(1-e^2)$ are the definitions, $\Gamma > 0$ is part of $\mathcal{C}$.

**Step S7.32: Image of the chart.** By Theorem 7.10 and Proposition 7.8(c) the elements of $\mathsf{Q}(\ell,g,L,G)$ are $(L,G,L^2,e(L,G))$. Openness: $a(\cdot), e(\cdot)$ are continuous on the open set $\mathcal{C}$. □

**Lemma 7.12** (Unperturbed Hamiltonian in Delaunay variables)**.** *Let $\mathcal{H}^{\mathrm{K}}(q,p) = E(q,p) - \Gamma(q,p) = \frac12|p|_2^2 + q_1p_0 - q_0p_1 - 1/|q|_2$ for $q\neq 0$ (the restricted Hamiltonian at $\mu = 0$ with the term $\mu/r_2$ removed). Then $\mathcal{H}^{\mathrm{K}}\circ\mathsf{Q}(\ell,g,L,G) = -\dfrac{1}{2L^2} - G$ on $\mathcal{D}_{\mathbb{R}}$. For collision-free $z = (q,p)$ at $\mu = 0$ one has $\mathcal{H}_0^{\mathrm{R3}}(z) = \mathcal{H}^{\mathrm{K}}(z)$ (the second primary enters with the coefficient $\mu = 0$).*
*Source: Chi05 (3.43) at $\varepsilon = 0$, p. 16 (the rotating-frame term $-G$); ours.*

*Proof.* **Step S7.33: Formula.** $q_1p_0 - q_0p_1 = -\Gamma(q,p)$ by definition, so $\mathcal{H}^{\mathrm{K}} = E - \Gamma$; then Proposition 7.8(b),(c). The second statement: at $\mu = 0$ the definition of $\mathcal{H}_0^{\mathrm{R3}}$ has $(1-0)/r_1 = 1/|q|$ and $0/r_2 = 0$. □

**Proposition 7.13** (Mean values)**.** *For $(L,G)\in\mathcal{C}_0$, $g\in\mathbb{R}$, $e = e(L,G)$, $a = L^2$, and $(q,p) = \mathsf{Q}(\ell,g,L,G)$, $r = |q|_2$:*

$$\frac{1}{2\pi}\int_0^{2\pi}\frac{d\ell}{r} = \frac1a, \qquad \frac{1}{2\pi}\int_0^{2\pi}|p|_2^2\,d\ell = \frac1a \ \ (\textit{virial}), \qquad \frac{1}{2\pi}\int_0^{2\pi} r\,d\ell = a\Big(1+\frac{e^2}{2}\Big).$$

*Source: standard (time averages of the Kepler problem; Chi05 does not state them); ours, as needed by track P.*

*Proof.* **Step S7.34: Change of variable.** $\ell \mapsto u = u_{\mathbb{R}}(\ell, e)$ is increasing and $C^1$ with $u(0) = 0$, $u(2\pi) = 2\pi$, and $d\ell = \mathsf{D}\, du$ (Proposition 7.3(b),(c)); by (X7), $\int_0^{2\pi} \phi(u(\ell))\, d\ell = \int_0^{2\pi} \phi(u)\mathsf{D}(u)\, du$ for continuous $\phi$, $\mathsf{D}(u) = 1 - e\cos u$.

**Step S7.35: Integrals.** $r = a\mathsf{D}$ (Proposition 7.8): $\int \frac{d\ell}{r} = \int_0^{2\pi} \frac{\mathsf{D}\, du}{a\mathsf{D}} = \frac{2\pi}{a}$. $\int r\, d\ell = a\int_0^{2\pi} \mathsf{D}^2 du = a(2\pi + \pi e^2)$ (using $\int \cos = 0$, $\int \cos^2 = \pi$). The virial identity: $\frac{1}{2}|p|^2 = E + 1/r$ with $E = -\frac{1}{2a}$ gives $\frac{1}{2\pi}\int |p|^2 d\ell = -\frac{1}{a} + \frac{2}{a} = \frac{1}{a}$. □

**Proposition 7.14** (Scaling to a general Kepler strength)**.** *For $k > 0$ let $E_k(q,p) = \frac{1}{2}|p|^2 - k/|q|$ and $\mathsf{S}_k(Q,P) = (Q/k, kP)$. Then $\mathsf{S}_k$ is linear and symplectic ($D\mathsf{S}_k^\top \Omega_0 D\mathsf{S}_k = \Omega_0$), $E_k \circ \mathsf{S}_k = k^2 E_1$, and if $(Q,P)(\tau)$ solves Hamilton's equations of $E_1$ then $(q,p)(t) = \mathsf{S}_k(Q,P)(k^2 t)$ solves those of $E_k$. The semi-major axis of the $q$-orbit is $a_k = a/k$, and the time averages over the mean anomaly (Proposition 7.13) are*

$$\left\langle \frac{1}{|q|_2} \right\rangle = \frac{1}{a_k}, \qquad \left\langle |p|_2^2 \right\rangle = \frac{k}{a_k}, \qquad E_k = -\frac{k}{2a_k}.$$

*(Mass $m$ in the planar problem $H = |p|^2/(2m) - Gm_0 m/|q|$: take $p = mv$; this is the dictionary used by track P.)*
*Source: standard; ours.*

*Proof.* **Step S7.36: Algebra.** $E_k(Q/k, kP) = \frac{k^2}{2}|P|^2 - k \cdot k/|Q| = k^2 E_1(Q,P)$; $D\mathsf{S}_k = \mathrm{diag}(k^{-1}, k^{-1}, k, k)$ gives the symplectic identity as in Lemma 6.7(a).

**Step S7.37: Flow and averages.** Naturality as in Step S7.23 (valid for any linear symplectic map): $D\mathsf{S}_k\, X_{E_k \circ \mathsf{S}_k} = X_{E_k} \circ \mathsf{S}_k$, i.e. $k^2 D\mathsf{S}_k\, X_{E_1} = X_{E_k} \circ \mathsf{S}_k$; so if $\frac{d}{d\tau}(Q,P) = X_{E_1}$ then $\frac{d}{dt}\mathsf{S}_k(Q,P)(k^2 t) = k^2 D\mathsf{S}_k X_{E_1} = X_{E_k}$. Averages: $\frac{1}{|q|} = k/|Q|$ has mean $k/a = 1/a_k$; $|p|^2 = k^2|P|^2$ has mean $k^2/a = k/a_k$; $E_k = k^2 E_1 = -k^2/(2a) = -k/(2a_k)$. □

7.5. **A common complex neighbourhood for the Delaunay chart.** Fix a nonempty compact set $\mathcal{K} \subset \mathcal{C}_0$ (the *safe action region*). Define

$$e_- = \min_{\mathcal{K}} e > 0, \quad e_1 = \max_{\mathcal{K}} e < 1, \quad e_2 = \tfrac{1+e_1}{2}, \quad m_L = \min_{\mathcal{K}} L > 0, \quad m_G = \min_{\mathcal{K}} G > 0,$$

$$r_{\min} = \min_{\mathcal{K}} L^2(1-e) > 0, \qquad \rho_{\max} = \max_{\mathcal{K}} L^2(1+e),$$

and $\delta_K = \delta_K(e_1)$, $\mathcal{P}(e_1)$ from Proposition 7.3. For $\delta > 0$: $\mathcal{K}_\delta = \{(L,G) \in \mathbb{C}^2 : \exists (L',G') \in \mathcal{K},\ |(L,G) - (L',G')|_\infty < \delta\}$ (open) and

$$\mathcal{U}(\delta) = \{(\ell, g, L, G) \in \mathbb{C}^4 :\ |\mathrm{Im}\,\ell| < \delta,\ |\mathrm{Im}\, g| < \delta,\ (L,G) \in \mathcal{K}_\delta\} \qquad (\delta \le \delta_K).$$

**Lemma 7.15** (Eccentricity on a complex neighbourhood)**.** *There is $\delta_1 \in (0, \min\{\delta_K, m_L, m_G\}/2]$, depending only on $\mathcal{K}$, such that for $(L,G) \in \mathcal{K}_{\delta_1}$ with nearest $(L',G') \in \mathcal{K}$: $\mathrm{Re}\, L > \frac{1}{2}m_L$, $L \neq 0$, $|G^2/L^2 - G'^2/L'^2| < \tau := \min\{e_-^2/2,\ e_2^2 - e_1^2\}$, so that $1 - G^2/L^2 \in D(e(L',G')^2,\ e_-^2/2) \subset \{\mathrm{Re}\, z > 0\}$ and $e(L,G) := \sqrt{1 - G^2/L^2}$ (principal branch) is holomorphic on $\mathcal{K}_{\delta_1}$, real positive at real $(L,G) \in \mathcal{C}_0 \cap \mathcal{K}_{\delta_1}$, with $e^2 = 1 - G^2/L^2$ and $|e(L,G)| < e_2$. Real points of $\mathcal{K}_{\delta_1}$ lie in $\mathcal{C}_0$. Hence $\mathsf{Q}(\ell, g, L, G) = \mathsf{Q}_{\mathrm{el}}(\ell, g, L, e(L,G), G/L)$ is holomorphic on $\mathcal{U}(\delta_1)$ (Proposition 7.7: $(\ell, e) \in \mathcal{P}(e_1)$ since $|\mathrm{Im}\,\ell| < \delta_K$, $|e| < e_2$), $2\pi$-periodic in $\ell$ and in $g$, and real on the real points of $\mathcal{U}(\delta_1)$, where it agrees with the real orbit map of Definition 7.6.*
*Source: ours.*

*Proof.* **Step S7.38: Uniform continuity.** $(L,G) \mapsto G^2/L^2$ is continuous where $L \neq 0$, hence uniformly continuous on the compact set $\{(L,G) \in \mathbb{C}^2 : \mathrm{dist}_\infty((L,G), \mathcal{K}) \le \min\{\delta_K, m_L, m_G\}/2\}$, on which $|L| \ge m_L/2$. So some $\delta_1$ as stated exists; the inequality $\mathrm{Re}\, L > \frac{1}{2}m_L$ and the reality assertion hold because $|L - L'| < \delta_1 \le m_L/2$ and likewise for $G$ ($G > m_G/2 > 0$ at real points).

**Step S7.39: Branches.** $|(1-G^2/L^2)-e(L',G')^2|<\tau\le e_-^2/2$ and $e(L',G')^2\ge e_-^2$, so $\mathrm{Re}(1-G^2/L^2)\ge e_-^2/2>0$; apply (X1). $|e|^2=|1-G^2/L^2|<e_1^2+\tau\le e_2^2$. For real points: $0<e^2$ and $e^2=1-G^2/L^2<1$ since $G\neq 0$, so $(L,G)\in\mathcal{C}_0$ ($L>0, G>0, G<L$). The remaining statements follow from Proposition 7.7, and from $u=u_{\mathbb{R}}$ on real points. □

7.6. **The exact restricted Hamiltonian in Delaunay variables.**

**Definition 7.16** (Potential of the shift)**.** Let $\mathsf{S}=\{z\in\mathbb{C}:\mathrm{Re}\,z>0\}$ and $\Xi=\{(q,\mu)\in\mathbb{C}^2\times\mathbb{C}:\ \vartheta(q),\ \vartheta(q+\mu\mathbf{e}),\ \vartheta(q-(1-\mu)\mathbf{e})\in\mathsf{S}\}$ (open). On $\Xi$:

$$w(q)=\vartheta(q)^{-1/2},\qquad \Psi(q,\mu)=(1-\mu)\,\vartheta(q+\mu\mathbf{e})^{-1/2}+\mu\,\vartheta(q-(1-\mu)\mathbf{e})^{-1/2},\qquad \Psi(q,0)=w(q),$$

holomorphic by (X1). For real $q$ and $\mu$, $(q,\mu)\in\Xi$ iff $q$ is collision free for $\mu$ in the sense of Definition 1.9 ($q+\mu\mathbf{e}=q-a_1(\mu)$, $q-(1-\mu)\mathbf{e}=q-a_2(\mu)$), and then $w(q)=1/\operatorname{rad}(q)$, $\Psi(q,\mu)=\frac{1-\mu}{\operatorname{rad}(q-a_1(\mu))}+\frac{\mu}{\operatorname{rad}(q-a_2(\mu))}$.

**Lemma 7.17** (The shift potential is analytic with the factor $\mu$ removed)**.** *Let $0<r_{\min}\le\rho_{\max}<1$, and choose $\varepsilon_0>0$ with $\varepsilon_0(\varepsilon_0+2\rho_{\max})<r_{\min}^2$ and $\varepsilon_0(\varepsilon_0+4)<(1-\rho_{\max})^2$ (so $\varepsilon_0<\frac14$); put*

$$\mu_2=\varepsilon_0/2,\qquad c_*=\min\{r_{\min}^2-\varepsilon_0(\varepsilon_0+2\rho_{\max}),\ (1-\rho_{\max})^2-\varepsilon_0(\varepsilon_0+4)\}>0,$$
$$\mathcal{O}=\{q\in\mathbb{C}^2:\ \exists q'\in\mathbb{R}^2,\ r_{\min}\le|q'|_2\le\rho_{\max},\ |q-q'|_2<\varepsilon_0/2\}.$$

*Then $\mathcal{O}\times\bar D(0,\mu_2)\subset\Xi$; on it $\mathrm{Re}\,\vartheta(\cdot)\ge c_*$ for the three arguments, so $|w|,|\vartheta(\cdot)^{-1/2}|\le c_*^{-1/2}$; and there is a holomorphic function $f:\mathcal{O}\times D(0,\mu_2)\to\mathbb{C}$ with*

$$\mu\,f(q,\mu)=w(q)-\Psi(q,\mu),\qquad f(q,\mu)=-\int_0^1\partial_\mu\Psi(q,t\mu)\,dt,\qquad |f(q,\mu)|\le C_{\mathrm{sh}}:=\frac{6}{\varepsilon_0\sqrt{c_*}}.$$

*For $\mu=0$: $f(q,0)=\dfrac{1}{|q|}+\dfrac{q_0}{|q|^3}-\dfrac{1}{|q-\mathbf{e}|}$ (written with $\vartheta^{1/2}$ for complex $q$). For real $q\in\mathcal{O}$ and real $\mu$ with $|\mu|<\mu_2$ the point $(q,\mu)$ is collision free and $f(q,\mu)$ is real. $C_{\mathrm{sh}}$, $\mu_2$ depend only on $(r_{\min},\rho_{\max})$.*

*Source: ours; replaces the truncated Chi05 (3.38)–(3.43), p. 13–16.*

*Proof.* **Step S7.40: Positivity of the radicands.** Let $q\in\mathcal{O}$, $q'$ as in the definition, $|\mu|\le\mu_2$. For $z,z'\in\mathbb{C}^2$: $|\vartheta(z)-\vartheta(z')|=|(z-z')\cdot(z+z')|\le|z-z'|_2\big(|z-z'|_2+2|z'|_2\big)$ (bilinear Cauchy–Schwarz). (i) $z=q$, $z'=q'$: $|z-z'|<\varepsilon_0/2$, so $\mathrm{Re}\,\vartheta(q)\ge\vartheta(q')-\varepsilon_0(\varepsilon_0+2\rho_{\max})\ge r_{\min}^2-\varepsilon_0(\varepsilon_0+2\rho_{\max})$. (ii) $z=q+\mu\mathbf{e}$, $z'=q'$: $|z-z'|\le|q-q'|+|\mu|<\varepsilon_0$, same bound. (iii) $z=q-(1-\mu)\mathbf{e}$, $z'=q'-\mathbf{e}$ (real, $\vartheta(z')=|q'-\mathbf{e}|^2\ge(1-|q'|)^2\ge(1-\rho_{\max})^2$, $|z'|\le2$): $|z-z'|<\varepsilon_0$, so $\mathrm{Re}\,\vartheta(z)\ge(1-\rho_{\max})^2-\varepsilon_0(\varepsilon_0+4)$. Hence all three lie in $\{\mathrm{Re}\ge c_*\}$, so $\mathcal{O}\times\bar D(0,\mu_2)\subset\Xi$ and the bounds $|\vartheta^{-1/2}|\le(\mathrm{Re}\,\vartheta)^{-1/2}\le c_*^{-1/2}$ hold. For real $q,\mu$ the radicands are real, hence $>0$.

**Step S7.41: Bound of the numerator.** $|\Psi|\le(1+\mu_2)c_*^{-1/2}+\mu_2c_*^{-1/2}\le\frac32c_*^{-1/2}$ ($\mu_2<\frac18$), $|w|\le c_*^{-1/2}$; so $g_q(\mu):=w(q)-\Psi(q,\mu)$ satisfies $|g_q|\le3c_*^{-1/2}$ on $\bar D(0,\mu_2)$, and $g_q(0)=0$. (The disc is contained in the open slice $\{\mu:(q,\mu)\in\Xi\}$, so $g_q$ is holomorphic near the closed disc.)

**Step S7.42: Division by $\mu$.** Define $f(q,\mu)=-\int_0^1\partial_\mu\Psi(q,t\mu)\,dt$; it is holomorphic on $\mathcal{O}\times D(0,\mu_2)$ by (X3) (the disc is star-shaped and $\Xi$ contains $\{q\}\times\bar D(0,\mu_2)$), and $\mu f(q,\mu)=g_q(\mu)$ by the fundamental theorem of calculus applied to $t\mapsto\Psi(q,t\mu)$. By the Schwarz lemma (X4), $|g_q(\mu)|\le\frac{|\mu|}{\mu_2}3c_*^{-1/2}$, hence for $\mu\neq0$: $|f|\le3c_*^{-1/2}/\mu_2=6/(\varepsilon_0\sqrt{c_*})$, and at $\mu=0$ by continuity.

**Step S7.43: Value at $\mu=0$.** $f(q,0)=-\partial_\mu\Psi(q,0)=-\big[-w(q)+\partial_\mu(\vartheta(q+\mu\mathbf{e})^{-1/2})|_0+\vartheta(q-\mathbf{e})^{-1/2}\big]$ with $\partial_\mu\vartheta(q+\mu\mathbf{e})^{-1/2}|_0=-q_0\vartheta(q)^{-3/2}$; so $f(q,0)=w(q)+q_0\vartheta(q)^{-3/2}-\vartheta(q-\mathbf{e})^{-1/2}$.

**Step S7.44: Reality.** For real $(q,\mu)$ with $\mu\neq0$, $f=(w-\Psi)/\mu$ is real; for $\mu=0$ the formula of Step S7.43 is real. □

**Proposition 7.18** (Exact restricted Hamiltonian in Delaunay variables)**.** *Let $\mathcal{K} \subset \mathcal{C}_0$ be nonempty and compact with $\rho_{\max} = \max_{\mathcal{K}} L^2(1+e) < 1$ (the condition $a_+(1+e_+) < 1$ of Theorem 1.19), and let $r_{\min}$, $\varepsilon_0, \mu_2, c_*, C_{\mathrm{sh}}$ be as in Lemma 7.17 for $(r_{\min}, \rho_{\max})$. There exist $\delta_2 \in (0, \delta_1]$ and a holomorphic function*

$$F:\ \mathcal{U}(\delta_2) \times D(0,\mu_2) \to \mathbb{C}, \qquad (w,\mu) \mapsto F_\mu(w) = f\big(q(w),\mu\big), \quad q(w) = \text{the position part of } \mathsf{Q}(w),$$

*such that, with constants depending on $\mathcal{K}$ only (not on $\mu$):*

*(i)* (uniform bound) $|F_\mu(w)| \le C_F := C_{\mathrm{sh}} = 6/(\varepsilon_0\sqrt{c_*})$ *on* $\mathcal{U}(\delta_2) \times D(0,\mu_2)$*;*

*(ii)* (periodicity, reality) $F_\mu(\ell + 2\pi, g, L, G) = F_\mu(\ell, g + 2\pi, L, G) = F_\mu(w)$*;* $F_\mu(w) \in \mathbb{R}$ *for real* $w \in \mathcal{U}(\delta_2)$ *and real* $\mu \in (-\mu_2, \mu_2)$*;*

*(iii)* (exactness) *for every real* $w = (\ell, g, L, G) \in \mathcal{U}(\delta_2)$ *and real* $\mu$ *with* $|\mu| < \mu_2$*:* $(L, G) \in \mathcal{C}_0$*, the point* $z = \mathsf{Q}(w)$ *is collision free (*`CollisionFree` $\mu$ $z$*), and*

$$\mathcal{H}^{\mathrm{R3}}_\mu\big(\mathsf{Q}(w)\big) = -\frac{1}{2L^2} - G + \mu\, F_\mu(w);$$

*(iv)* (the neighbourhood contains the safe region) *every real* $(\ell, g, L, G)$ *with* $(L, G) \in \mathcal{K}$ *lies in* $\mathcal{U}(\delta_2)$*.*

*Here $\mathcal{H}^{\mathrm{R3}}_\mu$ is the frozen restricted Hamiltonian (Definition 1.10); $F_\mu$ is the* exact *barycentric perturbation, including the displacement of both primaries ($a_1(\mu) = (-\mu, 0)$, $a_2(\mu) = (1-\mu, 0)$) and the indirect term $q_0/|q|^3$ at $\mu = 0$.*

*Source: ours; the model of Chi05 (3.38)–(3.43), pp. 13–16 is heliocentric and truncated and is not used.*

*Proof.* **Step S7.45: The real shell.** For real $w$ with $(L', G') \in \mathcal{K}$: $r = |q(w)|_2 \in [L'^2(1-e'), L'^2(1+e')] \subset [r_{\min}, \rho_{\max}]$ (Proposition 7.8(a)).

**Step S7.46: Choice of $\delta_2$.** Put $\delta_* = \delta_1/2$. The closure of $\{w \in \mathcal{U}(\delta_*) : 0 \le \operatorname{Re}\ell, \operatorname{Re} g \le 2\pi\}$ is compact and contained in $\mathcal{U}(\delta_1)$, on which $\mathsf{Q}$ is holomorphic (Lemma 7.15); by periodicity in $\ell, g$ and uniform continuity there is $\delta_2 \le \delta_*$ with: for every $w \in \mathcal{U}(\delta_2)$ there is a real $w' = (\operatorname{Re}\ell, \operatorname{Re} g, L', G')$ with $(L', G') \in \mathcal{K}$ and $|\mathsf{Q}(w) - \mathsf{Q}(w')|_2 < \varepsilon_0/2$ (the position parts). Then $q(w) \in \mathcal{O}$, because $q' = q(w')$ is a real point with $r_{\min} \le |q'| \le \rho_{\max}$ by Step S7.45. (This is the only place where $\delta_2$ is shrunk, and it depends only on $\mathcal{K}$.)

**Step S7.47: Holomorphy and bound.** $F_\mu(w) = f(q(w), \mu)$ is a composition of holomorphic maps ($q(w) \in \mathcal{O}$, $\mu \in D(0,\mu_2)$), with the bound $C_{\mathrm{sh}}$ of Lemma 7.17: (i). Periodicity of $q(\cdot)$ gives (ii) together with reality of $f$ for real points of $\mathcal{O}$ and real $\mu$ (Lemma 7.17) and reality of $\mathsf{Q}$ on real points. (iv): $\delta_2 > 0$ and $\mathcal{K} \subset \mathcal{K}_{\delta_2}$.

**Step S7.48: Exactness.** Let $w$ real in $\mathcal{U}(\delta_2)$, $\mu$ real, $|\mu| < \mu_2$. $(L, G) \in \mathcal{C}_0$ by Lemma 7.15; $q = q(w)$ is a real point of $\mathcal{O}$, so $(q, \mu) \in \Xi$ with real positive radicands, i.e. collision free, and $\operatorname{rad}(q - a_1(\mu)) = \sqrt{\vartheta(q + \mu\mathbf{e})}$, $\operatorname{rad}(q - a_2(\mu)) = \sqrt{\vartheta(q - (1-\mu)\mathbf{e})}$ (`radius`, `radiusSq`), so $\mathcal{H}^{\mathrm{R3}}_\mu(q,p) = \frac12|p|^2 + q_1p_0 - q_0p_1 - \Psi(q,\mu) = \mathcal{H}^{\mathrm{R3}}_0(q,p) + \big[w(q) - \Psi(q,\mu)\big] = -\frac{1}{2L^2} - G + \mu f(q,\mu)$, using $\mathcal{H}^{\mathrm{R3}}_0 = \frac12|p|^2 + q_1p_0 - q_0p_1 - w(q)$ with $w(q) = 1/|q|_2$ (Lemma 7.12, valid for the real $w$ since $\mathsf{Q}(w) \in \mathcal{C}$ for $(L, G) \in \mathcal{C}_0$) and Lemma 7.17. □

**Corollary 7.19** (Period-one form: the data of I-HEAD)**.** *Let $\mathcal{K}, \delta_2, \mu_2, C_F, F$ be as in Proposition 7.18, $s_0 = \delta_2/(2\pi)$, $r_0 = 2\pi\delta_2$, and $B \subset 2\pi\mathcal{K} = \{2\pi(L,G) : (L,G) \in \mathcal{K}\}$ an arbitrary subset. Let $\mathsf{S}$ be the conversion of Definition 6.6, $\tilde h = h_{(2\pi)^2, 1/(2\pi)}$, and for real $\mu$, $|\mu| < \mu_2$,*

$$\tilde f_\mu(\theta, I) = \mu\, F_\mu\big(\mathsf{S}(\theta, I)\big) = \mu\, F_\mu\big(2\pi\theta_1, 2\pi\theta_2, \tfrac{I_1}{2\pi}, \tfrac{I_2}{2\pi}\big), \qquad (\theta, I) \in T_{s_0} \times B_{r_0}.$$

*Then (a) $\mathsf{S}(T_{s_0} \times B_{r_0}) \subset \mathcal{U}(\delta_2)$; (b) $\tilde h$ is holomorphic on $B_{r_0}$ with $\operatorname{Re} I_1 > \pi m_L > 0$ there, and real at real $I$; (c) $\tilde f_\mu$ is holomorphic on $T_{s_0} \times B_{r_0}$, 1-periodic in $\theta$ (in each variable), real at real $\theta$ and*

*real $I \in B_{r_0}$ for real $\mu$, and $|\tilde f_\mu| \le \varepsilon := |\mu|\, C_F$; (d) (exactness) for every real $\theta$ and real $I \in B_{r_0}$:*

$$\mathcal{H}^{\mathrm{R3}}_\mu\big(\mathsf{Q}(\mathsf{S}(\theta, I))\big) = \tilde h(I) + \tilde f_\mu(\theta, I) = \texttt{realHamiltonian}\ \tilde h\ \tilde f_\mu\ (\theta, I),$$

*with $\mathsf{Q}\circ\mathsf{S}$ symplectic ($D(\mathsf{Q}\circ\mathsf{S})^{\top}\Omega_0\, D(\mathsf{Q}\circ\mathsf{S}) = \Omega_0$); (e) all of $s_0, r_0, \varepsilon/|\mu|$ depend on $\mathcal{K}$ only. In particular the hypotheses on $(h, f, \varepsilon)$ of Theorem 1.16 (with $h = \tilde h$, $f = \tilde f_\mu$, $B \subset 2\pi\mathcal{K}$) other than the nondegeneracy are satisfied with the smallness parameter $\varepsilon = \mu C_F$, $\mu \in [0, \mu_2)$.*
*Frozen Lean declaration: `KAM.realHamiltonian, KAM.complexStrip, KAM.actionNbhd`*
*Source: ours.*

*Proof.* **Step S7.49: Domains.** If $\theta \in T_{s_0}$ then $|\mathrm{Im}\, 2\pi\theta_i| < \delta_2$. If $I \in B_{r_0}$, there is $J \in B$, $J = 2\pi(L', G')$ with $(L', G') \in \mathcal{K}$, $\|I - J\| < 2\pi\delta_2$, so $|I/(2\pi) - (L', G')|_\infty < \delta_2$, i.e. $I/2\pi \in \mathcal{K}_{\delta_2}$: (a). By Lemma 7.15, $\mathrm{Re}(I_1/2\pi) > m_L/2$, so $I_1 \neq 0$.

**Step S7.50: Properties.** (b): Lemma 6.7(b) and Corollary 6.8's domain argument. (c): composition of the holomorphic $F_\mu$ with the linear $\mathsf{S}$; periodicity: $\theta \mapsto \theta + k$, $k \in \mathbb{Z}^2$, becomes $(\ell, g) \mapsto (\ell, g) + 2\pi k$; reality and the bound: Proposition 7.18(i),(ii) and $|\mu| \le \mu_2$ (the factor $\mu$).

**Step S7.51: Exactness.** Proposition 7.18(iii) at $w = \mathsf{S}(\theta, I)$ and $-\frac{1}{2L^2} - G = h_{1,1}(L, G) = \tilde h(I)$ (Lemma 6.7(b), $\kappa = \Omega = 1$); `realHamiltonian` is the real part of the same number, which is real. Symplecticity: Proposition 7.9, Lemma 6.7(a), and the chain rule for the product of symplectic matrices. □

**Proposition 7.20** (Transfer of Delaunay solutions to the physical problem)**.** *In the setting of Corollary 7.19, let $\mu \in \mathbb{R}$ with $|\mu| < \mu_2$, $J \subset \mathbb{R}$, $\tilde z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with* $\mathrm{Sol}(\texttt{realHamiltonian}\ \tilde h\ \tilde f_\mu,\ B,\ J,\ \tilde z)$ *(Definition 1.8). Then $z := \mathsf{Q}\circ\mathsf{S}\circ\tilde z$ satisfies* $\mathrm{RSol}_\mu(J, z)$*; moreover for $t \in J$, with $\tilde z(t) = (\theta, I)$:* $\mathrm{rad}(q(t)) = L^2(1 - e\cos u) \in [a(1-e), a(1+e)]$, *$a = L^2 = I_1^2/(2\pi)^2$ etc. (the elements of $z(t)$ are $(L, G) = I/(2\pi)$), and $\mathcal{H}^{\mathrm{R3}}_\mu(z(t)) = \texttt{realHamiltonian}\,\tilde h\ \tilde f_\mu(\tilde z(t))$.*
*Frozen Lean declaration: `KAM.Iso.IsRestrictedSolutionOn, KAM.Iso.IsHamiltonianSolutionOn`*
*Source: ours (the transfer step of D-I-PHYSICAL).*

*Proof.* **Step S7.52: Collision freedom and real-analyticity.** For $t \in J$, $I = \tilde z(t)_2 \in B$, so $I/2\pi \in \mathcal{K}$ and $\mathsf{S}\tilde z(t)$ is real with $(L, G) \in \mathcal{K}$, in $\mathcal{U}(\delta_2)$ (Proposition 7.18(iv)); $z(t) = \mathsf{Q}(\mathsf{S}\tilde z(t))$ is collision free by (iii) there; so $z(t) \in W_\mu$.

**Step S7.53: Agreement of Hamiltonians on an open set.** Let $\mathcal{V} = \{(\theta, I) \in \mathbb{R}^4 : \exists J \in B,\ \|I - J\| < r_0\}$, open, containing $\tilde z(t)$. By Corollary 7.19(d), $\texttt{realHamiltonian}\,\tilde h\ \tilde f_\mu = \mathcal{H}^{\mathrm{R3}}_\mu \circ \mathsf{Q}\circ\mathsf{S}$ on $\mathcal{V}$ (real arguments $I \in B_{r_0}$: note the identity is used with $B_{r_0}$, not only $B$); both sides are real-analytic there, so the real derivatives and the Hamiltonian fields coincide at $\tilde z(t)$.

**Step S7.54: Chain rule.** $\tilde z$ is differentiable at $t$ with $\tilde z'(t) = X_{\texttt{realHamiltonian}\tilde h\tilde f_\mu}(\tilde z(t)) = X_{\mathcal{H}_\mu \circ \mathsf{Q}\circ\mathsf{S}}(\tilde z(t))$. By Proposition 7.9 and Lemma 6.7(a), $\frac{d}{dt} z = D(\mathsf{Q}\circ\mathsf{S})\tilde z' = X_{\mathcal{H}^{\mathrm{R3}}_\mu}(z(t)) = V_\mu(z(t))$ by Lemma 7.1. The formulas for $\mathrm{rad}(q)$ and the energy are Proposition 7.8(a) and Corollary 7.19(d). □

### 7.7. **Nondegeneracy.**

**Proposition 7.21** (Isoenergetic nondegeneracy of the Delaunay Hamiltonian)**.** *In the setting of Corollary 7.19,* $\det \texttt{borderedHessian}\,\tilde h\, I = 3/I_1^4 > 0$ *for every real $I \in B_{r_0}$, indeed for every real $I$ with $I_1 > 0$; in conventional variables $\mathsf{B}_{h_{1,1}}(L, G) = \mathsf{K}_{1,1}(L)$ with $\det = 3/L^4$. Thus $\tilde h$ satisfies the nondegeneracy hypothesis of Theorem 1.16 on all real points of $B_{r_0}$, for every $B \subset 2\pi\mathcal{K}$.*
*Source: Chi05 p. 17, the display after (3.43) (conventional variables, $\kappa = \Omega = 1$); the period-one version is Part 6.*

*Proof.* **Step S7.55: Reality of the domain.** Real $I \in B_{r_0}$ have $I_1 > 0$: $\operatorname{Re} I_1/2\pi > m_L/2 > 0$ (Corollary 7.19(b)).

**Step S7.56: Determinant.** Lemma 6.5 and Lemma 6.7(d) with $\kappa = \Omega = 1$; the conventional statement is Lemma 6.5 at $\kappa = \Omega = 1$. □

7.8. **Summary of outputs, forward references, errata.**

*Remark* 7.22 (Outputs consumed by other parts).

- *Part 8 (proof of I-R3BP; forward reference).* Lemma 7.11 (the region $\mathcal{N}$ as the $\mathsf{Q}$-image of an action box), Theorem 7.10 (every $z_0 \in \mathcal{N}$ has Delaunay coordinates), Corollary 7.19 (data of I-HEAD with $\varepsilon = C_F\mu$), Proposition 7.21, Proposition 7.20, Proposition 7.8(a) and Proposition 7.18(iii). To use them with the frozen hypothesis $a_+(1+e_+) < 1$, Part 8 takes $\mathcal{K} \supset$ the closure of the $(L, G)$-image of $\mathcal{N}$ enlarged as in finding I04 of the Part-0 review, still satisfying $\max_{\mathcal{K}} L^2(1+e) < 1$ and $e < 1$, $e > 0$.
- *Track P.* Proposition 7.3 (Kepler's equation near $e = 0$), Proposition 7.7 (orbit maps holomorphic across $e = 0$), Proposition 7.8, Proposition 7.13, Proposition 7.14; the normalisation of Part 6.

*Remark* 7.23 (Lean totality). Every division and square root in this part is guarded: $L > 0$ on $\mathcal{C}_0$ and $L \neq 0$ on $\mathcal{K}_{\delta_1}$ (Lemma 7.15); $\mathsf{D} = 1 - e\cos u \neq 0$ (Proposition 7.3(a)); $e(L, G)$ uses a principal root of a number with positive real part; $\vartheta(\cdot)^{-1/2}$ is taken on $\{\mathrm{Re} > 0\}$; the radii of the primaries are positive on $W_\mu$ and on $\mathcal{O} \times D(0, \mu_2)$ (Lemma 7.17). The only `fderiv` is that of $\mathcal{H}_\mu^{\mathrm{R3}} \circ \mathsf{Q} \circ \mathsf{S}$ and of $\tilde{h}$ at points where they are real-analytic (Proposition 7.20, Lemma 6.5).

*Remark* 7.24 (Suspected defects of the source, for the errata ledger). (1) Chi05 p. 11: "The (Kepler) equation $\lambda = u - e\sin u$ ... can be inverted, for $|e|$ small enough": no radius is given and no domain in $\lambda$; not an error of the displayed expansions, a gap, filled by Proposition 7.3. (2) Chi05 (3.38)–(3.43), pp. 13–16: the model is the *heliocentric*, time-dependent system ($x^{(2)} - x^{(1)}$ with the circular motion of $P_1$ about $P_0$) with the primary at the origin, and the parameter $\varepsilon = m_1/m_0^{2/3}$ rescales mass and length; the frozen restricted Hamiltonian is the barycentric one with $\mu$ and total mass one, in which the primaries sit at $-\mu, 1-\mu$ and the indirect term appears as the $\mu$-dependence of the first primary's position. The two are related but different; this blueprint derives $F_\mu$ from the frozen $\mathcal{H}_\mu^{\mathrm{R3}}$ directly (Proposition 7.18).